Handwritten Stochastic Calculus Notes

Transcribed from handwritten chapter notes

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1 Brownian Motion and Its First Properties

1.1 Crash Course on Stochastic Processes

All random variables, if not specified, are defined on the probability space

$$(\Omega, \mathcal{F}, \mathbb{P}).$$

Definition 1.1. *We say that $(X_t)_{t\in T}$ is a stochastic process on the measurable space $(X, \mathcal{X})$ if, for every $t \in T$, X_t is a random variable, i.e.*

$$\omega \longmapsto X_t(\omega) \quad \textit{is } \mathcal{F}/\mathcal{X}\textit{-measurable.}$$

Remark 1.2. We are particularly interested in the case

$$T = \mathbb{R}_+ \qquad \text{and} \qquad X = \mathbb{R} \text{ or } \mathbb{R}^d$$

endowed with their Borel σ-fields $\mathcal{B}(\mathbb{R})$ or $\mathcal{B}(\mathbb{R}^d)$.

Remark 1.3. We characterize the distribution of a stochastic process through its finite-dimensional distributions.

Definition 1.4. (Distribution of a stochastic process through finite-dimensional distributions.)

Let $(X_t)_{t\in T}$ be a stochastic process valued in $(X, \mathcal{X})$. We call fidi *any distribution ν_I^X, for $I \subset T$ finite, on $(X^I, \mathcal{X}^{\otimes I})$ that corresponds to the distribution of*

$$(X_{t_1}, \dots, X_{t_n})$$

if $I = \{t_1, \dots, t_n\}$.

Then the distribution of (X_t) is the collection of all fidis, i.e.

$$\{\nu_I^X : I \subset T \textit{ finite}\}.$$

Corollary 1.5. *Let $(X_t)_{t\in T}$ and $(Y_t)_{t\in T}$ be two stochastic processes both valued in $(X, \mathcal{X})$. The processes $(X_t)_t$ and $(Y_t)_t$ have the same distribution as processes if*

$$\forall n,\ \forall t_1, \dots, t_n \in T \textit{ pairwise distinct}, \qquad (X_{t_1}, \dots, X_{t_n}) \overset{(d)}{=} (Y_{t_1}, \dots, Y_{t_n}).$$

Definition 1.6. *Let $(X_t^1)_{t\in T^1}$ and $(X_t^2)_{t\in T^2}$ be two stochastic processes valued in $(X^1, \mathcal{X}^1)$ and $(X^2, \mathcal{X}^2)$. We say that they are independent if, for every finite family of times $t_1^1, \dots, t_n^1 \in T^1$ and $t_1^2, \dots, t_m^2 \in T^2$,*

$$(X_{t_1^1}^1, \dots, X_{t_n^1}^1) \quad \textit{is independent of} \quad (X_{t_1^2}^2, \dots, X_{t_m^2}^2).$$

1.2 Definitions and First Properties

Definition 1.7. *We say that $(B_t)_{t\geq 0}$, a real stochastic process, is a (1D-standard) Brownian motion if*

(i) it is almost surely continuous,

(ii) $B_0 = 0$ a.s.,

(iii) for every n and every $0 \leq t_1 < \cdots < t_n$,

$$B_{t_1},\ B_{t_2} - B_{t_1},\ \dots,\ B_{t_n} - B_{t_{n-1}}$$

are independent random variables,

(iv) for every $s, t \geq 0$ with $s < t$,

$$B_t - B_s \sim \mathcal{N}(0, t-s).$$

Remark 1.8. Some authors define a Brownian motion as a real process satisfying (ii)–(iv), i.e. not supposing the continuity condition (i). Starting from $(\widetilde{B}_t)_{t\geq 0}$ satisfying (ii) to (iv), there exists a modification $(B_t)_{t\geq 0}$ that is continuous and even $(1/2-\varepsilon)$-Hölder for every $\varepsilon \in (0, 1/2)$.

Definition 1.9. *Let $(X_t)_{t\in T}$ and $(\widetilde{X}_t)_{t\in T}$ be two stochastic processes. They are said to be modifications of each other if*

$$\forall t \in T, \qquad \mathbb{P}(X_t = \widetilde{X}_t) = 1.$$

Theorem 1.10. *Let $(\widetilde{B}_t)_{t\geq 0}$ satisfy* (ii) *to* (iv)*. Then there exists a modification $(B_t)_{t\geq 0}$ that is $(1/2-\varepsilon)$-Hölder for every $\varepsilon \in (0, 1/2)$.*

Note that if T is not countable, the previous definition does not imply

$$\mathbb{P}\big(\forall t \in T,\ X_t = \widetilde{X}_t\big) = 1.$$

Thus one calls for the following definition.

Definition 1.11. *Let $(X_t)_{t\in T}$ and $(\widetilde{X}_t)_{t\in T}$ be two stochastic processes. They are said indistinguishable if*

$$\mathbb{P}\big(\forall t \in T,\ X_t = \widetilde{X}_t\big) = 1.$$

Note that if $(X_t)_{t\geq 0}$ and $(\widetilde{X}_t)_{t\geq 0}$ are continuous processes a.s. and if $(X_t)_{t\in\mathbb{Q}_+}$ and $(\widetilde{X}_t)_{t\in\mathbb{Q}_+}$ are modifications of each other, then $(X_t)_{t\geq 0}$ and $(\widetilde{X}_t)_{t\geq 0}$ are indistinguishable.

Remark 1.12. We define similarly $(B_t)_{t\in[0,T]}$ for any T. We simply restrict the times in (iii).

1.3 Definition 1 Naturally Characterizes the Distribution of $(B_t)_{t\geq 0}$

Proposition 1.13. *$(B_t)_{t\geq 0}$ is a 1-D Brownian motion if and only if it is a.s. continuous and, for every n and every $t_1 < \cdots < t_n$,*

$$(B_{t_1}, \ldots, B_{t_n})$$

is an n-dimensional Gaussian vector with zero mean and covariance matrix Σ with components

$$\Sigma_{ij} = \mathrm{Cov}(B_{t_i}, B_{t_j}) = t_i \wedge t_j.$$

Proof. Let G be a random variable. Then we know that

$$G \sim \mathcal{N}_n(0, \mathrm{diag}(a_1, \ldots, a_n)) \iff MG \sim \mathcal{N}_n(0, \Sigma_a),$$

where

$$M = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 1 & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 1 & \cdots & 1 & 1 \end{pmatrix} \in \mathbb{R}^{n\times n},$$

and Σ_a is the matrix with (i,j)-components

$$\sum_{\ell=1}^{i\wedge j} a_\ell.$$

The proof is then a simple application of this result for

$$G = (B_{t_1}, \ldots, B_{t_n} - B_{t_{n-1}}),$$

noting that

$$MG = (B_{t_1}, \dots, B_{t_n})$$

and

$$\sum_{\ell=1}^{i\wedge j}(t_\ell - t_{\ell-1}) = t_i \wedge t_j, \qquad \text{with } t_0 = 0.$$

Let us show the equivalence above. Note that we only need to show the forward implication, since M is invertible. Then the result just follows from the fact that

$$MG \sim \mathcal{N}\big(0, M\,\mathrm{diag}(a_1, \dots, a_n)M^\top\big),$$

where the components of $M\,\mathrm{diag}(a_1, \dots, a_n)M^\top$ are equal to

$$\begin{aligned}\mathrm{Cov}((MG)_i, (MG)_j) &= \mathbb{E}\left[\left(\sum_{\ell=1}^{i} G_\ell\right)\left(\sum_{\ell=1}^{j} G_\ell\right)\right] \\ &= \sum_{\ell=1}^{i\wedge j} a_\ell.\end{aligned}$$

□

Remark 1.14. The proposition is equivalent to the fact that $(B_t)_{t\geq 0}$ is a 1-D Brownian motion if and only if it is a continuous Gaussian process with mean $t \mapsto 0$ and covariance function $(s,t) \mapsto s \wedge t$ according to the following definition.

Definition 1.15. *$(X_t)_{t\in T}$ is a Gaussian process with mean function $t \mapsto m(t)$ and covariance function $(s,t) \mapsto \gamma(s,t)$ if, for every n and every pairwise distinct $t_1, \dots, t_n \in T$, $(X_{t_1}, \dots, X_{t_n})$ is a Gaussian vector with mean and covariance specified by*

$$\mathbb{E}[X_{t_i}] = m(t_i) \qquad \textit{and} \qquad \mathrm{Cov}(X_{t_i}, X_{t_j}) = \gamma(t_i, t_j).$$

Proposition 1.16. *Let $(B_t)_{t\geq 0}$ be a Brownian motion. Then the following processes are also Brownian motions:*

(a) $(-B_t)_{t\geq 0}$,

(b) $(\widetilde{B}_t)_{t\geq 0}$ *defined by*

$$\widetilde{B}_t = \begin{cases} tB_{1/t}, & t > 0, \\ 0, & t = 0, \end{cases}$$

(c) for every $a > 0$,

$$\left(\frac{1}{\sqrt{a}} B_{at}\right)_{t\geq 0},$$

(d) for every $T > 0$,

$$(B_T - B_{T-t})_{t\in[0,T]}.$$

Proof. The only difficult part is (b). Indeed, (a), (c) and (d) just consist in checking the definition.

Regarding (b), denote by

$$\widetilde{B}_t = \begin{cases} tB_{1/t}, & t > 0, \\ 0, & t = 0. \end{cases}$$

Then, for $0 \leq t_1 < \cdots < t_n$,

$$G = (\widetilde{B}_{t_1},\ \widetilde{B}_{t_2} - \widetilde{B}_{t_1},\ \dots,\ \widetilde{B}_{t_n} - \widetilde{B}_{t_{n-1}})$$

is a linear transformation of

$$(B_{1/t_1}, \dots, B_{1/t_n}),$$

with the convention $B_{1/t_i} = 0$ if $t_i = 0$. Therefore G is Gaussian with mean zero, and a covariance computation shows that it satisfies the same conditions as in the definition of a Brownian motion.

It remains to show (i), and more precisely the continuity at 0. Then $(\widetilde{B}_t)_{t\geq 0}$ satisfies (ii) to (iv) in the definition, so it admits a continuous modification $(\overline{B}_t)_{t\geq 0}$. Since $(\widetilde{B}_t)_{t\geq 0}$ is continuous a.s. on $\mathbb{Q}_+^*$ and $\mathbb{Q}_+^*$ is countable, one gets

$$\mathbb{P}\left(\sup_{t\in\mathbb{Q}_+^*} |\overline{B}_t - \widetilde{B}_t| > 0\right) \leq \sum_{t\in\mathbb{Q}_+^*} \mathbb{P}(\overline{B}_t \neq \widetilde{B}_t) = 0.$$

Hence $\overline{B}_t = \widetilde{B}_t$ for every $t > 0$ a.s., which implies

$$\lim_{t\downarrow 0} \widetilde{B}_t = \lim_{t\downarrow 0} \overline{B}_t = 0.$$

□

Definition 1.17. *We say that $(B_t)_{t\geq 0}$, a stochastic process valued in $\mathbb{R}^d$, is a (d-standard) Brownian motion if*

(i) it is almost surely continuous,

(ii) $B_0 = 0$ a.s.,

(iii) for every n and every $0 \leq t_1 < \cdots < t_n$,

$$B_{t_1},\ B_{t_2} - B_{t_1},\ \dots,\ B_{t_n} - B_{t_{n-1}}$$

are independent random variables,

(iv) for every $s, t \geq 0$ with $s < t$,

$$B_t - B_s \sim \mathcal{N}(0, (t-s)I_d).$$

Using the definition introduced previously we immediately get the following.

Proposition 1.18. *Let $\{(B_t^i)_{t\geq 0} : i = 1, \dots, d\}$ be d independent 1-D standard Brownian motions. Then*

$$B_t = (B_t^1, \dots, B_t^d)_{t\geq 0}$$

is a d-dimensional standard Brownian motion. Conversely, if $(B_t)_{t\geq 0}$ is a d-dimensional Brownian motion, then its components are d independent 1-D standard Brownian motions.

1.4 Construction of the Brownian Motion and Wiener Space

Theorem 1.19. *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space supporting a double indexed sequence of i.i.d. random variables $(G_{n,m})_{n,m\in\mathbb{N}^2}$ with distribution $\mathcal{N}(0,1)$. Then there exists a Brownian motion $(B_t)_{t\in[0,1]}$ defined on $(\Omega, \mathcal{F}, \mathbb{P})$.*

Proof. We define a sequence of processes

$$\{(X_m(t))_{t\in[0,1]} : m \in \mathbb{N}\}$$

such that, almost surely, this sequence of functions is Cauchy in the Banach space $C([0,1], \mathbb{R})$ endowed with the uniform norm

$$f \longmapsto \sup_{t\in[0,1]} |f_t| = \|f\|_\infty.$$

Then we show that the limit is a Brownian motion on $[0,1]$.

We define

$$X_0(t) = t\, G_{1,1},$$

and for $n \geq 0$,

$$X_{n+1}(t) = (2^{n+1}t - \lfloor 2^{n+1}t \rfloor) X_{n+1}\left(\frac{\lceil 2^{n+1}t \rceil}{2^{n+1}}\right) + (\lceil 2^{n+1}t \rceil - 2^{n+1}t) X_{n+1}\left(\frac{\lfloor 2^{n+1}t \rfloor}{2^{n+1}}\right).$$

More explicitly, for $k \in \{0, \dots, 2^{n+1}\}$,

$$X_{n+1}\left(\frac{k}{2^{n+1}}\right) = \begin{cases} X_n\left(\dfrac{k}{2^{n+1}}\right), & \text{if } k \equiv 0 \ [2], \\ X_n\left(\dfrac{k}{2^{n+1}}\right) + \dfrac{G_{k,n}}{2^{(n+1)/2}}, & \text{otherwise.} \end{cases}$$

In other words, we define by induction the values

$$\{X_n(k/2^n)\}$$

and consider $(X_n(t))_{t\in[0,1]}$ as the linear interpolation of these points.

We show that $\{(X_n(t))_{t\in[0,1]} : n \in \mathbb{N}^+\}$ is Cauchy. We define

$$A_n = \left\{ \|X_{n+1} - X_n\|_\infty \geq 2^{-n/4} \right\}.$$

Then, by the dyadic construction,

$$A_n \subset \bigcup_{k=0}^{2^n-1} \left\{ |G_{k,n}| \geq 2^{-n/4}\, 2^{(n+1)/2} \right\}.$$

which implies by Markov's inequality

$$\begin{aligned} \mathbb{P}(A_n) &\leq \left(2^{-n/4+(n+2)/2}\right)^{-8} \sum_k \mathbb{E}[|G_{k,n}|^8] \\ &\leq 2^{-2n-8} \times 2^n \times \mathbb{E}[|G_{1,1}|^8] \leq 2^{-n-8}. \end{aligned}$$

Hence

$$\sum_n \mathbb{P}(A_n) < \infty.$$

Therefore, by Borel–Cantelli,

$$\|X_{n+1} - X_n\|_\infty \leq 2^{-n/4}$$

for every n large enough almost surely. So $\{X_n : n \in \mathbb{N}\}$ is Cauchy in $C([0,1], \mathbb{R})$ and converges uniformly to $(X(t))_{t\in[0,1]}$ almost surely.

Claim. $(X(t))_{t\in[0,1]}$ is a Brownian motion on $[0,1]$.

Since $X_n(0) = 0$ for every n, we have $X(0) = 0$. Further, $(X(t))_t$ is continuous as a uniform limit of continuous functions. It remains to show that for every k and every

$$0 \leq t_1 < \cdots < t_k \leq 1,$$

$$(X(t_1), \dots, X(t_k))$$

is a zero-mean Gaussian vector with covariance

$$\mathrm{Cov}(X(t_i), X(t_j)) = t_i \wedge t_j.$$

Let $t_1 < \cdots < t_k \leq 1$. Then

$$(X_n(t_1), \ldots, X_n(t_k)) \to (X(t_1), \ldots, X(t_k))$$

as $n \to \infty$. Therefore, since $(X_n(t_1), \ldots, X_n(t_k))$ is Gaussian, it converges in L^p for every p, and $(X(t_1), \ldots, X(t_k))$ is Gaussian.

It remains to identify the mean and covariance. By convergence in L^1 and L^2,

$$\mathbb{E}[X(t_i)] = \lim_{n\to\infty} \mathbb{E}[X_n(t_i)] = 0$$

and

$$|\mathrm{Cov}(X(t_i), X(t_j)) - \mathrm{Cov}(X_n(t_i), X_n(t_j))| \to 0.$$

So it remains to calculate

$$\lim_{n\to\infty} \mathrm{Cov}(X_n(t_i), X_n(t_j)).$$

However, one shows by induction that for dyadic points,

$$\mathrm{Cov}\left(X_n\left(\frac{\ell_1}{2^n}\right), X_n\left(\frac{\ell_2}{2^n}\right)\right) = \frac{\ell_1 \wedge \ell_2}{2^n}, \qquad \forall \ell_1, \ell_2.$$

If this is true, then for general $s \leq t$ one gets

$$\mathrm{Cov}(X_n(s), X_n(t)) \to s \qquad \text{as } n \to \infty,$$

hence

$$\mathrm{Cov}(X(s), X(t)) = s = s \wedge t.$$

To complete the induction proof of the dyadic covariance identity, note first that it is true for $n = 0$. Suppose it holds for n.

Case 1. ℓ and $k \equiv 0$ [2]. This is clear by the induction hypothesis.

Case 2. $\ell \equiv 1$ [2] and $k \equiv 0$ [2]. Using that

$$X_{n+1}\left(\frac{\ell}{2^{n+1}}\right) = \frac{X_n\left(\frac{\ell-1}{2^n}\right) + X_n\left(\frac{\ell+1}{2^n}\right)}{2} + \frac{G_{\ell,n+1}}{2^{(n+1)/2}},$$

and that $G_{\ell,n+1}$ is independent of $X_n(k/2^{n+1})$, we get

$$\mathrm{Cov}\left(X_{n+1}\left(\frac{\ell}{2^{n+1}}\right), X_n\left(\frac{k}{2^{n+1}}\right)\right) = \frac{1}{2}\left\{\frac{\ell-1}{2^n} \wedge \frac{k}{2^{n+1}} + \frac{\ell+1}{2^n} \wedge \frac{k}{2^{n+1}}\right\} = \frac{\ell \wedge k}{2^{n+1}},$$

distinguishing $k \geq \ell + 1$ or $k = \ell - 1$, and recalling that $k \neq \ell$ here.

Case 3. Similarly, one treats the case $k \equiv 1$ [2], $\ell \equiv 0$ [2].

Case 4. When $k \equiv \ell \equiv 1$ [2], using the previous identity for both k and ℓ, and using that $\{X_n(\ell'/2^n)\}_{\ell'}$ is independent of $\{G_{\ell,n}\}_\ell$, we obtain

$$\mathrm{Cov}\left(X_{n+1}\left(\frac{\ell}{2^{n+1}}\right), X_{n+1}\left(\frac{k}{2^{n+1}}\right)\right) = \frac{1}{2^{n+3}}\left\{(\ell-1) \wedge (k-1) + (\ell+1) \wedge (k-1) + (\ell-1) \wedge (k+1) + (\ell+1) \wedge \right.$$

which equals $(\ell \wedge k)/2^{n+1}$. □

Corollary 1.20. *There exists a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ on which a Brownian motion is defined.*

Proof. By the previous result, we can easily define a probability space such that there is a sequence $\{(B_t^k)_{t\in[0,1]} : k \in \mathbb{N}\}$ of independent Brownian motions on $[0,1]$. We then define

$$B_t = B^{\lfloor t \rfloor}_{t-\lfloor t \rfloor} + \sum_{k=0}^{\lfloor t \rfloor - 1} B_1^k.$$

It is easy to verify that $(B_t)_t$ is indeed a Brownian motion. □

1.5 Canonical Space and Canonical Process

Wiener Space and Measure

We consider the space of continuous functions

$$\mathcal{W}^d = \{w : \mathbb{R}_+ \to \mathbb{R}^d : w \text{ continuous}\}.$$

Since $\mathbb{R}_+$ is not compact, $\mathcal{W}^d$ is not separable when endowed with the uniform topology, i.e. with the topology associated with the uniform norm. Instead, we consider the distance

$$d_{\mathcal{W}}(w, \widetilde{w}) = \sum_{n\geq 0} 2^{-n} \frac{\|w - \widetilde{w}\|_{\infty,n}}{1 + \|w - \widetilde{w}\|_{\infty,n}},$$

where

$$\|w - \widetilde{w}\|_{\infty,n} = \sup_{[0,n]} \|w - \widetilde{w}\|.$$

It is an easy exercise that $d_{\mathcal{W}}$ is a distance. In addition, $(\mathcal{W}^d, d_{\mathcal{W}})$ is a Polish space.

Since a Brownian motion is a.s. continuous, it induces a potential distribution on $(\mathcal{W}^d, d_{\mathcal{W}})$ endowed with its Borel σ-field $\mathcal{W}^d$. We can verify that it is indeed the case.

Theorem 1.21. *Let $(B_t)_{t\geq 0}$ be a Brownian motion on $(\Omega, \mathcal{F}, \mathbb{P})$. Consider a version that is continuous. Then*

$$\omega \in \Omega \longmapsto (B_t(\omega))_{t\geq 0}$$

is $\mathcal{F}/\mathcal{W}^d$-measurable. Therefore its distribution, denoted by

$$\mu^{\mathcal{W}}(A) = \mathbb{P}\big((B_t)_{t\geq 0} \in A\big), \qquad \forall A \in \mathcal{W}^d,$$

is well-defined as a measure on $(\mathcal{W}^d, \mathcal{W}^d)$. It is called the Wiener measure.

Proof. This result naturally extends to any process that is almost surely continuous. Consider the canonical process on $\mathcal{W}^d$ defined, for every t, by

$$W_t : w \in \mathcal{W}^d \longmapsto w_t \in \mathbb{R}^d.$$

It is a continuous map from $(\mathcal{W}^d, d_{\mathcal{W}})$ to $\mathbb{R}^d$ for every t, hence measurable. Then $(W_t)_{t\geq 0}$, seen as a process on $(\mathcal{W}^d, \mathcal{W}^d, \mu^{\mathcal{W}})$, is a d-dimensional Brownian motion.

The key lemma used in the proof is

$$\mathcal{W}^d = \sigma(W_t : t \geq 0).$$

□

Remark 1.22. To prove the key lemma, denote for fixed w and r,

$$F(\widetilde{w}) = d_{\mathcal{W}}(w, \widetilde{w}).$$

Then

$$B(w,r) = F^{-1}((0,r)).$$

Hence it is enough to prove that F is $\sigma(W_t : t \geq 0)/\mathcal{B}(\mathbb{R}_+)$-measurable.

But, by density of $\mathbb{Q}$ and continuity,

$$\|w - \widetilde{w}\|_{\infty,n} = \sup_{[0,n]\cap\mathbb{Q}} \|w - \widetilde{w}\|.$$

Therefore, for every n, the map $\widetilde{w} \mapsto \|w - \widetilde{w}\|_{\infty,n}$ is $\sigma(W_t : t \geq 0)/\mathcal{B}(\mathbb{R}_+)$-measurable, and so is F. It follows by definition and the fact that a countable union of measurable sets is measurable that

$$\mathcal{F} = \sigma(W_t : t \geq 0).$$

Definition 1.23. *A stochastic process indexed by T is a family of random variables*

$$\{Y_t\}_{t\in T}$$

indexed by T.

A path associated with $\omega \in \Omega$ is the function

$$t \longmapsto Y_t(\omega).$$

We suppose to ease the presentation that Y_t is X-valued for every t, where $(X, \mathcal{X})$ is a measurable space. However, results and definitions extend when each Y_t is valued in $(X_t, \mathcal{X}_t)$.

Question. Can we see the process as a random variable? We need to endow

$$X^T = \{t \mapsto x_t\}$$

with a σ-field. To this end, we introduce the canonical process.

Definition 1.24. *For every t, we denote*

$$X_t : (x_s)_{s\in T} \longmapsto x_t.$$

The process $(X_t)_{t\in T}$ is called the canonical process.

The product σ-field on X^T is defined as

$$\mathcal{X}^{\otimes T} = \sigma(X_t : t \in T),$$

that is, the smallest σ-field such that X_t is $\mathcal{X}^{\otimes T}/\mathcal{X}$-measurable for every t.

Theorem 1.25. *Any stochastic process on X defines a random variable from $(\Omega, \mathcal{F})$ to $(X^T, \mathcal{X}^{\otimes T})$.*

Now if μ is a probability measure on $(X^T, \mathcal{X}^{\otimes T})$, then $(X_t)_{t\in T}$ is a stochastic process under μ, with marginals distributed according to the pushforwards

$$(X_{t_1}, \ldots, X_{t_n})_{\#}\mu.$$

Any such distributions are called *fidis*.

Question. Are these families of distributions characterizations of a measure μ? More generally, how can one ensure existence and uniqueness of μ?

Definition 1.26. *For every $J \subset I \subset T$, we define*

$$X_I^J : (x_t)_{t\in I} \in X^I \longmapsto (x_t)_{t\in J} \in X^J.$$

If $I = T$, X_T^J is denoted by X^J.

- We define a *cylinder* $A \subset X^T$ as any subset such that there exists $A_J \subset X^J$, with $J \subset T$ finite, for which
$$A = (X^J)^{-1}(A_J).$$
- A *rectangular cylinder* is a cylinder for which A_J in the previous definition can be written as $\prod_{j\in J} A_j$ with $A_j \in \mathcal{X}$.

Set C_X for cylinders and C_X^R for rectangular cylinders.

Example 1.27. If $X = \mathbb{R}$ and $T = \mathbb{N}$, then

$$B_1(0,1) \times \mathbb{R} \times \mathbb{R} \times \cdots$$

and

$$[0,1]^d \times \mathbb{R} \times \mathbb{R} \times \cdots$$

are rectangular cylinders.

Proposition 1.28. *1)*

$$\mathcal{X}^{\otimes T} = \sigma(C_X) = \sigma(C_X^R).$$

2) If $\mu, \nu \in \mathcal{P}(X^T, \mathcal{X}^{\otimes T})$, then

$$\mu = \nu \quad \Longleftrightarrow \quad \mu(A) = \nu(A) \ \ \forall A \in C_X^R.$$

3) If $(Y_t)_t$ and $(Z_t)_t$ are stochastic processes valued in X, then

$$\mathcal{L}(Y_t)_{t\in T} = \mathcal{L}(Z_t)_{t\in T} \quad \Longleftrightarrow \quad \forall J \subset T \text{ finite}, \ \mathcal{L}\big((Y_t)_{t\in J}\big) = \mathcal{L}\big((Z_t)_{t\in J}\big).$$

Proof. Points 2) and 3) follow from the π-λ theorem.

For 1), since $C_X^R \subset \mathcal{X}^{\otimes T}$, we clearly have

$$\sigma(C_X^R) \subset \mathcal{X}^{\otimes T}.$$

Conversely, by definition, for every $A \in \mathcal{X}$ and every $t \in T$,

$$(X^t)^{-1}(A) \in C_X^R.$$

Hence every X_t is $\sigma(C_X^R)/\mathcal{X}$-measurable, which implies

$$\mathcal{X}^{\otimes T} \subset \sigma(C_X^R).$$

This yields

$$\mathcal{X}^{\otimes T} = \sigma(C_X^R) = \sigma(C_X).$$

□

Remark 1.29. Point 3) means that for pairwise distinct $t_1, \dots, t_n \in T$,

$$(Y_{t_1}, \dots, Y_{t_n}) \overset{(d)}{=} (Z_{t_1}, \dots, Z_{t_n}).$$

We conclude that a distribution on $(X^T, \mathcal{X}^{\otimes T})$ is characterized by its fidis, i.e. by the family

$$\{X^J_\# \mu : J \subset T \text{ finite}\}.$$

Now, based on some fidis, can we define a distribution on $(X^T, \mathcal{X}^{\otimes T})$?

Definition 1.30. *Let a collection $\{\mu_J : J \subset T \text{ finite}\}$ be such that each $\mu_J \in \mathcal{P}(X^J)$. It is said to be* compatible *if for every $J \subset I \subset T$, with J, I finite,*

$$(X^J_I)_\# \mu_I = \mu_J.$$

Theorem 1.31 (Kolmogorov extension theorem)**.** *Suppose that $(X, \mathcal{X})$ is Polish. Let $\{\mu_I : I \subset T,\ I \text{ finite}\}$ be a collection of fidis on X that is compatible. Then there exists a measure μ on $(X^T, \mathcal{X}^{\otimes T})$ such that for every finite I,*

$$\mu_I = (X^I)_\# \mu.$$

Proof. Admitted. □

1.6 Independence

Theorem 1.32. *Let $(X_t)_{t\in T}$ and $(Y_t)_{t\in T}$ be two stochastic processes valued in X and Y respectively. Then, seen as random variables valued in $(X^T, \mathcal{X}^{\otimes T})$ and $(Y^T, \mathcal{Y}^{\otimes T})$ respectively, these processes are independent if and only if for every $t_1, \dots, t_n$ pairwise distinct and every n,*

$$(X_{t_1}, \dots, X_{t_n}) \perp (Y_{t_1}, \dots, Y_{t_n}).$$

Proof. Sketch. The fidis of the joint process $(X_t, Y_t)_{t\in T}$ are of the form

$$\nu^X_I \otimes \nu^Y_I,$$

where ν^X_I (respectively ν^Y_I) are the fidis of $(X_t)_t$ (respectively $(Y_t)_t$). The proof is then completed using the characterization of distributions on path spaces based on their fidis. □

1.7 Kolmogorov Extension Theorem

Note that another proof consists in following the same lines as this result, invoking the π-λ theorem.

Theorem 1.33. *Let $(X_t)_{t\in I}$ be a stochastic process valued in a complete space (X, d_X)[1] and $I \subset \mathbb{R}$ an interval. Suppose that there exist $p \geq 1$, $\varepsilon > 0$ and $C > 0$ such that*

$$\mathbb{E}[d_X(X_t, X_s)^p] \leq C|t - s|^{1+\varepsilon}.$$

Then there exists a modification $(\widetilde{X}_t)_{t\in I}$ that is locally α-Hölder for every $\alpha \in (0, \varepsilon/p)$, i.e. for every $\alpha \in (0, \varepsilon/p)$, every $[a, b] \subset I$ and every $\omega \in \Omega$, there exists $A(\omega, \alpha, a, b)$ such that for every $s, t \in [a, b]$,

$$d(\widetilde{X}_t, \widetilde{X}_s) \leq A|s - t|^\alpha$$

and

$$\mathbb{E}[A^p] < \infty.$$

Proof. Admitted. □

[1] $\mathcal{X} = \mathcal{B}(X)$ here.

2 Function Properties of the Brownian Motion: Markov and Quadratic Variation

Before starting the next chapter, let us mention the following result implying that two processes with the same distribution give the same distribution on $\mathcal{W}^d$.

Theorem 2.1. *If $(X_t)_{t\geq 0}$ and $(Y_t)_{t\geq 0}$ are two almost surely continuous processes valued in $\mathbb{R}^d$ with the same distribution, then their distributions on $\mathcal{W}^d$ are equal.*

In all this chapter, $(B_t)_{t\geq 0}$ is a standard 1-dimensional Brownian motion and we denote by $(\mathcal{F}_t)_{t\geq 0}$ its completed filtration, i.e.

$$\mathcal{F}_t = \sigma(B_s : s \leq t) \vee \mathcal{N},$$

where $\mathcal{N}$ is the set of negligible sets.

2.1 Markov Property

Definition 2.2. *Let $(X_t)_{t\geq 0}$ be a stochastic process valued in $\mathbb{R}^d$ and $(\mathcal{G}_t)_{t\geq 0}$ be a filtration, i.e. a family of σ-fields such that $\mathcal{G}_s \subset \mathcal{G}_t$ for $s \leq t$. (X_t) is said to satisfy the Markov property with respect to $(\mathcal{G}_t)$ if, for every t and every $s \leq t$, the conditional distribution of X_t given $\mathcal{G}_s$ is equal to the conditional distribution of X_t given X_s. For any measurable bounded function $f : \mathbb{R}^d \to \mathbb{R}$,*

$$\mathbb{E}[f(X_t) \mid \mathcal{G}_s] = \mathbb{E}[f(X_t) \mid X_s].$$

Theorem 2.3. *For any $s \geq 0$, the process $(B_{t+s} - B_s)_{t\geq 0}$ is a Brownian motion independent of $\mathcal{F}_s$.*

Proof. We have to show that $(B_{t+s} - B_s)_{t\geq 0}$ is independent of $(B_u)_{u\leq s}$. It is enough to prove the factorization of expectations for finite families of times and bounded measurable test functions, and to rewrite the left-hand side as a function of

$$(B_{s_1},\, B_{s_2} - B_{s_1},\, \dots,\, B_s - B_{s_p},\, B_{s+t_1} - B_s, \dots)$$

and conclude by independence of Brownian increments. □

Proposition 2.4. *For every $s \geq 0$, $(B_t - B_s)_{t\geq s}$ is independent of*

$$\mathcal{F}_{s+} := \bigcap_{u>s} \mathcal{F}_u.$$

Therefore $(B_t)_{t\geq 0}$ satisfies the Markov property with respect to $(\mathcal{F}_{t+})_{t\geq 0}$.

Proof. We have to show that for any $A \in \mathcal{F}_{s+}$, $s \geq 0$, any $s \leq t_1 < \cdots < t_m$, and any $\varphi_1, \dots, \varphi_m$ continuous and bounded,

$$\mathbb{E}\left[\mathbf{1}_A \prod_{i=1}^m \varphi_i(B_{t_i} - B_s)\right] = \mathbb{P}(A)\, \mathbb{E}\left[\prod_{i=1}^m \varphi_i(B_{t_i} - B_s)\right]. \tag{1}$$

Note that without loss of generality we can suppose $t_1 = s$. Let $u > s$ be such that $t_1 = s < u < t_2$. Then $A \in \mathcal{F}_u$. Therefore, since $(B_t - B_s)_{t\geq s}$ is independent of $\mathcal{F}_u$ and $B_u - B_s$ is $\mathcal{F}_u$-measurable, we get

$$\begin{aligned}\mathbb{E}\left[\mathbf{1}_A \prod_{i=1}^m \varphi_i(B_{t_i} - B_s)\right] &= \mathbb{E}[\mathbf{1}_A\, \psi(B_u - B_s)\, \rho_1] \\ &= \mathbb{E}[\mathbf{1}_A\, \psi(B_u - B_s)]\, \rho(0),\end{aligned}$$

where

$$\psi(b) = \mathbb{E}\left[\prod_{i=2}^{m} \varphi_i(B_{t_i} - B_u + b)\right].$$

Note that ψ is continuous since the φ_i are continuous and bounded and by equal integrability. Since $B_u \to B_s$ almost surely as $u \downarrow s$, by dominated convergence and the continuity of ψ, we get

$$\begin{aligned}\mathbb{E}\left[\mathbf{1}_A \prod_{i=1}^{m} \varphi_i(B_{t_i} - B_s)\right] &= \mathbb{E}[\mathbf{1}_A\, \psi(0)\, \varphi_1(0)] \\ &= \mathbb{E}[\mathbf{1}_A]\, \psi(0)\, \varphi_1(0) \\ &= \mathbb{E}[\mathbf{1}_A]\, \mathbb{E}\left[\prod_{i=1}^{m} \varphi_i(B_{t_i} - B_s)\right].\end{aligned}$$

Using $\psi(0)\varphi_1(0) = \mathbb{E}[\prod_{i=1}^{m} \varphi_i(B_{t_i} - B_s)]$ completes the proof. □

Corollary 2.5. *$(B_t)_{t\geq 0}$ satisfies the Markov property with respect to $(\mathcal{F}_{t+})_{t\geq 0}$. In particular, for any measurable bounded φ, and any t, s,*

$$\mathbb{E}[\varphi(B_t) \mid \mathcal{F}_{s+}] = P_{t-s}\varphi(B_s),$$

where $(P_u)_{u\geq 0}$ is defined, for every measurable bounded $\varphi : \mathbb{R} \to \mathbb{R}$, by

$$P_u\varphi(x) = \mathbb{E}[\varphi(x + B_u)].$$

Proof. Let $s \leq t$. Then, for any bounded measurable φ,

$$\mathbb{E}[\varphi(B_t) \mid \mathcal{F}_{s+}] = \mathbb{E}[\varphi(B_t - B_s + B_s) \mid \mathcal{F}_{s+}].$$

Since $B_t - B_s$ is independent of $\mathcal{F}_{s+}$ and B_s is $\mathcal{F}_{s+}$-measurable, we get

$$\mathbb{E}[\varphi(B_t) \mid \mathcal{F}_{s+}] = \psi(B_s), \qquad \psi(b) = \mathbb{E}[\varphi(B_{t-s} + b)] = P_{t-s}\varphi(b).$$

As $\sigma(B_s) \subset \mathcal{F}_{s+}$, the tower property gives

$$\begin{aligned}\mathbb{E}[\varphi(B_t) \mid B_s] &= \mathbb{E}[\mathbb{E}[\varphi(B_t) \mid \mathcal{F}_{s+}] \mid B_s] \\ &= \psi(B_s).\end{aligned}$$

So $(B_t)_{t\geq 0}$ is Markov. □

$(P_t)_{t\geq 0}$ is defined as a family of operators from the set of measurable bounded functions to itself.

Definition 2.6. *Let $(X, \mathcal{X})$ be a measurable space. On $X \times \mathcal{X} \to [0, 1]$, Q is a Markov kernel if*

(a) for every $x \in X$, $A \mapsto Q(x, A)$ is a probability measure;

(b) for every $A \in \mathcal{X}$, $x \mapsto Q(x, A)$ is $\mathcal{X}/\mathcal{B}(\mathbb{R})$-measurable.

Definition 2.7. *We define the Markov semigroup $(\widetilde{P}_t)_{t\geq 0}$ of (B_t) as the collection of Markov kernels*

$$\widetilde{P}_t(x, A) = \mathbb{P}(x + B_t \in A).$$

$(\widetilde{P}_t)$ will also be denoted by (P_t), and

$$P_{t+s}\varphi = P_t P_s \varphi.$$

Proof. Let s, t, x and φ be given.

$$\begin{aligned} P_{t+s}\varphi(x) &= \mathbb{E}[\varphi(x + B_{t+s})] \\ &= \mathbb{E}[\varphi(x + B_{t+s} - B_s + B_s)] \\ &= \mathbb{E}[\mathbb{E}[\varphi(x + B_{t+s} - B_s + B_s) \mid \mathcal{F}_s]] \\ &= \mathbb{E}[P_t\varphi(x + B_s)] \\ &= P_s(P_t\varphi)(x). \end{aligned}$$

□

We denote by

$$C_0(\mathbb{R}) = \{f \in C(\mathbb{R}, \mathbb{R}) : \lim_{x \to \pm\infty} f(x) = 0\}$$

and

$$C_c^2(\mathbb{R}) = \{f \in C^2(\mathbb{R}, \mathbb{R}) : f \text{ has compact support}\}.$$

Proposition 2.8. *(i) for every $\varphi \in C(\mathbb{R}, \mathbb{R})$ bounded, and every t,*

$$P_t\varphi \in C(\mathbb{R}, \mathbb{R}) \text{ bounded};$$

(ii) for every $\varphi \in C_0(\mathbb{R})$ and every t,

$$P_t\varphi \in C_0(\mathbb{R}) \quad \text{and} \quad \lim_{t \to 0^+} \|P_t\varphi - \varphi\|_\infty = 0;$$

(iii) for every $\varphi \in C_c^2(\mathbb{R})$, every t and every $x \in \mathbb{R}$,

$$\frac{P_t\varphi(x) - \varphi(x)}{t} \xrightarrow[t \downarrow 0]{} \frac{1}{2}\varphi''(x);$$

(iv) for every $\varphi \in C_c^2(\mathbb{R})$, setting for any t, x,

$$u_t(x) = P_t\varphi(x),$$

$(t, x) \mapsto u_t(x)$ is a solution of the heat equation

$$\partial_t u_t(x) = \frac{1}{2}\partial_x^2 u_t(x), \qquad t \in (0, +\infty),\ x \in \mathbb{R}.$$

Proof. (i) follows by dominated convergence.

For (ii), let $\varphi \in C_0(\mathbb{R})$ and $t \geq 0$. Then

$$P_t\varphi(x) = \mathbb{E}[\varphi(x + B_t)].$$

Showing that $\lim_{x \to \infty} \varphi(x) = 0$ is easy and left to the reader. Let us show

$$\lim_{t \to 0^+} \|P_t\varphi - \varphi\|_\infty = 0.$$

Fix $\varepsilon > 0$. For every $x \in \mathbb{R}$,

$$|P_t\varphi(x) - \varphi(x)| = \left| \int \psi(z)(\varphi(x + t^{1/2}z) - \varphi(x))\, dz \right|,$$

where ψ denotes the density of $\mathcal{N}(0, 1)$. Then choose $k_\varepsilon \geq 0$ such that

$$\int_{[-k_\varepsilon, k_\varepsilon]^c} \psi(z)\, dz \leq \frac{\varepsilon}{4\|\varphi\|_\infty}.$$

Split the estimate into the tail part and the compact part. For t small enough,

$$\|P_t\varphi - \varphi\|_\infty \leq \varepsilon.$$

□

Corollary 2.9. *For every $A \in \mathcal{F}_{0+}$, $\mathbb{P}(A) \in \{0,1\}$.*

Proof. Since the previous result shows that $(B_t - B_0)_{t\geq 0}$ is independent of $\mathcal{F}_{0+}$, and $B_0 = 0$, we have $\mathcal{F}_\infty$ independent of $\mathcal{F}_{0+}$. So for $A \in \mathcal{F}_{0+} \subset \mathcal{F}_\infty$,

$$\mathbb{P}(A) = \mathbb{E}[\mathbf{1}_A] = \mathbb{E}[\mathbf{1}_A^2] = \mathbb{P}(A)^2,$$

hence $\mathbb{P}(A) \in \{0,1\}$. □

Exercise 2.10. Show that almost surely

$$\liminf_{t\to\infty} \frac{B_t}{\sqrt{t}} = -\infty.$$

Deduce that

$$\limsup_{t\to\infty} \frac{B_t}{\sqrt{t}} = +\infty \qquad \text{and} \qquad \limsup_{t\to 0} \frac{B_t}{\sqrt{t}} = +\infty, \quad \liminf_{t\to 0} \frac{B_t}{\sqrt{t}} = -\infty.$$

Solution sketch. The last limits follow by considering both $(B_t)_{t\geq 0}$ and $(tB_{1/t})_{t\geq 0}$. Let us show the first claim. For $l > 0$, let

$$A_n^l = \left\{ \frac{B_n}{\sqrt{n}} \leq -l \right\}, \qquad A^l = \bigcap_{n\geq 1} \bigcup_{m\geq n} A_m^l.$$

As $\mathcal{F}_{1/n} \subset \mathcal{F}_{0+}$ for all n, we have $A^l \in \mathcal{F}_{0+}$ after applying the same argument to the process $(t^{1/2}B_{1/t})_{t>0}$. Hence $\mathbb{P}(A^l) \in \{0,1\}$. Moreover,

$$\mathbb{P}(A^l) \geq \mathbb{P}(A_n^l) = \mathbb{P}(G \leq -l) > 0, \qquad G \sim \mathcal{N}(0,1),$$

so $\mathbb{P}(A^l) = 1$. Since this holds for every $l > 0$, we obtain

$$\liminf_{n\to\infty} \frac{B_n}{\sqrt{n}} = -\infty \quad \text{a.s.}$$

and therefore also

$$\liminf_{t\to\infty} \frac{B_t}{\sqrt{t}} = -\infty \quad \text{a.s.}$$

□

We can even strengthen a bit the Markov property and show that $(B_t)_{t\geq 0}$ is strong Markov. To this end, we need to introduce stopping times.

Definition 2.11. *Let $(\mathcal{F}_t)_{t\geq 0}$ be a filtration on $(\Omega, \mathcal{F}, \mathbb{P})$. A random variable τ valued in $\mathbb{R}_+ \cup \{+\infty\}$ is a stopping time with respect to $(\mathcal{F}_t)$ if*

$$\forall t, \qquad \{\tau \leq t\} \in \mathcal{F}_t.$$

Example 2.12. For $a \in \mathbb{R}$,

$$\tau_a = \inf\{t \geq 0 : B_t = a\}$$

with the convention $\inf \varnothing = +\infty$, is a stopping time with respect to $(\mathcal{F}_t)$. Indeed,

$$\{\tau_a \leq t\} = \left\{ \sup_{r\in[0,t]\cap\mathbb{Q}} |B_r - a| = 0 \right\} \in \mathcal{F}_t.$$

As a corollary, we see that

$$S_t = \sup_{r \le t} |B_r|$$

is $(\mathcal{F}_t)$-adapted, i.e. for every t, S_t is $\mathcal{F}_t$-measurable.

Definition 2.13. *We define the σ-field of a stopping time τ with respect to $(\mathcal{F}_t)_{t\ge 0}$ as*

$$\mathcal{F}_\tau = \{A \cap \{\tau \le t\} \in \mathcal{F}_t,\ \forall t\},$$

where $\mathcal{F}_\infty = \sigma(\mathcal{F}_t : t \ge 0)$. It is indeed a σ-field.

Remark 2.14.

$$\{\tau \le t\} \in \mathcal{F}_\tau \qquad \forall t,$$

and also $\{\tau < \infty\} \in \mathcal{F}_\tau$. In addition, if Z is $\mathcal{F}_\tau$-measurable and bounded, then for every N,

$$Z\mathbf{1}_{\{\tau \le N\}}$$

is $\mathcal{F}_N$-measurable.

Theorem 2.15. *Let τ be a $(\mathcal{F}_t)$-stopping time. Then, on $\{\tau < \infty\}$, the process*

$$(B_{t+\tau} - B_\tau)_{t\ge 0}$$

is a Brownian motion independent of $\mathcal{F}_{\tau+}$, where

$$\mathcal{F}_{\tau+} = \{A \in \mathcal{F}_\infty :\ A \cap \{\tau \le t\} \in \mathcal{F}_{t+}\ \forall t \ge 0\}.$$

Proof. We have to show that for every $A \in \mathcal{F}_{\tau+}$, every $0 \le t_1 < \cdots < t_m$, and every $\varphi_1, \dots, \varphi_m$ continuous and bounded,

$$\mathbb{E}\left[\mathbf{1}_A \mathbf{1}_{\{\tau<\infty\}} \prod_{i=1}^m \varphi_i(B_{t_i+\tau} - B_\tau)\right] = \mathbb{P}(A \cap \{\tau < \infty\}) \mathbb{E}\left[\prod_{i=1}^m \varphi_i(B_{t_i})\right]. \tag{2}$$

To this end, we use the same strategy as above, discretizing τ. Define

$$\tau_N = \frac{\lfloor 2^N \tau \rfloor + 1}{2^N}.$$

Then

$$\mathbf{1}_{\{\tau<\infty\}} \prod_{i=1}^m \varphi_i(B_{t_i+\tau} - B_\tau) = \lim_{N\to\infty} \mathbf{1}_{\{\tau<\infty\}} \prod_{i=1}^m \varphi_i(B_{t_i+\tau_N} - B_{\tau_N})$$

almost surely, and dominated convergence gives

$$\mathbb{E}\left[\mathbf{1}_A \mathbf{1}_{\{\tau<\infty\}} \prod_{i=1}^m \varphi_i(B_{t_i+\tau} - B_\tau)\right] = \lim_{N\to\infty} \mathbb{E}\left[\mathbf{1}_A \mathbf{1}_{\{\tau<\infty\}} \prod_{i=1}^m \varphi_i(B_{t_i+\tau_N} - B_{\tau_N})\right].$$

Now for fixed N,

$$\mathbf{1}_A \mathbf{1}_{\{\tau<\infty\}} = \sum_{k\ge 1} \mathbf{1}_{A_{k,N}}, \qquad A_{k,N} := A \cap \left\{\frac{k}{2^N} < \tau \le \frac{k+1}{2^N}\right\}.$$

Since $A_{k,N} \in \mathcal{F}_{(k+1)/2^N}$ and $\tau_N = (k+1)/2^N$ on $A_{k,N}$, the ordinary Markov property at time $(k+1)/2^N$ yields

$$\begin{aligned}\mathbb{E}\left[\mathbf{1}_A \mathbf{1}_{\{\tau<\infty\}} \prod_{i=1}^m \varphi_i(B_{t_i+\tau_N} - B_{\tau_N})\right] &= \sum_{k\ge 1} \mathbb{P}(A_{k,N})\, \mathbb{E}\left[\prod_{i=1}^m \varphi_i(B_{t_i})\right] \\ &= \mathbb{P}(A \cap \{\tau < \infty\})\, \mathbb{E}\left[\prod_{i=1}^m \varphi_i(B_{t_i})\right].\end{aligned}$$

Using the convergence above shows (2). □

Example 2.16. Consider, for $a \in \mathbb{R}$,

$$\tau_a = \inf\{t \geq 0 : \ B_t = a\}.$$

By the strong Markov property, $(\tau_a)_{a\geq 0}$ is a process with independent increments, i.e. for $0 \leq a_1 < \cdots < a_m$,

$$(\tau_{a_1}, \tau_{a_2} - \tau_{a_1}, \ldots, \tau_{a_m} - \tau_{a_{m-1}})$$

has independent components. Moreover,

$$\tau_b - \tau_a \overset{\text{dist}}{=} \tau_{b-a} \qquad \forall a \leq b.$$

Indeed, on $\{\tau_a < \infty\}$,

$$\tau_b = \inf\{t \geq 0 : \ B_t = b\} = \tau_a + \inf\{s \geq 0 : \ B_{s+\tau_a} - B_{\tau_a} = b - a\}.$$

Applying the strong Markov property at time τ_a gives the claim for $m = 2$, and the general case follows by induction. In particular, for every $c > 0$, the process $(\tau_{ca})_{a\geq 0}$ has the same law as $(c^2\tau_a)_{a\geq 0}$. Together with the previous properties and monotonicity, this shows that $(\tau_a)_{a\geq 0}$ is a stable subordinator of order $1/2$.

Using the strong Markov property, we can also show the reflection principle.

Theorem 2.17. *Recall that we define*

$$S_t = \sup_{r\in[0,t]} B_r, \qquad t \geq 0.$$

Then for every $b \leq a$ and $t > 0$,

$$\mathbb{P}(S_t \geq a, \ B_t \leq b) = \mathbb{P}(B_t \geq 2a - b).$$

As a consequence, S_t has the same distribution as $|B_t|$.

Proof. This is easy as an exercise. (see TD sheet) □

Remark 2.18. The result does not imply that $(S_t)_{t\geq 0}$ as a process has the same distribution as $(|B_t|)_{t\geq 0}$. This is very wrong (why?). But the two processes have the same time marginals.

Definition 2.19. *We say that a process $(X_t)_{t\geq 0}$ satisfies the strong Markov property with respect to the filtration $(\mathcal{F}_t)$ if for every stopping time τ and every bounded measurable function f,*

$$\mathbb{E}\Big[\mathbf{1}_{\{\tau<\infty\}} f(X_{t+\tau}) \mid \mathcal{F}_{\tau+}\Big] = \mathbf{1}_{\{\tau<\infty\}} \psi(t, X_\tau),$$

where

$$\psi(t, x_0) = \mathbb{E}[f(X_t) \mid X_0 = x_0]$$

for any possible initial point x_0.

Proposition 2.20. *Any Brownian motion is strong Markov with respect to its filtration. More precisely, for any bounded measurable g,*

$$\mathbb{E}\Big[\mathbf{1}_{\{\tau<\infty\}} g(B_{t+\tau}) \mid \mathcal{F}_{\tau+}\Big] = \mathbf{1}_{\{\tau<\infty\}} P_t g(B_\tau).$$

Proof. Exercise. □

2.2 Regularity of Paths of $(B_t)_{t\geq 0}$

Proposition 2.21. *Let $t > 0$ be fixed and let $(t_k^n)_{k=0}^{k_n}$, $n \in \mathbb{N}$, be a sequence of meshes of $[0,t]$, that is,*

$$\forall n, \qquad 0 = t_0^n < \cdots < t_{k_n}^n = t,$$

and

$$\max_{1\leq k\leq k_n} |t_k^n - t_{k-1}^n| \xrightarrow[n\to\infty]{} 0.$$

Then, in L^2,

$$\sum_{k=1}^{k_n} \left(B_{t_k^n} - B_{t_{k-1}^n}\right)^2 \xrightarrow[n\to\infty]{} t.$$

If $t_k^n = tk/2^n$, the convergence also holds almost surely.

Proof. Denote by

$$Y_k^n = \left(B_{t_k^n} - B_{t_{k-1}^n}\right)^2 - (t_k^n - t_{k-1}^n), \qquad Y^n = \sum_{k=1}^{k_n} Y_k^n.$$

Since the Y_k^n are independent,

$$\mathrm{Var}(Y^n) = \sum_{k=1}^{k_n} \mathrm{Var}(Y_k^n).$$

$$\mathrm{Var}(Y^n) \leq 3\sum_{k=1}^{k_n} |t_k^n - t_{k-1}^n|^2 \leq 3t \max_{1\leq k\leq k_n} |t_k^n - t_{k-1}^n| \xrightarrow[n\to\infty]{} 0.$$

Hence $Y^n \to 0$ in L^2, which concludes the proof of the first part.

Assume now $t_k^n = tk/2^n$. By scaling we may suppose without loss of generality that $t = 1$. Then, using the estimate above and Chebyshev's inequality,

$$\mathbb{P}(|Y^n| \geq 1/n) \leq n^2 2^{-n},$$

up to the harmless universal constant from the variance bound. Therefore

$$\sum_n \mathbb{P}(|Y^n| \geq 1/n) < \infty,$$

and Borel–Cantelli implies $Y^n \to 0$ almost surely. □

Corollary 2.22. *The Brownian motion $(B_t)_{t\geq 0}$ has infinite variation on every interval.*

Proof. Let $a < b$. We denote by $\mathcal{V}([a,b])$ the set of meshes of $[a,b]$. For a function $f : [a,b] \to \mathbb{R}$, define the total variation

$$\|f\|_{\mathrm{TV},[a,b]} = \sup\left\{ \sum_{k=1}^{m} |f(t_k) - f(t_{k-1})| : \ (t_k)_{k=0}^m \in \mathcal{V}([a,b]) \right\}.$$

Then almost surely

$$\|B\|_{\mathrm{TV},[a,b]} = +\infty.$$

Indeed, if $\|B\|_{\mathrm{TV},[a,b]} < \infty$, then for the dyadic mesh

$$t_k^n = a + \frac{(b-a)k}{2^n},$$

we would have

$$\sum_k (B_{t_k^n} - B_{t_{k-1}^n})^2 \leq \sup_k |B_{t_k^n} - B_{t_{k-1}^n}| \, \|B\|_{\mathrm{TV},[a,b]}.$$

Since $(B_t)_{t\geq 0}$ is almost surely continuous,

$$\sup_k |B_{t_k^n} - B_{t_{k-1}^n}| \xrightarrow[n\to\infty]{} 0 \quad \text{a.s.}$$

This would imply

$$\sum_k (B_{t_k^n} - B_{t_{k-1}^n})^2 \xrightarrow[n\to\infty]{} 0,$$

which contradicts the previous proposition. □

3 Measurability, Filtrations and Stopping Times

Here we gather some technical conditions and results which, while they may seem overly complicated at first sight, are important to consider if we want to construct and manipulate continuous-time processes. We start with filtrations satisfying the usual conditions. We consider throughout this chapter a fixed probability space $(\Omega, \mathcal{F}, \mathbb{P})$ on which all random variables are defined.

3.1 Filtrations and Usual Conditions

Definition 3.1. *A filtration $(\mathcal{F}_t)_{t\geq 0}$ on $(\Omega, \mathcal{F}, \mathbb{P})$ is an increasing family of sub-σ-fields of $\mathcal{F}$, that is,*

$$s \leq t \implies \mathcal{F}_s \subset \mathcal{F}_t.$$

We recall the definition of the right-continuous filtration associated with $(\mathcal{F}_t)_{t\geq 0}$.

Definition 3.2. *(a) Let $(\mathcal{F}_t)_{t\geq 0}$ be a filtration. We define the right-continuous filtration associated with $(\mathcal{F}_t)$ as the family of σ-fields $(\mathcal{F}_{t+})_{t\geq 0}$ given by*

$$\mathcal{F}_{t+} = \bigcap_{s>t} \mathcal{F}_s.$$

(b) The filtration $(\mathcal{F}_t)_{t\geq 0}$ is said to be right-continuous if

$$\forall t, \qquad \mathcal{F}_t = \mathcal{F}_{t+}.$$

One interpretation is that $(\mathcal{F}_t)$ is right-continuous if the information associated with $(\mathcal{F}_t)$ evolves smoothly with time and does not contain new information coming from an infinitesimal time change.

Proposition 3.3. *Let $(\mathcal{F}_t)_{t\geq 0}$ be a filtration.*

(a) $(\mathcal{F}_{t+})_{t\geq 0}$ is a filtration.

(b) $(\mathcal{F}_{t+})_{t\geq 0}$ is right-continuous.

Proof. Exercise. □

Example 3.4. (1) Consider $\mathcal{F}_t = \{\varnothing, \Omega\}$ while $\mathcal{F}_0 = \sigma(Z)$ for some random variable Z. Then $(\mathcal{F}_t)$ is not right-continuous.

(2) Consider $(B_t)_{t\geq 0}$ a Brownian motion and, for $a > 0$,

$$\tau_a = \inf\{t \geq 0 : \ B_t = a\}, \qquad \sigma_a = \inf\{t \geq 0 : \ B_t > a\}.$$

We have seen in chapter 2 that τ_a is a stopping time with respect to the natural filtration $(\mathcal{F}_t^B)$ of (B_t), where $\mathcal{F}_t^B = \sigma(B_s : s \leq t)$. It is easy to verify that σ_a is an $(\mathcal{F}_t^B)$-stopping time:

$$\{\sigma_a \leq t\} = \bigcap_{\varepsilon \in \mathbb{Q}_+} \left\{ \sup_{r \in [0,t+\varepsilon] \cap \mathbb{Q}} B_r > a \right\} \in \mathcal{F}_{t+}^B.$$

On the other hand, σ_a is not an $(\mathcal{F}_t^B)$-stopping time, since

$$\{\sigma_a \leq t\} = \left\{ \sup_{r\in[0,t]} B_r > a \right\} \cup \left\{ \sup_{r\in[0,t]} B_r = a \text{ and } \exists s > 0 : \ B_{t+s} > a \right\}.$$

The second event corresponds to the fact that (B_t) has local maxima greater than a in any right neighborhood. Since almost surely (B_t) crosses strictly above a whenever it hits a, the two times coincide almost surely, and therefore σ_a is complete for the completed natural filtration. More generally, for large classes of processes including Brownian motion, the completed natural filtration $(\overline{\mathcal{F}}_t)$ is right-continuous. However, for general processes, this is not always the case, so to show that some first hitting times are stopping times we must consider right-continuous and complete filtrations.

Definition 3.5. *Let $(\mathcal{F}_t)_{t\geq 0}$ be a filtration.*

(a) We say that $(\mathcal{F}_t)$ is complete if all negligible sets are included in $\mathcal{F}_0$.

(b) We define the completed filtration $(\mathcal{F}_t^c)_{t\geq 0}$ by

$$\mathcal{F}_t^c = \sigma(\mathcal{F}_t \cup \mathcal{N}),$$

where $\mathcal{N}$ denotes the negligible sets.

Proposition 3.6. *Let $(\mathcal{F}_t)_{t\geq 0}$ be a filtration.*

(a) $(\mathcal{F}_t^c)_{t\geq 0}$ is complete.

(b) If $(\mathcal{F}_t)$ is right-continuous, then $(\mathcal{F}_t^c)$ satisfies the usual conditions, that is, it is complete and right-continuous.

We define the usual augmentation of $(\mathcal{F}_t)$ as the complete and right-continuous filtration $(\mathcal{F}_{t+}^c)_{t\geq 0}$.

Proof. Exercise. □

Exercise 3.7. Using the Brownian example above, show that $\sigma_a = \tau_a$ almost surely.

Solution sketch. We already know that $\sigma_a \geq \tau_a$ almost surely. On the other hand, the event $\{\sigma_a > \tau_a\}$ is contained in

$$\left\{\exists t \geq 0 : \sup_{s\in[t,t+\varepsilon]} B_s = a\right\},$$

which has probability 0 by the Markov property together with the fact that Brownian motion crosses any level immediately after hitting it. □

Exercise 3.8. Show that τ is an $(\mathcal{F}_t)$-stopping time if and only if

$$\forall t, \qquad \{\tau < t\} \in \mathcal{F}_t.$$

Solution sketch. If τ is a stopping time, then

$$\{\tau < t\} = \bigcup_{n\geq 1} \left\{\tau \leq t - \frac{1}{n}\right\} \in \mathcal{F}_t.$$

Conversely, if $\{\tau < t\} \in \mathcal{F}_t$ for every t, then

$$\{\tau \leq s\} = \bigcap_{n\geq 1} \left\{\tau < s + \frac{1}{n}\right\} \in \mathcal{F}_{s+},$$

and in particular $\{\tau \leq s\} \in \mathcal{F}_s$ when the filtration is right-continuous. □

Exercise 3.9. Show that τ is an $(\mathcal{F}_{t+})_{t\geq 0}$-stopping time if and only if $\tau \wedge t$ is $\mathcal{F}_t$-measurable for every t.

In this course, if not specified otherwise, we will always consider filtered probability spaces satisfying the usual conditions.

3.2 Measurability and Stochastic Processes

As already seen in chapter 2, the filtration generated by a stopping time is natural. Dealing with processes, one would like to know that if (X_t) is a stochastic process, then $X_\tau \mathbf{1}_{\{\tau<\infty\}}$ is $\mathcal{F}_\tau$-measurable. This is not always the case, which is why we need the following concepts.

Definition 3.10. *Let $(X_t)_{t\geq 0}$ be a stochastic process valued in a measurable space $(E,\mathcal{E})$, and let $(\mathcal{F}_t)$ be a filtration.*

(a) We say that (X_t) is measurable if

$$(t,\omega)\mapsto X_t(\omega)$$

is $\mathcal{B}([0,\infty))\otimes\mathcal{F}/\mathcal{E}$-measurable.

(b) We say that (X_t) is adapted to $(\mathcal{F}_t)$ if

$$X_t \text{ is } \mathcal{F}_t/\mathcal{E}\text{-measurable for every } t.$$

(c) We say that (X_t) is progressively measurable if for every $t>0$,

$$(\omega,s)\in\Omega\times[0,t]\mapsto X_s(\omega)$$

is $\mathcal{F}_t\otimes\mathcal{B}([0,t])/\mathcal{E}$-measurable.

It is almost immediate that every progressively measurable process is measurable and adapted.

Proposition 3.11. *If $(X_t)_{t\geq 0}$ is right-continuous, then it is measurable.*

Proof. Just remark that X_t is the pointwise limit of the measurable processes

$$X_t^{(n)} = X_{\lceil 2^n t\rceil/2^n},$$

since (X_t) is right-continuous. □

Proposition 3.12. *If $(X_t)_{t\geq 0}$ is right-continuous and adapted to $(\mathcal{F}_t)$, then it is progressively measurable.*

Proof. Let $t>0$. We can write the process on $[0,t]$ as the pointwise limit of the simple processes

$$X_s^{(n)} = \sum_{k\geq 0} X_{k/2^n}\,\mathbf{1}_{(k/2^n,\,(k+1)/2^n]}(s),$$

truncated to $[0,t]$. Each $X^{(n)}$ is $\mathcal{F}_t\otimes\mathcal{B}([0,t])$-measurable because

$$\{(\omega,s) : X_s^{(n)}(\omega)\in A\}$$

is a finite union of sets of the form

$$[0,t]\times\{X_{k/2^n}\in A\} \quad \text{or} \quad \left(\frac{k}{2^n},\frac{k+1}{2^n}\right]\times\{X_{k/2^n}\in A\},$$

which belong to $\mathcal{B}([0,t])\otimes\mathcal{F}_t$. □

Definition 3.13. *We say that $A\in\mathcal{B}(\mathbb{R}_+)\otimes\mathcal{F}$ is progressively measurable with respect to $(\mathcal{F}_t)$ if*

$$(\omega,s)\mapsto \mathbf{1}_A(s,\omega)$$

is progressively measurable.

Proposition 3.14. *(1) The set of progressively measurable subsets of $\mathcal{B}(\mathbb{R}_+) \otimes \mathcal{F}$ is a σ-field.*

(2) $(X_t)_{t\geq 0}$ is progressively measurable if and only if

$$(t, \omega) \mapsto X_t(\omega)$$

is measurable with respect to the progressive σ-field.

Proof. (1) Left as an exercise.

(2) If (X_t) is progressively measurable, then for any $A \in \mathcal{E}$,

$$(\omega, t) \mapsto \mathbf{1}_{\{X_t(\omega)\in A\}}$$

is progressively measurable, which means

$$\{(\omega, t) : X_t(\omega) \in A\}$$

belongs to the progressive σ-field. Conversely, if these inverse images are progressive for every $A \in \mathcal{E}$, then the mapping $(t, \omega) \mapsto X_t(\omega)$ is measurable by the usual criterion.

□

Remark 3.15. $(X_t)_{t\geq 0}$ may be measurable without any continuity assumption and still fail to be progressive with respect to a given filtration.

3.3 Stopping Times and Measurability

Definition 3.16. *Let τ be an $(\mathcal{F}_t)$-stopping time. We define*

$$\mathcal{F}_{\tau-} = \{A \in \mathcal{F} : \ A \cap \{\tau < t\} \in \mathcal{F}_t \ \forall t \geq 0\}.$$

This is also a sub-σ-field of $\mathcal{F}$. The following facts are repeatedly useful:

- $\mathcal{F}_{\tau-} \subset \mathcal{F}_{\tau+}$.
- If $(\mathcal{F}_t)$ is right-continuous, then $\mathcal{F}_\tau = \mathcal{F}_{\tau+}$.
- If $\tau \leq t \in \mathbb{R}_+$, then $\mathcal{F}_\tau \subset \mathcal{F}_t$ and $\mathcal{F}_{\tau+} \subset \mathcal{F}_{t+}$.
- If we denote $\widetilde{\mathcal{F}}_t = \mathcal{F}_{t+}$, then $\widetilde{\mathcal{F}}_\tau = \mathcal{F}_{\tau+}$.
- τ is $\mathcal{F}_\tau$-measurable.
- For any $A \in \mathcal{F}_\tau$,
$$\tau^A := \mathbf{1}_A \tau + \infty \mathbf{1}_{A^c}$$
is a stopping time if and only if $A \in \mathcal{F}_\tau$.

Proposition 3.17. *Let σ and τ be two $(\mathcal{F}_t)$-stopping times.*

(a) If $\sigma \leq \tau$, then $\mathcal{F}_\sigma \subset \mathcal{F}_\tau$ and $\mathcal{F}_{\sigma+} \subset \mathcal{F}_{\tau+}$.

(b) $\sigma \wedge \tau$ and $\sigma \vee \tau$ are $(\mathcal{F}_t)$-stopping times, and

$$\mathcal{F}_{\sigma\wedge\tau} = \mathcal{F}_\sigma \cap \mathcal{F}_\tau.$$

Moreover,

$$\{\sigma < t\}, \ \{\sigma = \tau\}, \ \{\sigma \leq \tau\} \in \mathcal{F}_{\sigma\wedge\tau}.$$

(c) $\sigma + t$ is an $(\mathcal{F}_t)$-stopping time for every $t \geq 0$.

Proof. Exercise. □

Theorem 3.18. *Let $(\tau_n)_{n\geq 1}$ be a non-increasing sequence of $(\mathcal{F}_t)$-stopping times and set $\tau = \inf_n \tau_n$. Then τ is an $(\mathcal{F}_t)$-stopping time and*

$$\mathcal{F}_{\tau+} = \bigcap_{n\geq 1} \mathcal{F}_{\tau_n+}.$$

In particular, if $(\mathcal{F}_t)$ is right-continuous, then

$$\mathcal{F}_\tau = \bigcap_{n\geq 1} \mathcal{F}_{\tau_n}.$$

Proof. We have

$$\{\tau < t\} = \bigcap_{n\geq 1} \{\tau_n < t\} \in \mathcal{F}_t \qquad \forall t,$$

so τ is a stopping time by the previous exercise. If $A \in \bigcap_n \mathcal{F}_{\tau_n+}$, then for every t,

$$A \cap \{\tau < t\} = \bigcap_{n\geq 1} \big(A \cap \{\tau_n < t\}\big) \in \mathcal{F}_t,$$

hence $A \in \mathcal{F}_{\tau+}$. The reverse inclusion follows from $\tau \leq \tau_n$. □

We can now state the main result of this section.

Theorem 3.19. *Let $(X_t)_{t\geq 0}$ be an $(\mathcal{F}_t)$-progressively measurable process valued in $\mathbb{R}^d$, and let τ be an $(\mathcal{F}_t)$-stopping time. Then*

$$X_\tau \mathbf{1}_{\{\tau<\infty\}}$$

is $\mathcal{F}_\tau$-measurable.

Proof. Let $A \in \mathcal{B}(\mathbb{R}^d)$. Then

$$\{X_\tau \mathbf{1}_{\{\tau<\infty\}} \in A\} \cap \{\tau \leq t\} = \{X_{\tau\wedge t} \in A\} \cap \{\tau \leq t\}.$$

Also, the mapping

$$(\omega, s) \mapsto (\tau(\omega) \wedge t,\ X_s(\omega))$$

is measurable from $\mathcal{F}_t \otimes \mathcal{B}([0,t])$ to $\mathcal{B}([0,t]) \otimes \mathcal{B}(\mathbb{R}^d)$, since $\tau \wedge t$ is $\mathcal{F}_t$-measurable and $(X_s)_{s\leq t}$ is progressively measurable. Thus

$$\omega \mapsto X_{\tau(\omega)\wedge t}(\omega)$$

is $\mathcal{F}_t/\mathcal{B}(\mathbb{R}^d)$-measurable, and therefore

$$\{X_\tau \mathbf{1}_{\{\tau<\infty\}} \in A\} \cap \{\tau \leq t\} \in \mathcal{F}_t$$

for every t, proving the claim. □

We conclude this chapter with results that justify why we consider filtrations satisfying the usual conditions.

Theorem 3.20. *If $(X_t)_{t\geq 0}$ is $(\mathcal{F}_t)$-progressively measurable and $(\mathcal{F}_t)$ satisfies the usual conditions, then for any $A \in \mathcal{E}$,*

$$\tau_A = \inf\{t \geq 0 :\ X_t \in A\}$$

is an $(\mathcal{F}_t)$-stopping time whenever A is open or closed and (X_t) is right-continuous.

Proof. Admitted. □

Theorem 3.21. *Let $(\mathcal{F}_t)$ be a complete filtration and let $(X_t)_{t\geq 0}$ be a right-continuous process valued in $\mathbb{R}^d$. Let $A \in \mathcal{B}(\mathbb{R}^d)$, and define*

$$\tau_A = \inf\{t \geq 0 : \ X_t \in A\}.$$

(a) If A is open, then τ_A is an $(\mathcal{F}_t)$-stopping time.

(b) If A is closed, then τ_A is an $(\mathcal{F}_t)$-stopping time.

Proof. Left as an exercise. □

4 Continuous-Time Martingales and Submartingales

We consider a filtered probability space

$$(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P}).$$

4.1 Definitions

Definition 4.1. *We say that a real-valued process $(M_t)_{t\geq 0}$ is an $(\mathcal{F}_t)_{t\geq 0}$-martingale (respectively a submartingale) if*

- *$M_t \in L^1$ for every t, i.e. $\mathbb{E}[|M_t|] < \infty$;*
- *for all $s \leq t$,*
$$\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s \qquad (\textit{respectively } \geq M_s).$$

Remark 4.2. We only consider one-dimensional martingales here, but most results have obvious multi-dimensional analogues. If $(\mathcal{F}_t)$ is not implicit, one should also require that (M_t) be $(\mathcal{F}_t)$-adapted.

Example 4.3. We say that $(B_t)_{t\geq 0}$ is a Brownian motion with filtration $(\mathcal{F}_t)_{t\geq 0}$ if

- $B_0 = 0$;
- for every s, $(B_{t+s} - B_s)_{t\geq 0}$ is a Brownian motion independent of $\mathcal{F}_s$;
- $(B_t)_{t\geq 0}$ is $(\mathcal{F}_t)$-adapted.

One can easily verify that $(B_t)_{t\geq 0}$ is a usual Brownian motion if and only if $(B_t)_{t\geq 0}$ is a Brownian motion with its natural filtration $(\mathcal{F}^B_t)$, $\mathcal{F}^B_t = \sigma(B_r : r \leq t)$.

With this definition, the following processes are $(\mathcal{F}_t)$-martingales:

(a) $(B_t)_{t\geq 0}$;

(b) $(B_t^2 - t)_{t\geq 0}$;

(c) for every $u \in \mathbb{R}$,
$$\left(\exp(uB_t - u^2 t/2)\right)_{t\geq 0}.$$

Indeed, for (a),
$$\mathbb{E}[B_t \mid \mathcal{F}_s] = \mathbb{E}[(B_t - B_s) + B_s \mid \mathcal{F}_s] = B_s,$$
using $B_t - B_s \perp \mathcal{F}_s$ and that B_s is $\mathcal{F}_s$-measurable.

For (b),
$$\begin{aligned}\mathbb{E}[B_t^2 - t \mid \mathcal{F}_s] &= \mathbb{E}[(B_t - B_s)^2 \mid \mathcal{F}_s] + 2\mathbb{E}[(B_t - B_s)B_s \mid \mathcal{F}_s] + \mathbb{E}[B_s^2 \mid \mathcal{F}_s] - t \\ &= (t - s) + B_s^2 - t \\ &= B_s^2 - s.\end{aligned}$$

For (c),
$$\begin{aligned}\mathbb{E}\Big[e^{uB_t - u^2 t/2} \mid \mathcal{F}_s\Big] &= \mathbb{E}\Big[e^{u(B_t - B_s)} e^{uB_s - u^2 t/2} \mid \mathcal{F}_s\Big] \\ &= e^{uB_s - u^2 t/2}\, \mathbb{E}[e^{u(B_t - B_s)} \mid \mathcal{F}_s] \\ &= e^{uB_s - u^2 t/2}\, e^{u^2(t-s)/2} \\ &= e^{uB_s - u^2 s/2}.\end{aligned}$$

Proposition 4.4. *Let $(M_t)_{t\geq 0}$ be an $(\mathcal{F}_t)$-martingale. Then for every t,*

$$\sup_{s\in[0,t]} \mathbb{E}[|M_s|] \leq \mathbb{E}[|M_t|] < \infty.$$

Proof. By Jensen's inequality,

$$\mathbb{E}[|M_t| \mid \mathcal{F}_s] \geq |\mathbb{E}[M_t \mid \mathcal{F}_s]| = |M_s|.$$

So $(|M_s|)_{s\in[0,t]}$ is a submartingale and hence, for every $s \leq t$,

$$\mathbb{E}[|M_s|] \leq \mathbb{E}[|M_t|].$$

This completes the proof. □

Exercise 4.5. Show that the same proposition is true for non-negative submartingales.

4.2 Martingales and Continuity

In the sequel, we extend results from discrete martingale theory to the continuous-time setting. For this, we rely on additional continuity assumptions.

Theorem 4.6 (Doob inequality)**.** *Let $(M_t)_{t\geq 0}$ be an $(\mathcal{F}_t)$-martingale. Then, for every t and every $p > 1$,*

$$\mathbb{E}\left[\sup_{s\in[0,t]} |M_s|^p\right] \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}[|M_t|^p].$$

Proof. For $n \in \mathbb{N}$, consider the discrete-time process

$$M_q^{(n)} = M_{tq/2^n}, \qquad q = 0, \ldots, 2^n.$$

Since (M_t) is an $(\mathcal{F}_t)$-martingale, $(M_q^{(n)})_{q=0}^{2^n}$ is a $(\mathcal{F}_{tq/2^n})$-martingale. Applying the discrete Doob inequality gives

$$\mathbb{E}\left[\sup_{q\in\{0,\ldots,2^n\}} |M_{tq/2^n}|^p\right] \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}[|M_t|^p].$$

Since (M_t) is right-continuous, almost surely

$$\sup_{q\in\{0,\ldots,2^n\}} |M_{tq/2^n}| \uparrow \sup_{s\in[0,t]} |M_s|$$

as $n \to \infty$. Fatou's lemma and monotone convergence complete the proof. □

Theorem 4.7 (Discrete Doob inequality)**.** *Let $(\widetilde{M}_n)_{n\in\mathbb{N}}$ be a $(\mathcal{G}_n)_{n\in\mathbb{N}}$-martingale, that is, a $(\mathcal{G}_n)$-adapted process such that for all n,*

$$\mathbb{E}[\widetilde{M}_{n+1} \mid \mathcal{G}_n] = \widetilde{M}_n \qquad \text{and} \qquad \mathbb{E}[|\widetilde{M}_n|] < \infty.$$

Then, for every n and every $p > 1$,

$$\mathbb{E}\left[\sup_{0\leq q\leq n} |\widetilde{M}_q|^p\right] \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}[|\widetilde{M}_n|^p].$$

Theorem 4.8. *Let $(\widetilde{M}_n)_{n\in\mathbb{N}}$ be a $(\mathcal{G}_n)$-submartingale and let T be a $(\mathcal{G}_n)$-stopping time. Then $(\widetilde{M}_{n\wedge T})_{n\in\mathbb{N}}$ is also a $(\mathcal{G}_n)$-submartingale. In particular, for every n, $(\widetilde{M}_{n\wedge T})_{0\leq q\leq n}$ is a discrete submartingale.*

Theorem 4.9 (Doob maximal inequality)**.** *Let $(M_t)_{t\geq 0}$ be a right-continuous $(\mathcal{F}_t)$-submartingale. Then for every $\lambda > 0$ and every T,*

$$\lambda\,\mathbb{P}\left(\sup_{t\in[0,T]} M_t > \lambda\right) \leq 3\,\mathbb{E}[|M_T|].$$

Proof. The proof is the extension of the discrete case. □

Theorem 4.10. *Let $(\widetilde{M}_n)_{n\in\mathbb{N}}$ be a discrete-time submartingale with respect to $(\mathcal{G}_n)$. Then for every $\lambda > 0$ and every N,*

$$\lambda\,\mathbb{P}\left(\sup_{q\leq N} \widetilde{M}_q > \lambda\right) \leq 3\,\mathbb{E}[|\widetilde{M}_N|].$$

Proof. We consider the sequence of random variables

$$M_q^{(n)} = M_{(q+1)/2^n \wedge t}, \qquad q \in \left\{0, \ldots, \lfloor 2^n t \rfloor\right\}.$$

Applying the discrete case gives, for $\lambda > 0$,

$$\lambda\,\mathbb{P}\left(\sup_{q\in\{0,\ldots,\lfloor 2^n t\rfloor\}} |M_{q/2^n}| > \lambda\right) \leq 3\,\mathbb{E}[|M_t|].$$

Since

$$\sup_{q\in\{0,\ldots,\lfloor 2^n t\rfloor\}} |M_{q/2^n}| \to \sup_{s\in[0,t]} |M_s|$$

as $n \to \infty$ when (M_t) is right-continuous, dominated convergence completes the proof. □

Theorem 4.11. *Let $(\mathcal{F}_t)_{t\geq 0}$ be a right-continuous filtration. If $(M_t)_{t\geq 0}$ is an $(\mathcal{F}_t)$-submartingale and*

$$t \mapsto \mathbb{E}[M_t]$$

is right-continuous, then there exists a right-continuous with left limits modification of (M_t) which is again a submartingale.

4.3 Convergence of Martingales and Optional Stopping

Theorem 4.12. *Let $a > 0$, $b > a$, and let $(\widetilde{M}_q)_{q\in\mathbb{N}}$ be a discrete submartingale. Denote by*

$$N_{a,b} = \sum_q \mathbf{1}_{[a,b]}(\widetilde{M}_q)$$

the number of upcrossings of $[a, b]$. Then

$$\mathbb{E}[N_{a,b}] \leq \sup_q \frac{\mathbb{E}[(\widetilde{M}_q - a)_+]}{b-a}.$$

Theorem 4.13. *Let $I = \mathbb{R}_+$ or $\mathbb{N}$, and let $(X_i)_{i\in I}$ be a family of real random variables. Suppose that $X_i \to X$ in probability and that $(X_i)_{i\in I}$ is uniformly integrable, i.e.*

$$\lim_{A\to\infty} \sup_{i\in I} \mathbb{E}[\mathbf{1}_{\{|X_i|\geq A\}}|X_i|] = 0.$$

Then $X_i \to X$ in L^1.

Remark 4.14. This condition is automatically satisfied when $(M_t)_t$ is a martingale with uniform L^p bounds for some $p > 1$.

Theorem 4.15. *Let $(M_t)_{t\geq 0}$ be a right-continuous $(\mathcal{F}_t)$-submartingale. Assume*

$$\sup_{t\geq 0} \mathbb{E}[M_t^+] < \infty.$$

Then there exists $M_\infty \in L^1$ such that

$$M_t \xrightarrow[t\to\infty]{} M_\infty \qquad \text{almost surely.}$$

Proof. Consider the dyadic grid submartingales

$$M_q^{(n)} = M_{q/2^n}, \qquad \mathcal{G}_q^{(n)} = \mathcal{F}_{q/2^n}.$$

Fix $a > 0$, $b > a$. Then, denoting by $N_{a,b}^{(n)}$ the number of upcrossings of $[a, b]$ by $(M_q^{(n)})_{q\in\mathbb{N}}$, we have

$$\mathbb{E}[N_{a,b}^{(n)}] \leq \sup_q \frac{\mathbb{E}[(M_q^{(n)} - a)_+]}{b-a} \leq \sup_t \frac{\mathbb{E}[(M_t - a)_+]}{b-a}.$$

Let D be the set of dyadic rational numbers in $\mathbb{R}_+$. Since D is countable, the dyadic skeletons imply that

$$\{t \in \mathbb{R}_+ : \ M_t \in [a, b]\}$$

has only finitely many upcrossings almost surely. As D is dense and (M_t) is right-continuous, we obtain that for every $a < b$,

$$\{t \in \mathbb{R}_+ : \ M_t \in [a, b]\}$$

has finitely many upcrossings almost surely. Hence

$$\lim_{t\to\infty} M_t = M_\infty$$

exists almost surely in $\mathbb{R} \cup \{\pm\infty\}$. To show that $M_\infty \in L^1$, note first by Fatou's lemma that

$$\mathbb{E}[M_\infty^+] \leq \liminf_{t\to\infty} \mathbb{E}[M_t^+] < \infty.$$

Also, since (M_t) is a submartingale and $t \mapsto \mathbb{E}[M_t]$ is increasing,

$$\mathbb{E}[M_t^-] \leq \mathbb{E}[M_t^+] - \mathbb{E}[M_t]$$

is uniformly bounded, so by Fatou again $\mathbb{E}[M_\infty^-] < \infty$. Therefore $M_\infty \in L^1$. □

Corollary 4.16. *Let $(M_t)_{t\geq 0}$ be a right-continuous supermartingale. Then there exists M_∞ such that*

$$M_t \to M_\infty \qquad \text{almost surely,}$$

and

$$\mathbb{E}[M_\infty] \leq \mathbb{E}[M_0].$$

Proof. This is a direct application of the previous result to $(-M_t)_{t\geq 0}$. □

Definition 4.17. *We say that a (sub-)martingale (M_t) is closed if there exists $M_\infty \in L^1$ such that*

$$M_t = \mathbb{E}[M_\infty \mid \mathcal{F}_t] \qquad \forall t.$$

Corollary 4.18. *Suppose $(M_t)_{t\geq 0}$ is a (sub-)martingale such that*

$$\sup_t \mathbb{E}[|M_t|^p] < \infty$$

for some $p > 1$. Then (M_t) converges almost surely and in L^p.

Proof. By the previous result there exists $M_\infty \in L^1$ such that $M_t \to M_\infty$ almost surely. In addition, Fatou's lemma implies $M_\infty \in L^p$. Doob's inequality gives $\sup_t |M_t| \in L^p$, and dominated convergence then completes the proof. □

Before giving the next result, we recall uniform integrability.

Definition 4.19. *A family $\{X_i : i \in I\}$ of random variables is uniformly integrable if*

$$\lim_{A\to\infty} \sup_{i\in I} \mathbb{E}[\mathbf{1}_{\{|X_i|\geq A\}}|X_i|] = 0.$$

Theorem 4.20. *Let $\{X_i : i \in \mathbb{N}\}$ be a uniformly integrable family of random variables. Suppose that $X_i \to X$ in probability. Then $X_i \to X$ in L^1.*

Proof. Exercise. □

Exercise 4.21. Let $(X_i)_{i\in I}$ be uniformly integrable. Show that (X_i) is equi-continuous in the sense that for every $\varepsilon > 0$, there exists $\delta > 0$ such that for every A with $\mathbb{P}(A) \leq \delta$,

$$\sup_{i\in I} \mathbb{E}[\mathbf{1}_A|X_i|] \leq \varepsilon.$$

Solution sketch. Fix $\varepsilon > 0$. Since (X_i) is uniformly integrable, there exists M such that

$$\sup_i \mathbb{E}[\mathbf{1}_{\{|X_i|\geq M\}}|X_i|] \leq \varepsilon/2.$$

If $\mathbb{P}(A) \leq \delta$, then

$$\mathbb{E}[\mathbf{1}_A|X_i|] \leq \varepsilon/2 + \mathbb{E}[\mathbf{1}_A\mathbf{1}_{\{|X_i|<M\}}|X_i|] \leq \varepsilon/2 + M\delta.$$

Choosing $\delta = \varepsilon/(2M)$ gives the claim. □

Exercise 4.22. Let $(X_i)_{i\in I}$ be uniformly integrable and let $(\mathcal{G}_j)_{j\in J}$ be a family of sub-σ-fields of $\mathcal{F}$. Show that

$$\{\mathbb{E}[X_i \mid \mathcal{G}_j] : i \in I,\ j \in J\}$$

is uniformly integrable.

Solution sketch. By the previous exercise, there exists $\delta > 0$ such that

$$\mathbb{P}(A) \leq \delta \implies \sup_i \mathbb{E}[\mathbf{1}_A|X_i|] \leq \varepsilon.$$

Set $Y_{i,j} = \mathbb{E}[X_i \mid \mathcal{G}_j]$. Then, by Jensen's inequality,

$$\mathbb{E}[\mathbf{1}_{\{|Y_{i,j}|\geq R\}}|Y_{i,j}|] \leq \mathbb{E}[\mathbf{1}_{\{|Y_{i,j}|\geq R\}}\mathbb{E}[|X_i| \mid \mathcal{G}_j]]$$

and by Markov's inequality,

$$\mathbb{P}(|Y_{i,j}| \geq R) \leq R^{-1}\mathbb{E}[|Y_{i,j}|] \leq R^{-1}\mathbb{E}[|X_i|].$$

Using the L^1-boundedness of uniformly integrable families, choose $R = \delta^{-1}\sup_i \mathbb{E}[|X_i|]$ and apply the previous exercise to conclude. □

Definition 4.23. *We say that $(M_t)_{t\geq 0}$ is a closed $(\mathcal{F}_t)$-martingale if there exists $M_\infty \in L^1$ such that for every t,*

$$M_t = \mathbb{E}[M_\infty \mid \mathcal{F}_t].$$

Theorem 4.24. *Suppose that $(M_t)_{t\geq 0}$ is a $(\mathcal{F}_t)$-(sub-)martingale which is uniformly integrable. Then*

(a) $M_t \to M_\infty$ *almost surely and in* L^1*;*

(b) (M_t) *is closed.*

Proof. (a) Since (M_t) is uniformly integrable, the family $\{M_t : t \geq 0\}$ is bounded in L^1, hence by the previous convergence theorem it converges almost surely. The theorem on uniformly integrable families then implies that the convergence also holds in L^1.

(b) For $t \leq T$,

$$M_t = \mathbb{E}[M_T \mid \mathcal{F}_t].$$

For every $A \in \mathcal{F}_t$,

$$\mathbb{E}[\mathbf{1}_A M_t] = \mathbb{E}[\mathbf{1}_A M_T].$$

Since $(\mathbf{1}_A M_T)_{T \geq t}$ converges to $\mathbf{1}_A M_\infty$ and is uniformly integrable, it also converges in L^1, which gives

$$\mathbb{E}[\mathbf{1}_A M_t] = \mathbb{E}[\mathbf{1}_A M_\infty].$$

Thus

$$M_t = \mathbb{E}[M_\infty \mid \mathcal{F}_t].$$

□

Theorem 4.25. *If* $(M_t)_{t \geq 0}$ *is closed, then it is uniformly integrable.*

Proof. This is an immediate application of the fact that for any uniformly integrable family $\{X_i\}$ and any family of sub-σ-fields $\{\mathcal{G}_j\}$, the family $\{\mathbb{E}[X_i \mid \mathcal{G}_j] : i, j\}$ is uniformly integrable. □

Theorem 4.26. *Let* $(\mathcal{F}_{-n})_{n \geq 0}$ *be a filtration indexed by* $\mathbb{Z}_-$*, i.e.*

$$\mathcal{F}_{-n} \subset \mathcal{F}_{-(n+1)} \qquad \forall n \geq 0.$$

Let $(M_{-n})_{n \geq 0}$ *be an associated backward submartingale:*

$$\mathbb{E}[M_{-(n+1)} \mid \mathcal{F}_{-n}] \geq M_{-n} \qquad \forall n \geq 0.$$

Suppose that

$$\sup_{n \geq 0} \mathbb{E}[|M_{-n}|] < \infty.$$

Then (M_{-n}) *converges as* $n \to \infty$ *almost surely and in* L^1 *to*

$$M_{-\infty} = \mathbb{E}[M_0 \mid \mathcal{F}_{-\infty}], \qquad \mathcal{F}_{-\infty} = \bigcap_{n \geq 0} \mathcal{F}_{-n}.$$

Proof. Left as an exercise. Hint: consider the family

$$\left(\mathcal{F}_{-n \wedge N}\right)_{n \in \{0, \ldots, N\}}, \qquad \left(M_{-n \wedge N}\right)_{n \in \{0, \ldots, N\}},$$

and apply the forward convergence theorem. □

Deduce from it that the number of intervals belonging to $[a, b]$ is finite for every $a < b$.

We now extend optional stopping to continuous time. We first recall the discrete-time statements.

Theorem 4.27 (Optional stopping theorem, discrete uniformly integrable case)**.** *Consider* $(\widetilde{M}_n)_{n \in \mathbb{N}}$ *a discrete-time (sub-)martingale. Assume that it is uniformly integrable. Then, for any stopping times* τ *and* σ *such that* $\tau \geq \sigma$*,*

$$\mathbb{E}[\widetilde{M}_\tau \mid \mathcal{G}_\sigma] \geq \widetilde{M}_\sigma$$

almost surely, with equality in the martingale case.

Theorem 4.28 (Optional stopping theorem, discrete bounded case)**.** *Consider $(\widetilde{M}_n)_{n\in\mathbb{N}}$ a discrete-time (sub-)martingale. Let τ and σ be two bounded stopping times such that $\tau \geq \sigma$. Then*

$$\mathbb{E}[\widetilde{M}_\tau \mid \mathcal{G}_\sigma] \geq \widetilde{M}_\sigma$$

almost surely, with equality in the martingale case.

Theorem 4.29 (Optional stopping theorem, continuous uniformly integrable case)**.** *Consider $(M_t)_{t\geq 0}$ a càdlàg $(\mathcal{F}_t)$-(sub-)martingale. Assume that it is uniformly integrable. Then, for any stopping times τ and σ such that $\tau \geq \sigma$,*

$$\mathbb{E}[M_\tau \mid \mathcal{F}_\sigma] \geq M_\sigma$$

almost surely, with equality in the martingale case.

Proof. 1. We first show that $M_t \in L^1$ for every stopping time t, and that for

$$\sigma_n = \frac{\lceil 2^n \sigma \rceil}{2^n}$$

we have

$$M_{\sigma_n} \to M_\sigma$$

almost surely and in L^1. Consider then the discrete-time martingales

$$M_q^{(n)} = M_{q/2^n}, \qquad q \in \mathbb{N}.$$

Since (M_t) is uniformly integrable, so is each $(M_q^{(n)})_{q\in\mathbb{N}}$. Moreover σ_n is a stopping time for the dyadic filtration

$$\mathcal{F}_q^{(n)} = \mathcal{F}_{q/2^n},$$

because

$$\{\sigma_n \leq q\} \subset \{2^{-n}\sigma \leq q\} \in \mathcal{F}_q^{(n)}.$$

Applying the discrete optional stopping theorem gives

$$\mathbb{E}[M_{\sigma_n} \mid \mathcal{F}_{\tau_n}] \geq M_{\tau_n}, \tag{3}$$

with equality in the martingale case, for suitable dyadic approximations τ_n, σ_n of τ, σ. Setting

$$N_{-n} = M_{\sigma_n}, \qquad \mathcal{G}_{-n} = \mathcal{F}_{\sigma_n},$$

the sequence $(N_{-n})_{n\geq 0}$ is a backward martingale by (3), hence converges almost surely and in L^1. Since (M_t) is càdlàg, this limit is precisely M_σ.

2. We have $M_\tau, M_\sigma \in L^1$. Now consider the dyadic approximations

$$\sigma_n = \frac{\lceil 2^n \sigma \rceil}{2^n}, \qquad \tau_n = \frac{\lceil 2^n \tau \rceil}{2^n}.$$

Using the same argument as in step 1, and applying the optional stopping theorem in the discrete case, we obtain for every n,

$$\mathbb{E}[M_{\tau_n} \mid \mathcal{F}_{\sigma_n}] \geq M_{\sigma_n},$$

with equality in the martingale case. Therefore, for every $A \in \mathcal{F}_\sigma \subset \mathcal{F}_{\sigma_n}$,

$$\mathbb{E}[\mathbf{1}_A M_{\tau_n}] \geq \mathbb{E}[\mathbf{1}_A M_{\sigma_n}].$$

Using once more the right-continuity of (M_t) and passing to the limit in L^1 completes the proof. □

Theorem 4.30 (Optional stopping theorem, continuous bounded case)**.** *Consider $(M_t)_{t\geq 0}$ a càdlàg $(\mathcal{F}_t)$-(sub-)martingale. Let τ and σ be two bounded stopping times such that $\tau \geq \sigma$. Then*

$$\mathbb{E}[M_\tau \mid \mathcal{F}_\sigma] \geq M_\sigma$$

almost surely, with equality in the martingale case.

Proof. Exercise. □

Corollary 4.31. *Consider $(M_t)_{t\geq 0}$ a càdlàg $(\mathcal{F}_t)$-(sub-)martingale. Let τ and σ be two bounded stopping times such that $\tau \geq \sigma$. Then*

$$\mathbb{E}[M_\tau \mid \mathcal{F}_\sigma] \geq M_{\tau\wedge\sigma},$$

with equality in the martingale case.

Proof.

$$\begin{aligned}\mathbb{E}[M_\tau \mid \mathcal{F}_\sigma] &= \mathbf{1}_{\{\tau<\sigma\}}\mathbb{E}[M_\tau \mid \mathcal{F}_\sigma] + \mathbf{1}_{\{\sigma\leq\tau\}}\mathbb{E}[M_\tau \mid \mathcal{F}_\sigma] \\ &\geq \mathbf{1}_{\{\tau<\sigma\}}M_{\tau\wedge\sigma} + \mathbf{1}_{\{\sigma\leq\tau\}}M_\sigma \\ &= M_{\tau\wedge\sigma}.\end{aligned}$$

□

Corollary 4.32. *Stopped càdlàg (sub-)martingales are càdlàg (sub-)martingales.*

Proof. If τ is a stopping time, then $(M_{t\wedge\tau})_{t\geq 0}$ is a $(\mathcal{F}_t)$-(sub-)martingale. Apply the previous result with $\tau \wedge t$ and $\sigma = s$ for $s \leq t$. □

Example 4.33. Let $(B_t)_{t\geq 0}$ be a Brownian motion and

$$\tau_a = \inf\{t \geq 0 : \ B_t = a\}, \qquad a > 0.$$

Then the exponential martingale

$$M_t = \exp\left(\theta B_{t\wedge\tau_a} - \frac{\theta^2}{2}(t \wedge \tau_a)\right)$$

is bounded, hence converges in L^p and almost surely to

$$M_\infty = \exp\left(\theta a - \frac{\theta^2}{2}\tau_a\right).$$

Since $\mathbb{E}[M_\infty] = \mathbb{E}[M_0] = 1$, we deduce that

$$\mathbb{E}\left[e^{-\theta^2\tau_a/2}\right] = e^{-\theta a} \qquad \forall\theta > 0.$$

By symmetry, we also get

$$\mathbb{E}\left[e^{-\theta^2\tau_a/2}\right] = e^{-|\theta|a} \qquad \forall\theta \in \mathbb{R},$$

which yields the Laplace transform of τ_a.

Example 4.34. Let $(X_t, Y_t)_{t\geq 0}$ be a two-dimensional Brownian motion starting from $(0,0)$, and let

$$T = \inf\{t \geq 0 : \ Y_t = 1\}.$$

Then

$$\mathbb{E}[e^{i\theta T^{1/2} X_T}] = e^{-|\theta|}.$$

Since T is independent of $(X_t)_{t\geq 0}$, we get

$$X_T = T^{1/2} G, \qquad G \sim \mathcal{N}(0,1), \qquad G \perp T,$$

which implies

$$\mathbb{E}[e^{i\theta X_T}] = \mathbb{E}[e^{-\theta^2 T/2}] = e^{-|\theta|}.$$

Therefore X_T has the Cauchy distribution with density

$$\frac{1}{\pi(1+x^2)}.$$

Theorem 4.35 (Optional stopping theorem, closed case)**.** *Let $(M_t)_{t\geq 0}$ be a closed and right-continuous martingale with respect to $(\mathcal{F}_t)$. Then for every stopping times σ, τ such that $\sigma \leq \tau$ almost surely,*

$$\mathbb{E}[M_\tau \mid \mathcal{F}_\sigma] = M_\sigma \qquad \textit{almost surely.}$$

Proof. We use the proposition below, together with the fact that for any stopping time s,

$$\mathbb{E}[M_\infty \mid \mathcal{F}_s] = M_s.$$

Then

$$\mathbb{E}[M_\tau \mid \mathcal{F}_\sigma] = \mathbb{E}[\mathbb{E}[M_\infty \mid \mathcal{F}_\tau] \mid \mathcal{F}_\sigma] = \mathbb{E}[M_\infty \mid \mathcal{F}_\sigma] = M_\sigma.$$

□

Proposition 4.36. *If σ, τ are stopping times such that $\sigma \leq \tau$ almost surely, then*

$$\mathcal{F}_\sigma \subset \mathcal{F}_\tau.$$

Theorem 4.37. *Let $(\widetilde{M}_n)_{n\in\mathbb{N}}$ be a closed discrete-time martingale. Then for any stopping times σ, τ such that $\sigma \leq \tau$ almost surely,*

$$\mathbb{E}[\widetilde{M}_\tau \mid \mathcal{G}_\sigma] = \widetilde{M}_\sigma.$$

Theorem 4.38. *Let $(\widetilde{M}_n)_{n\in\mathbb{N}}$ be a discrete-time martingale. Then the following are equivalent:*

(a) $(\widetilde{M}_n)$ is uniformly integrable;

(b) $\widetilde{M}_n \to \widetilde{M}_\infty$ in L^1;

(c) $(\widetilde{M}_n)$ is closed, and more precisely

$$\widetilde{M}_n = \mathbb{E}[\widetilde{M}_\infty \mid \mathcal{G}_n].$$

Proof. The idea is to discretize time and use dyadic skeletons. Let

$$\sigma_n = \frac{\lceil 2^n \sigma \rceil}{2^n}.$$

Consider the martingales

$$M_q^{(n)} = M_{q/2^n}.$$

Then $(M_q^{(n)})_q$ is a closed martingale for the dyadic filtration, and σ_n is a stopping time for this dyadic filtration. By the discrete optional stopping theorem,

$$\mathbb{E}[M_\infty \mid \mathcal{F}_{\sigma_n}] = M_{\sigma_n}.$$

Since $\mathcal{F}_{\sigma_n} \supset \mathcal{F}_\sigma$ and $\sigma_n \downarrow \sigma$ almost surely, $M_{\sigma_n} \to M_\sigma$ almost surely by right-continuity. On the other hand, the family $(M_{\sigma_n})_n$ is uniformly integrable because

$$\mathbb{E}[\mathbf{1}_{\{|M_{\sigma_n}|\geq A\}}|M_{\sigma_n}|] \leq \mathbb{E}[\mathbf{1}_{\{|M_\infty|\geq A\}}|M_\infty|].$$

Hence $M_{\sigma_n} \to M_\sigma$ in L^1, which yields the claim. □

Exercise 4.39. Let $\{X_i : i \in \mathbb{N}\}$ be a uniformly integrable sequence of random variables. Suppose that $(X_i)_i$ converges to X in probability. Then $(X_i)_i$ converges in L^1.

Proof. First note that it is standard to show that there exists a subsequence converging to X almost surely. Fatou's lemma implies that $X \in L^1$.

Furthermore, let $\varepsilon > 0$. There exist N and $M \geq 1$ such that for all $i \geq N$,

$$\mathbb{P}(|X_i| \geq M) \leq \mathbb{E}[\mathbf{1}_{\{|X_i|\geq M\}}|X_i|] \leq \varepsilon/6$$

and

$$\mathbb{P}(|X| \geq M) \leq \mathbb{E}[\mathbf{1}_{\{|X|\geq M\}}|X|] \leq \varepsilon/6.$$

Now, using that $X_i \to X$ in probability,

$$\begin{aligned}\mathbb{E}[|X_i - X|] &\leq \mathbb{E}[|X_i - X|\,\mathbf{1}_{\{|X_i|\geq 2M\}\cup\{|X|\geq M\}}] + \mathbb{E}[|X_i - X|\,\mathbf{1}_{\{|X_i|\vee|X|\leq M\}}] \\ &\leq \frac{2}{3}\varepsilon + \frac{\varepsilon}{3} + 3M\,\mathbb{P}(|X_i - X| \geq \varepsilon/3),\end{aligned}$$

which yields $\limsup_i \mathbb{E}[|X_i - X|] \leq \varepsilon$ for every ε, hence the proof. □

Exercise 4.40. Let $(X_i)_{i\in I}$ be uniformly integrable. Show that $(X_i)_{i\in I}$ is equi-continuous, i.e.: for every $\varepsilon > 0$, there exists $\delta > 0$ such that for every $A \in \mathcal{F}$,

$$\mathbb{P}(A) \leq \delta \Longrightarrow \mathbb{E}[\mathbf{1}_A|X_i|] \leq \varepsilon.$$

Proof. Let $\varepsilon > 0$. Since $(X_i)_i$ is uniformly integrable, there exists M such that

$$\sup_i \mathbb{E}[\mathbf{1}_{\{|X_i|\geq M\}}|X_i|] \leq \varepsilon/2.$$

On the other hand, choosing δ so that $\mathbb{P}(A) \leq \delta$, we get for all i,

$$\mathbb{E}[\mathbf{1}_A|X_i|] \leq \varepsilon/2 + \mathbb{E}[\mathbf{1}_A\mathbf{1}_{\{|X_i|<M\}}|X_i|] \leq \varepsilon/2 + M\delta.$$

Choosing $\delta = \varepsilon/(2M)$ concludes the proof. □

Exercise 4.41. Let $(X_i)_{i\in I}$ be uniformly integrable and let $(\mathcal{G}_j)_{j\in J}$ be a family of sub-σ-fields of $\mathcal{F}$. Show that

$$\{\mathbb{E}[X_i \mid \mathcal{G}_j] : i \in I,\ j \in J\}$$

is uniformly integrable.

Proof. Since $(X_i)_i$ is uniformly integrable, it is L^1-bounded:

$$\sup_i \mathbb{E}[|X_i|] =: M < \infty.$$

By the previous exercise, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$\mathbb{P}(A) \leq \delta \Longrightarrow \mathbb{E}[\mathbf{1}_A |X_i|] \leq \varepsilon \qquad \forall i.$$

On the other hand, writing $Y_{i,j} := \mathbb{E}[X_i \mid \mathcal{G}_j]$, Jensen gives

$$\mathbb{E}[\mathbf{1}_{\{|Y_{i,j}| \geq R\}} |Y_{i,j}|] \leq \mathbb{E}\Big[\mathbf{1}_{\{|Y_{i,j}| \geq R\}} \mathbb{E}[|X_i| \mid \mathcal{G}_j]\Big].$$

Yet by Markov's inequality,

$$\mathbb{P}(|Y_{i,j}| \geq R) \leq R^{-1}\mathbb{E}[|Y_{i,j}|] \leq R^{-1}\mathbb{E}[|X_i|] \leq R^{-1}M.$$

Choosing $R = \delta^{-1}M$, the previous implication yields

$$\mathbb{E}[\mathbf{1}_{\{|Y_{i,j}| \geq R\}} |Y_{i,j}|] \leq \varepsilon \qquad \forall i, j.$$

□

5 Continuous Semi-Martingales

Here we develop a general theory of stochastic integrals with respect to stochastic processes. We first consider processes with bounded variation, second martingale-like processes, and finally processes which are sums of the two.

5.1 Bounded Variation Functions and Processes

Definition 5.1. *Let $I \subset \mathbb{R}$ be an interval. We say that a function $f : I \to \mathbb{R}$ is of bounded variation on $[a, b] \subset I$ if the total variation of f on $[a, b]$, defined by*

$$\mathrm{TV}([a,b], f) = \sup \left\{ \sum_{i=0}^{n-1} |f(t_{i+1}) - f(t_i)| : \ n \in \mathbb{N}, \ a = t_0 < \cdots < t_n = b \right\},$$

is finite.

Remark 5.2. If $I = [n, M]$, it is enough to check that $\mathrm{TV}([n, M], f) < \infty$.

Proposition 5.3. *Let $f : I \to \mathbb{R}$ be of finite variation on I. Then there exist two non-decreasing functions F_+, F_- such that*

$$f = F_+ - F_-.$$

Remark 5.4. The functions F_+ and F_- are not unique.

Proof. We first consider the case $I = [a, b]$. Set

$$F_+(t) = \mathrm{TV}([a,t], f), \qquad t \in [a,b].$$

By definition F_+ is non-decreasing. Let

$$F_- = F_+ - f.$$

Then for $s \le t$ and $\varepsilon > 0$,

$$\begin{aligned} F_-(t) - F_-(s) &= F_+(t) - F_+(s) - f(t) + f(s) \\ &\ge \mathrm{TV}([s,t], f) + f(s) - f(t) - \varepsilon \\ &\ge -\varepsilon, \end{aligned}$$

where we used the definition of total variation and the bound $\mathrm{TV}([s,t], f) \ge |f(t) - f(s)|$. Letting $\varepsilon \to 0$ shows that F_- is non-decreasing.

The cases $[a, +\infty)$ and $(-\infty, a]$ are similar. For $I = \mathbb{R}$, one may assume without loss of generality that $f(0) = 0$ and define

$$F_+(t) = \begin{cases} \mathrm{TV}([0,t], f), & t \ge 0, \\ -\mathrm{TV}([t,0], f), & t \le 0. \end{cases}$$

The same proof then applies. □

Based on this result, we can define a unique signed measure on I associated with a bounded variation function f.

Theorem 5.5. *Let $F : I \to \mathbb{R}$ be a non-decreasing function. Then there exists a unique finite measure on $(I, \mathcal{B}(I))$, denoted by dF, such that for all $s \le t$ in I,*

$$dF([s,t]) = F(t) - F(s).$$

Proof. Uniqueness is an easy consequence of the π-λ theorem.

For existence, we first consider the case where F is continuous. Define the generalized inverse

$$F^{-1}(u) = \inf\{t : \ F(t) \geq u\}, \qquad u \in F(I).$$

Let U be uniformly distributed on $F(I)$, and let π be the law of $F^{-1}(U)$. Then

$$\pi([s,t]) = \mathbb{P}(F^{-1}(U) \leq t) - \mathbb{P}(F^{-1}(U) \leq s) = \frac{F(t) - F(s)}{|F(I)|}.$$

Setting

$$dF = |F(I)|\, \pi$$

concludes the proof. The general càdlàg increasing case follows by the standard extension theorems. □

Definition 5.6. *Let f be a bounded variation function from I to $\mathbb{R}$. Let F_+ and F_- be two non-decreasing functions such that*

$$f = F_+ - F_-.$$

Set

$$df = dF_+ - dF_-.$$

This does not depend on the choice of F_+ and F_-, and is called the Stieltjes signed measure associated with f.

Proof. If $f = \widetilde{F}_+ - \widetilde{F}_-$ is another decomposition, then

$$dF_+ - dF_- = d\widetilde{F}_+ - d\widetilde{F}_-.$$

Indeed, both signed measures have the same values on intervals $[s,t]$, namely

$$f(t) - f(s),$$

and the π-λ theorem concludes. □

Theorem 5.7. *Let f be a bounded variation function on I. Then there exist two unique measures on $(I, \mathcal{B}(I))$, denoted by df^+ and df^-, such that*

(a)

$$df = df^+ - df^-;$$

(b) df^+ and df^- are mutually singular, i.e. there exists $N \in \mathcal{B}(I)$ such that

$$df^+(N) = 0, \qquad df^-(I \setminus N) = 0.$$

Proof. This is an easy consequence of the Hahn–Jordan decomposition theorem. A direct argument is as follows: write

$$df = \varphi\, d\lambda$$

for a dominating finite measure λ, set

$$df^+ = \mathbf{1}_{\{\varphi \geq 0\}} \varphi\, d\lambda, \qquad df^- = -\mathbf{1}_{\{\varphi \leq 0\}} \varphi\, d\lambda,$$

and check the two required properties. □

Theorem 5.8. *Let $\mu = \lambda_1 - \lambda_2$, where λ_1, λ_2 are finite measures on $(X, \mathcal{X})$. Then there exist two unique measures μ^+ and μ^- such that*

$$\mu = \mu^+ - \mu^-,$$

and μ^+ and μ^- are mutually singular.

Definition 5.9. *Let $f : I \to \mathbb{R}$ be of bounded variation. We define the total variation measure of f as the measure*

$$|df| = df^+ + df^-.$$

We call

$$|df|(I)$$

the total variation of f.

Definition 5.10. *Let $\mu = \lambda_1 - \lambda_2$, where λ_1, λ_2 are finite measures. We define the total variation measure of μ by*

$$|\mu| = \mu^+ + \mu^-.$$

Its total mass $|\mu|(X)$ is called the total variation of μ.

Exercise 5.11. Show that f has density with respect to $|df|$ given by $\mathbf{1}_N - \mathbf{1}_{N^c}$, where N is such that

$$df^-(N) = 0 \qquad \text{and} \qquad df^+(N^c) = 0.$$

Definition 5.12. *Let $f : I \to \mathbb{R}$ be of bounded variation. For any bounded Borel function $g : I \to \mathbb{R}$ such that*

$$\int |g|\,|df| < \infty,$$

we define

$$\int g\,df = \int g\,df^+ - \int g\,df^-.$$

Proposition 5.13. *Let $f : I \to \mathbb{R}$ be of bounded variation, $g : I \to \mathbb{R}$ measurable and bounded, and let $[a, b] \subset I$.*

(1)

$$\int_{[a,b]} g\,df = \int_{[a,b]} g\,df^+ - \int_{[a,b]} g\,df^-.$$

(2)

$$\left| \int_{[a,b]} g\,df \right| \le \int_{[a,b]} |g|\,|df|.$$

(3) If

$$G(t) = \int_{[a,t]} g\,df, \qquad t \in I,$$

then G is of bounded variation on I, with

$$dG = g\,df, \qquad dG^+ = g\,df^+ + g\,df^-, \qquad dG^- = g\,df^- + g\,df^+,$$

up to the obvious sign convention when g changes sign.

(4) If $f \in C^1(I)$, then

$$df = f'(t)\,dt \qquad \text{and} \qquad |df| = |f'(t)|\,dt.$$

Proof. (1) and (2) follow directly from the definition.

For (3), by definition

$$G(t) - G(s) = dG([s,t]) = \int_{[s,t]} g\, df,$$

so $dG = g\, df$.

For (4), one has

$$f(t) - f(s) = \int_s^t f'(u)\, du,$$

and similarly for the positive and negative variations. □

Proposition 5.14. *Let $f : I \to \mathbb{R}$ be of bounded variation and $g : I \to \mathbb{R}$ measurable and bounded. Then for every mesh $(t_i^n)_i$ of $[a,b]$ whose mesh size tends to 0,*

$$\int_{[a,b]} g\, df = \lim_{n\to\infty} \sum_i g(t_i^n)(f(t_{i+1}^n) - f(t_i^n)).$$

Remark 5.15 (Comparison with total variation). If $f : I \to \mathbb{R}$ is of bounded variation and $[a,b] \subset I$, then

$$|df|([a,b]) = \sup\left\{ \sum_i |f(t_{i+1}) - f(t_i)| :\ (t_i)_i \text{ mesh of } [a,b] \right\}.$$

Proof. The easy inequality is immediate from the definition of $|df|$. For the reverse inequality, normalize to $[0,1]$ and $|df|([0,1]) = 1$, then construct a martingale on dyadic partitions whose expectation gives the sums

$$S_n = \sum_i |f(i2^{-n}) - f((i-1)2^{-n})|$$

and show that $S_n \to 1$. This yields the claim. □

5.2 Stochastic Processes with Bounded Variation

We consider a filtered probability space

$$(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$$

satisfying the usual conditions, and all random variables are defined on this space.

Definition 5.16. *We say that $(A_t)_{t\geq 0}$ is a bounded variation process if, for every ω, $t \mapsto A_t(\omega)$ is of bounded variation on every compact interval, and (A_t) is $(\mathcal{F}_t)$-adapted.*

Proposition 5.17. *Let (A_t) be a bounded variation process and (H_t) a progressively measurable process such that for every t,*

$$\int_0^t |H_s|\, |dA_s| < \infty.$$

Then the process $(H \cdot A)_t$ defined by

$$(H \cdot A)_t = \int_0^t H_s\, dA_s$$

is again a bounded variation process.

Proof. By the previous deterministic results, it only remains to check adaptedness. Decompose

$$\int H\, dA = \int H_+\, dA_+ + \int H_-\, dA_- - \int H_+\, dA_- - \int H_-\, dA_+.$$

We may therefore assume that $H \geq 0$ and A is non-decreasing. Then for $t \geq 0$, every $u \in [0,t]$, and every $C \in \mathcal{F}_t$, the map

$$\omega \mapsto \int_0^u H_s(\omega) \mathbf{1}_C(\omega)\, dA_s(\omega)$$

is measurable, which implies that $(H \cdot A)_t$ is $\mathcal{F}_t$-measurable. □

5.3 Continuous Local Martingales

Definition 5.18. *We recall that we consider a filtered probability space*

$$(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$$

satisfying the usual conditions, and all random variables are defined on this space.

A stochastic process $(M_t)_{t\geq 0}$ is a local $(\mathcal{F}_t)$-martingale if there exists a sequence of $(\mathcal{F}_t)$-stopping times $(\tau_n)_{n\in\mathbb{N}}$ such that

- *for every n, $(M_{t\wedge\tau_n})_{t\geq 0}$ is a true $(\mathcal{F}_t)$-martingale;*
- *$\tau_n \uparrow +\infty$ almost surely as $n \to \infty$.*

Exercise 5.19. Show that:

(1) any continuous martingale is a local martingale;

(2) if (M_t) is a local martingale and τ a stopping time, then $(M_{t\wedge\tau})_{t\geq 0}$ is also a local martingale;

(3) if (τ_n) reduces (M_t), then for every $a > 0$, $(M_{t\wedge\tau_n\wedge a})_{t\geq 0}$ is a uniformly integrable martingale;

(4) if (τ_n) reduces (M_t) and (σ_n) is a non-decreasing sequence of stopping times such that $\sigma_n \to \infty$, then $(\sigma_n \wedge \tau_n)$ also reduces (M_t);

(5) if (M_t) and (N_t) are local martingales, then $(M_t + N_t)$ is of finite variation if and only if it is constant.

Theorem 5.20. *Let $(M_t)_{t\geq 0}$ be a continuous local martingale.*

(1) If $M_t \in L^1$ and $M_t \geq 0$ almost surely for every t, then (M_t) is a supermartingale.

(2) If for every t,

$$\mathbb{E}\left[\sup_{s\in[0,t]} |M_s|\right] < \infty,$$

then (M_t) is a true continuous martingale.

Proof. (1) Let (τ_n) be a reducing sequence for (M_t). Then

$$\mathbb{E}[M_{t\wedge\tau_n} \mid \mathcal{F}_s] = M_{s\wedge\tau_n} \qquad \forall s \leq t.$$

Since $M_t \geq 0$, Fatou's lemma gives, after letting $n \to \infty$,

$$\mathbb{E}[M_t \mid \mathcal{F}_s] \leq M_s,$$

so (M_t) is a supermartingale.

(2) The proof follows the same lines as above, using dominated convergence instead of Fatou's lemma. □

Corollary 5.21. *Let $(M_t)_{t\geq 0}$ be a continuous local martingale such that $M_0 \in L^1$. Then the sequence of stopping times*

$$\tau_n = \inf\{t \geq 0 : \ |M_t - M_0| = n\}$$

reduces (M_t).

Proof. For every n, $(M_{t\wedge\tau_n})_{t\geq 0}$ is a bounded local martingale, hence a true martingale by the previous theorem. In addition, continuity implies $\tau_n \to \infty$ almost surely. □

Theorem 5.22. *Let $(M_t)_{t\geq 0}$ be a local martingale of bounded variation. Then almost surely*

$$M_t = M_0 \qquad \forall t.$$

Proof. First assume that (M_t) is a bounded martingale of bounded variation on $[0,t]$, so that

$$|dM|([0,t]) < \infty.$$

Then for every mesh,

$$\mathbb{E}[M_t^2] = \sum_i \mathbb{E}\Big[(M_{t_{i+1}} - M_{t_i})^2\Big] \leq \sup_i |M_{t_{i+1}} - M_{t_i}|\,|dM|([0,t]),$$

which goes to 0 as the mesh goes to 0. Hence $M_t = M_0$ almost surely for every t.

If (M_t) is only a local martingale, consider

$$\tau_n = \inf\{t \geq 0 :\ |M_t| + |dM|([0,t]) = n\}.$$

Then (τ_n) reduces (M_t) and $(M_{t\wedge\tau_n})$ is a bounded martingale of bounded variation. Applying the first part shows that

$$M_{t\wedge\tau_n} = M_0 \qquad \forall n,\ \forall t,$$

and letting $n \to \infty$ concludes the proof. □

5.4 Quadratic Variation of Continuous Local Martingales

We already know that Brownian motion has quadratic variation $t \mapsto t$. We now extend this to any continuous local martingale.

Theorem 5.23. *Let (M_t) be a continuous local martingale. Then there exists an increasing continuous $(\mathcal{F}_t)$-adapted process $(\langle M\rangle_t)_{t\geq 0}$ such that*

$$M_t^2 - \langle M\rangle_t$$

is a local martingale and $\langle M\rangle_0 = 0$. Moreover, $\langle M\rangle$ is the unique increasing continuous adapted process with this property. Finally, for any sequence of meshes (t_i^n) of $[0,t]$,

$$\sum_i (M_{t_{i+1}^n} - M_{t_i^n})^2 \xrightarrow[n\to\infty]{\mathbb{P}} \langle M\rangle_t.$$

Proof. Admitted. □

Remark 5.24. $\langle M\rangle$ is called the bracket or quadratic variation of (M_t). In discrete time, one checks that if (M_n) is a martingale, then

$$H_n = \sum_{k=0}^{n-1} \mathbb{E}[(M_{k+1} - M_k)^2 \mid \mathcal{F}_k]$$

is a predictable increasing process such that $M_n^2 - H_n$ is a martingale.

Remark 5.25. For an adapted process (A_t) and a stopping time τ, we denote by $(A_t^\tau)_{t\geq 0}$ the stopped process

$$A_t^\tau = A_{t\wedge\tau}.$$

Theorem 5.26. *Let (M_t) be a continuous local martingale and τ a stopping time. Then (M_t^τ) is again a continuous local martingale and*

$$\langle M^\tau\rangle_t = \langle M\rangle_{t\wedge\tau} = \langle M\rangle_t^\tau.$$

Proof. That M^τ is a local martingale follows because any reducing sequence for (M_t) also reduces (M_t^τ), since

$$M_{t\wedge\tau\wedge\tau_n}$$

is a martingale. The identity for the brackets follows by checking that

$$(M_t^\tau)^2 - \langle M\rangle_{t\wedge\tau}$$

is a local martingale, hence by uniqueness it must be the bracket of M^τ. □

Definition 5.27. *A martingale (M_t) is said to be square-integrable if for every t,*

$$\mathbb{E}[M_t^2] < \infty.$$

Theorem 5.28. *Let $(M_t)_{t\geq 0}$ be a continuous local martingale with $M_0 \in L^2$. Then*

$$\mathbb{E}[\langle M\rangle_t] < \infty \qquad \forall t$$

if and only if (M_t) is a true continuous square-integrable martingale. In addition,

$$M_t^2 - \langle M\rangle_t$$

is then a true martingale, and

$$\mathbb{E}[\langle M\rangle_t] = \mathbb{E}[M_t^2] - \mathbb{E}[M_0^2].$$

Proof. Let (τ_n) be a sequence reducing both $(M_t^2 - \langle M\rangle_t)$ and (M_t). Then for every $s \leq t$,

$$\mathbb{E}[M_{t\wedge\tau_n}^2 - \langle M\rangle_{t\wedge\tau_n} \mid \mathcal{F}_s] = M_{s\wedge\tau_n}^2 - \langle M\rangle_{s\wedge\tau_n},$$

so in particular

$$\mathbb{E}[M_{t\wedge\tau_n}^2] = \mathbb{E}[\langle M\rangle_{t\wedge\tau_n}] + \mathbb{E}[M_0^2].$$

If (M_t) is square-integrable, Doob's inequality implies

$$\mathbb{E}\left[\sup_{s\in[0,t]} M_s^2\right] \leq 4\mathbb{E}[M_t^2] < \infty,$$

and monotone convergence then gives

$$\mathbb{E}[\langle M\rangle_t] \leq 4\mathbb{E}[M_t^2] < \infty.$$

Conversely, if $\mathbb{E}[\langle M\rangle_t] < \infty$, the same identity and Fatou's lemma imply $M_t \in L^2$, and one upgrades the local martingale to a true square-integrable martingale. □

Corollary 5.29. *If $(M_t)_{t\geq 0}$ is a continuous local martingale with $M_0 = 0$ and*

$$\langle M\rangle_t = 0 \qquad \forall t,$$

then $M_t = 0$ for every t.

Proof. By the previous theorem, for every t,

$$\mathbb{E}[M_t^2] = \mathbb{E}[\langle M\rangle_t] = 0.$$

So $M_t = 0$ almost surely for every t, and then for all t by continuity. □

Definition 5.30. *A martingale (M_t) is said to be bounded in L^2 if*

$$\sup_{t\geq 0} \mathbb{E}[M_t^2] < \infty.$$

Proposition 5.31. *Let (M_t) be a continuous martingale with $M_0 = 0$. Then*

$$\mathbb{E}[\langle M\rangle_\infty] < \infty$$

if and only if (M_t) is bounded in L^2. In that case, (M_t) converges in L^2 to some M_∞ and

$$\mathbb{E}[M_\infty^2] = \mathbb{E}[\langle M\rangle_\infty] = \sup_{t\geq 0} \mathbb{E}[M_t^2].$$

Moreover,

$$(M_t^2 - \langle M\rangle_t)_{t\geq 0}$$

is uniformly integrable.

Proof. Using the previous theorem, for every t,

$$\mathbb{E}[M_t^2] = \mathbb{E}[\langle M\rangle_t] \leq \mathbb{E}[\langle M\rangle_\infty].$$

So (M_t) is bounded in L^2 if $\mathbb{E}[\langle M\rangle_\infty] < \infty$. Conversely, by monotone convergence,

$$\mathbb{E}[\langle M\rangle_\infty] = \sup_{t\geq 0} \mathbb{E}[\langle M\rangle_t] = \sup_{t\geq 0} \mathbb{E}[M_t^2].$$

The equality of norms implies the L^2-convergence of (M_t), and then $(M_t^2 - \langle M\rangle_t)$ is uniformly integrable since it is a martingale converging in L^1. □

Exercise 5.32. If (M_t) is a continuous martingale, then

$$\lambda^2\langle M\rangle_t = \langle \lambda M\rangle_t.$$

We can see from this result that the bracket acts as a norm square. The polarization identity

$$2\langle x, y\rangle = \|x+y\|^2 - \|x\|^2 - \|y\|^2 \qquad \forall x, y \in \mathbb{R}^d$$

therefore suggests the following definition.

Definition 5.33. *Let (M_t) and (N_t) be two continuous local martingales. We define the bracket between (M_t) and (N_t) by*

$$\langle M, N\rangle_t := \frac{1}{2}\big(\langle M+N\rangle_t - \langle M\rangle_t - \langle N\rangle_t\big).$$

Remark 5.34. This definition is consistent with the previous one since

$$\langle M, M\rangle_t = \langle M\rangle_t.$$

Using the polarization identity again,

$$\langle M, N\rangle_t = \frac{1}{4}\big(\langle M+N\rangle_t - \langle M-N\rangle_t\big).$$

Proposition 5.35. *Let (M_t) and (N_t) be two continuous local martingales.*

(1) $(\langle M, N\rangle_t)_{t\geq 0}$ is the unique continuous finite variation process such that

$$M_tN_t - \langle M, N\rangle_t$$

is a continuous local martingale.

(2) For any t and any sequence of meshes (t_i^n) of $[0,t]$,

$$\sum_i (M_{t_{i+1}^n} - M_{t_i^n})(N_{t_{i+1}^n} - N_{t_i^n}) \xrightarrow[n\to\infty]{\mathbb{P}} \langle M, N\rangle_t.$$

(3) For any stopping time τ,

$$\langle M^\tau, N^\tau\rangle_t = \langle M, N\rangle_{t\wedge\tau}.$$

Proof. (1) follows from the decomposition

$$M_t N_t = \frac{1}{2}((M_t + N_t)^2 - M_t^2 - N_t^2)$$

and the uniqueness of the bracket.

(2) follows by summation from the identities for the quadratic variations of $M+N$, M, and N.

(3) follows from uniqueness again, together with the stopped-process property of the bracket. □

Proposition 5.36. *Let (M_t) and (N_t) be two continuous martingales bounded in L^2. Then*

$$\mathbb{E}[\langle M, N\rangle_\infty] = \mathbb{E}[M_\infty N_\infty].$$

Proof. We already know that

$$\mathbb{E}[\langle M\rangle_\infty] = \mathbb{E}[M_\infty^2].$$

Applying this identity to $M+N$, and expanding the square, gives the result. □

Proposition 5.37. *The mapping*

$$(M, N) \in \mathcal{M}_2^c \longmapsto \langle M, N\rangle$$

is bilinear and symmetric.

Definition 5.38. *Two continuous local martingales (M_t) and (N_t) are said to be orthogonal if*

$$\langle M, N\rangle_t = 0 \qquad \forall t,$$

or equivalently if $(M_t N_t)_{t\geq 0}$ is a continuous local martingale.

Theorem 5.39 (Kunita–Watanabe inequality)**.** *Let (M_t) and (N_t) be two continuous local martingales, and let (H_t) and (K_t) be two progressively measurable processes. Then*

$$\int_0^\infty |H_t K_t|\, |d\langle M, N\rangle_t| \leq \left(\int_0^\infty H_t^2\, d\langle M\rangle_t\right)^{1/2} \left(\int_0^\infty K_t^2\, d\langle N\rangle_t\right)^{1/2}.$$

Proof. Denote for $s \leq t$,

$$\langle M, N\rangle_s^t = \langle M, N\rangle_t - \langle M, N\rangle_s.$$

The first step is to show that

$$|\langle M, N\rangle_s^t| \leq \left(\langle M\rangle_s^t \langle N\rangle_s^t\right)^{1/2} \qquad \text{a.s.}$$

Indeed, for any mesh (t_i^n) of $[s,t]$,

$$\sum_i (M_{t_{i+1}^n} - M_{t_i^n})(N_{t_{i+1}^n} - N_{t_i^n})$$

is bounded in absolute value by Cauchy–Schwarz:

$$\left(\sum_i (M_{t_{i+1}^n} - M_{t_i^n})^2\right)^{1/2} \left(\sum_i (N_{t_{i+1}^n} - N_{t_i^n})^2\right)^{1/2}.$$

Passing to the limit in probability yields the claim.

The second step is to discretize and use the comparison with total variation. For dyadic meshes one gets

$$\sum_i |\langle M, N\rangle_{t_i^n}^{t_{i+1}^n}| \le \left(\sum_i \langle M\rangle_{t_i^n}^{t_{i+1}^n}\right)^{1/2} \left(\sum_i \langle N\rangle_{t_i^n}^{t_{i+1}^n}\right)^{1/2},$$

and therefore

$$|d\langle M, N\rangle| \le \left(d\langle M\rangle\, d\langle N\rangle\right)^{1/2}$$

in the sense of measures.

Finally, for any Borel set $A \subset \mathbb{R}_+$, denoting

$$\lambda = d|\langle M, N\rangle|, \qquad \mu = d\langle M\rangle, \qquad \nu = d\langle N\rangle,$$

the measure inequality gives

$$|\lambda(A)| \le \left(\mu(A)\nu(A)\right)^{1/2}.$$

Applying this with step-function approximations of H and K, then using a monotone class argument, concludes the proof. □

5.5 Continuous Semi-Martingales

Definition 5.40. *A process $(X_t)_t$ is said to be a semi-martingale if it can be decomposed as*

$$X_t = X_0 + M_t + A_t,$$

where $(M_t)_t \in \mathcal{M}_{\text{loc}}^c$ and $(A_t)_t$ is of bounded variation, with

$$M_0 = 0, \qquad A_0 = 0.$$

Remark 5.41. The decomposition above is unique up to indistinguishability. It is referred to as the canonical decomposition.

Definition 5.42. *Let $(X_t)_t$ and $(Y_t)_t$ be two semi-martingales with decompositions*

$$X_t = X_0 + M_t + A_t, \qquad Y_t = Y_0 + N_t + B_t,$$

with $(M_t)_t$ and $(N_t)_t$ in $\mathcal{M}_{\text{loc}}^c$. We define the bracket of $(X_t)_t$ and $(Y_t)_t$ as the process of bounded variation

$$\langle X, Y\rangle_t = \langle M, N\rangle_t,$$

and in particular

$$\langle X\rangle_t = \langle X, X\rangle_t.$$

Proposition 5.43. *Using the notation of the previous definition, for any mesh $\{t_i^n\}_{i=0}^n$, $n \in \mathbb{N}$, of $[0, t]$, we have*

$$\sum_{i=0}^{n-1} (X_{t_{i+1}^n} - X_{t_i^n})(Y_{t_{i+1}^n} - Y_{t_i^n}) \xrightarrow[n\to\infty]{\mathbb{P}} \langle X, Y\rangle_t.$$

Proof. Exercise. □

Proposition 5.44. *Let $(H_t)_t$ be a continuous adapted process and let $(X_t)_t$ and $(Y_t)_t$ be two semi-martingales. Then, for any mesh $\{t_i^n\}_{i=0}^n$, $n \in \mathbb{N}$, of $[0,t]$, for every t,*

$$\sum_i H_{t_i^n}(X_{t_{i+1}^n} - X_{t_i^n})(Y_{t_{i+1}^n} - Y_{t_i^n}) \xrightarrow[n\to\infty]{\mathbb{P}} \int_0^t H_s \, d\langle X, Y\rangle_s.$$

Proof. Note by interpolation that we only need to consider the case $(X_t)_t = (Y_t)_t$. Consider the decomposition

$$X_s = X_0 + M_s + V_s, \qquad \forall s.$$

Then

$$\left|\sum_i H_{t_i^n}(X_{t_{i+1}^n} - X_{t_i^n})^2 - \sum_i H_{t_i^n}(M_{t_{i+1}^n} - M_{t_i^n})^2\right| \le \sup_{[0,t]}|H_s| \left\{2\sup_j |M_{t_{j+1}^n} - M_{t_j^n}| \, TV(V,[0,t]) + \sup_j |V_{t_{j+1}^n} - V_t\right.$$

This converges to 0 $\mathbb{P}$-a.s. since $(M_t)_t$ is almost surely continuous and $(V_t)_t$ is of bounded variation.

Therefore it remains to consider $X_s = M_s$. Now, for m, n, $m \ge n$,

$$\left|\sum_i H_{t_i^n}(M_{t_{i+1}^m} - M_{t_i^m})^2 - \int_0^t H_s \, d\langle M\rangle_s\right| = T_0(n,m),$$

with

$$T_0(n,m) \le T_1(n,m) + T_2(n,m) + T_3(n),$$

where

$$T_1(n,m) = \left|\sum_j H_{t_j^n}\left\{\sum_{i:\, t_i^m, t_{i+1}^m \in [t_j^n, t_{j+1}^n]} (M_{t_{i+1}^m} - M_{t_i^m})^2 - \left[\langle M\rangle_{t_{j+1}^n} - \langle M\rangle_{t_j^n}\right]\right\}\right|,$$

$$T_2(n,m) = \left|\sum_j H_{t_j^n}\left[\langle M\rangle_{t_{j+1}^n} - \langle M\rangle_{t_j^n}\right] - \sum_j H_{t_j^m}\left[\langle M\rangle_{t_{j+1}^m} - \langle M\rangle_{t_j^m}\right]\right|,$$

and

$$T_3(n) = \left|\sum_j H_{t_j^n}\left[\langle M\rangle_{t_{j+1}^n} - \langle M\rangle_{t_j^n}\right] - \int_0^t H_s \, d\langle M\rangle_s\right|.$$

We analyze each term separately.

For $T_1(n,m)$, one gets

$$T_1(n,m) \le \sum_j \sup_{s\in[t_j^n, t_{j+1}^n]} |H_{t_j^n} - H_s| \sum_{i:\, t_i^m, t_{i+1}^m \in [t_j^n, t_{j+1}^n]} (M_{t_{i+1}^m} - M_{t_i^m})^2,$$

and we deduce, since

$$\sum_i (M_{t_{i+1}^m} - M_{t_i^m})^2 \to \langle M\rangle_t$$

in probability as $m \to \infty$, and since $(H_s)_s$ is almost surely continuous, that for every $\varepsilon > 0$,

$$\lim_{n\to\infty} \limsup_{m\to\infty} \mathbb{P}(T_1(n,m) \ge \varepsilon) = 0.$$

Regarding $T_2(n,m)$, we get for every n,

$$\lim_{m\to\infty} T_2(n,m) = 0 \qquad \text{in probability.}$$

Further, by the previous proposition on the integral of Stieltjes, since $(H_s)_s$ is supposed to be continuous,

$$T_3(n) \to 0 \qquad \text{as } n \to \infty \text{ in probability.}$$

Combining these results, for any $\varepsilon > 0$,

$$\limsup_{n\to\infty} \limsup_{m\to\infty} \mathbb{P}(T_0(n,m) \geq \varepsilon) \leq \limsup_{n\to\infty} \limsup_{m\to\infty} \mathbb{P}(T_1(n,m) \geq \varepsilon/2) + \mathbb{P}(T_3(n) \geq \varepsilon/2).$$

Taking $n \to \infty$ completes the proof. □

6 Stochastic Integrals

We fix a probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ satisfying the usual conditions.

6.1 L^2-Martingales

Definition 6.1. *We denote by $\mathcal{M}_{2,c}$ the set of bounded martingales bounded in L^2, continuous, and null at 0, quotient by the equivalence relation $(M_t) \sim (N_t)$ if and only if (M_t) and (N_t) are indistinguishable.*

Proposition 6.2. *If $(M_t)_t \in \mathcal{M}_{2,c}$, then for any stopping time τ, $(M_t^\tau)_t \in \mathcal{M}_{2,c}$.*

For $\lambda \in \mathbb{R}$, if $(M_t)_t$ and $(N_t)_t$ belong to $\mathcal{M}_{2,c}$, then $\langle M, N\rangle_\infty$ is well-defined in L^1, and

$$((M_t), (N_t)) \in \mathcal{M}_{2,c} \longmapsto \mathbb{E}[\langle M, N\rangle_\infty]$$

is well-defined on $\mathcal{M}_{2,c}$.

Proof. The first point follows from the identity

$$2\langle M, N\rangle_t = \langle M + N\rangle_t - \langle M\rangle_t - \langle N\rangle_t.$$

Since all processes in the right-hand side are continuous martingales, $\langle M, N\rangle$ is continuous. Furthermore,

$$2|\langle M, N\rangle_t| \leq \langle M + N\rangle_t + \langle M\rangle_t + \langle N\rangle_t \in L^1.$$

By the previous chapter, we can define

$$((M_t), (N_t)) \in \mathcal{M}_{2,c} \longmapsto \mathbb{E}[\langle M, N\rangle_\infty].$$

□

Theorem 6.3. *The map $\langle \cdot, \cdot \rangle_{\mathcal{M}_{2,c}}$ is a scalar product on $\mathcal{M}_{2,c}$, making it a Hilbert space.*

Proof. It is often lost that it is bilinear and positive, and if

$$\langle (M_t), (N_t)\rangle_{\mathcal{M}_{2,c}} = \mathbb{E}[\langle M, N\rangle_\infty] = 0,$$

then $\langle N\rangle_\infty = 0$. We have seen in the last chapter that $N_t = N_\infty$ almost surely. Since $N_0 = 0$ and $(N_t)_t$ is continuous, it follows that $N_t = 0$ almost surely.

It remains to show completeness. Let $\{(M_t^n) : n \in \mathbb{N}\}$ be a Cauchy sequence in $\mathcal{M}_{2,c}$ for $\langle \cdot, \cdot \rangle_{\mathcal{M}_{2,c}}$. Since

$$\|(M_t^n)\|^2_{\mathcal{M}_{2,c}} = \mathbb{E}[\langle M^n\rangle_\infty] = \mathbb{E}[(M_\infty^n)^2],$$

$(M_\infty^n)_n$ is a Cauchy sequence in L^2, hence there exists $M^\infty \in L^2$ such that $M_\infty^n \to M^\infty$ in L^2 and up to a subsequence almost surely.

Define $M_t = \mathbb{E}[M^\infty \mid \mathcal{F}_t]$. Doob's theorem shows that $(M_t)_t$ is right-continuous and converges in L^2 and almost surely to M^∞. It remains to show that a modification $(\widetilde{M}_t)_t$ of $(M_t)_t$ belongs to $\mathcal{M}_{2,c}$, and

$$\|\widetilde{M}^n - \widetilde{M}\|_{\mathcal{M}_{2,c}} \to 0.$$

First, note that by Doob's inequality,

$$\mathbb{E}\Big[\sup_t |M_t^n - M_t|^2\Big] \leq 4 \sup_t \mathbb{E}[|M_t^n - M_t|^2] = 4\mathbb{E}[|M_\infty^n - M^\infty|^2] \to 0.$$

Therefore, up to a subsequence, $(M_t^n)_t$ converges uniformly on compact sets almost surely to a right-continuous process. Let Ω_0 be such that $\mathbb{P}(\Omega_0) = 1$, and on Ω_0, $(M_t^n(\omega))_t$ converges. We set $\widetilde{M}_t = M_t$ on Ω_0 and $\widetilde{M}_t = 0$ otherwise. Clearly $(\widetilde{M}_t)_t \in \mathcal{M}_{2,c}$ and is a modification of $(M_t)_t$. Finally,

$$\mathbb{E}[\langle \widetilde{M}^n - \widetilde{M}\rangle_\infty] = \mathbb{E}[|\widetilde{M}_\infty^n - \widetilde{M}_\infty|^2] = \mathbb{E}[|M_\infty^n - M^\infty|^2] \to 0.$$

□

Definition 6.4. *Let $M \in \mathcal{M}_{2,c}$, and denote by $\mathcal{L}^2(M)$ the set of non-progressive processes $(H_t)_{t\geq 0}$ such that*

$$(\omega, t) \longmapsto H_t(\omega)$$

is $\mathcal{F}/\mathcal{B}(\mathbb{R}_+)$-measurable and

$$\mathbb{E}\Big[\int_0^\infty H_s^2\, d\langle M\rangle_s\Big] < \infty.$$

We define the quotient relation on $\mathcal{L}^2(M)$ by

$$(H_t)_t \sim (K_t)_t \Longleftrightarrow \exists A \in \mathcal{F},\ \mathbb{P}(A) = 1 \text{ such that } \mathbb{E}\Big[\int_0^\infty |H_s(\omega) - K_s(\omega)|^2\, d\langle M\rangle_s\Big] = 0.$$

We denote by $L^2(M) = \mathcal{L}^2(M)/\sim$.

Theorem 6.5. *For any $M \in \mathcal{M}_{2,c}$, $L^2(M)$ is a Hilbert space endowed with the scalar product*

$$\langle H, K\rangle_{L^2(M)} = \mathbb{E}\Big[\int_0^\infty H_s K_s\, d\langle M\rangle_s\Big].$$

Proof. The proof follows exactly the same lines as the proof that every L^2-space is a Hilbert space and is omitted. □

Theorem 6.6. *Let $M \in \mathcal{M}_{2,c}$. Then $\phi(H) = \int H\, dM$ defines an element $\phi(H) \in \mathcal{M}_{2,c}$ for every $H \in L^2(M)$, denoted by $(H \cdot M)_t$, such that*

$$\langle (H\cdot M), N\rangle_{\mathcal{M}_{2,c}} = \int_0^\infty H_s\, d\langle M, N\rangle_s, \qquad \forall N \in \mathcal{M}_{2,c}.$$

In addition, the map

$$(H_t)_t \in L^2(M) \longmapsto (H\cdot M)_t \in \mathcal{M}_{2,c}$$

is an isometry.

Proof. For $M \in \mathcal{M}_{2,c}$ and $H \in L^2(M)$, $(H\cdot M)_t$ is first defined by

$$\Big(\int_0^t H_s\, dM_s\Big)_t$$

when $(H_t)_t$ is elementary.

Consider $(H_t)_t$ elementary, of the form

$$H_t = \sum_i H_i\, \mathbf{1}_{(t_i, t_{i+1}]}(t), \qquad 0 = t_0 < \cdots < t_n < \infty,$$

with $H_i \in \mathcal{F}_{t_i}$. Then we define

$$I_t = \sum_i H_i(M_{t_{i+1}\wedge t} - M_{t_i\wedge t}).$$

It is easy to prove that $(I_t)_t \in \mathcal{M}_{2,c}$. For any $\tau \in \mathbb{R}_+$, one may induct on $i \in \{0, \ldots, n\}$ to show

$$\mathbb{E}[I_\tau^2] = \sum_i \mathbb{E}\Big[H_i^2\big(M_{t_{i+1}\wedge\tau} - M_{t_i\wedge\tau}\big)^2\Big].$$

Indeed,

$$\begin{aligned}\mathbb{E}[I_\tau^2] &= \mathbb{E}\Big[H_n^2\big(M_{t_{n+1}\wedge\tau} - M_{t_n\wedge\tau}\big)^2\Big] \\ &\quad + 2\sum_{j<i} \mathbb{E}\Big[H_i H_j\big(M_{t_{i+1}\wedge\tau} - M_{t_i\wedge\tau}\big)\big(M_{t_{j+1}\wedge\tau} - M_{t_j\wedge\tau}\big)\Big].\end{aligned}$$

Conditioning on $\mathcal{F}_{t_i}$, we get that the cross term is 0. In addition, since $H_i \in \mathcal{F}_{t_i}$, by uniqueness and the identity

$$\mathbb{E}[(M_t - M_s)^2 \mid \mathcal{F}_s] = \mathbb{E}[\langle M\rangle_t - \langle M\rangle_s \mid \mathcal{F}_s],$$

we get

$$\mathbb{E}[I_\tau^2] = \mathbb{E}\left[\sum_i H_i^2(\langle M\rangle_{t_{i+1}\wedge\tau} - \langle M\rangle_{t_i\wedge\tau})\right] = \mathbb{E}\left[\int_0^\tau H_s^2\, d\langle M\rangle_s\right].$$

This shows the claimed isometry in the elementary case.

If $(N_t)_t \in \mathcal{M}_{2,c}$, then

$$I_t^N = \int_0^t H_s\, d\langle M, N\rangle_s = \sum_i H_i\Big(\langle M, N\rangle_{t_{i+1}\wedge t} - \langle M, N\rangle_{t_i\wedge t}\Big),$$

and one deduces easily that for every t,

$$\mathbb{E}\Big[I_t N_t - \int_0^t H_s\, d\langle M, N\rangle_s\Big] = 0.$$

So

$$\Big(I_t N_t - \int_0^t H_s\, d\langle M, N\rangle_s\Big)_t$$

is a martingale, and therefore

$$\langle I, N\rangle_t = \int_0^t H_s\, d\langle M, N\rangle_s, \qquad \forall t.$$

Equivalently, for every t,

$$\mathbb{E}\Big[I_t N_t - \int_0^t H_s\, d\langle M, N\rangle_s \,\Big|\, \mathcal{F}_s\Big] = I_s N_s - \int_0^s H_u\, d\langle M, N\rangle_u,$$

which is the martingale identity.

We first define the map on the set $\mathcal{E}$ of elementary processes. Then, by the previous remark, the map

$$(H_t)_t \in \mathcal{E} \longmapsto (I_t^H)_t$$

is an isometry on $\mathcal{E}$, hence can be uniquely extended by density to the closure of $\mathcal{E}$, which is dense in $L^2(M)$.

Let $H \in L^2(M)$.

Uniqueness. Suppose there are two processes (I_t^1) and (I_t^2) such that for every $(N_t)_t$,

$$\langle I^j, N\rangle_t = \int_0^t H_s\, d\langle M, N\rangle_s, \qquad \forall t.$$

Taking $N = I^1 - I^2$, we get

$$\langle I^1 - I^2\rangle_t = 0, \qquad \forall t,$$

so $(I_t^1) \sim (I_t^2)$ in $\mathcal{M}_{2,c}$.

Existence. We define a linear continuous form φ_H on $\mathcal{M}_{2,c}$ by

$$\varphi_H(N) = \langle (H \cdot M), N \rangle_{\mathcal{M}_{2,c}} \qquad \forall N \in \mathcal{M}_{2,c}.$$

Using the statement already proved, we set

$$\varphi_H(N) = \mathbb{E}\Big[\int_0^\infty H_s \, d\langle M, N \rangle_s \Big].$$

This is well-defined by the Kunita–Watanabe inequality and Cauchy–Schwarz:

$$|\varphi_H(N)| \le \mathbb{E}\Big[\int_0^\infty H_s^2 \, d\langle M \rangle_s \Big]^{1/2} \mathbb{E}[\langle N \rangle_\infty]^{1/2} \le C_H \|N\|_{\mathcal{M}_{2,c}}.$$

If

$$\psi^{(N)} = \langle (H \cdot M), N \rangle_{\mathcal{M}_{2,c}}, \qquad N \in \mathcal{M}_{2,c},$$

then

$$\psi^{(N)} = \mathbb{E}\left[\int_0^\infty H_s \, d\langle M, N \rangle_s \right],$$

and this is continuous by Kunita–Watanabe and Cauchy–Schwarz. Hence, by the Riesz theorem, there exists a unique $(H \cdot M) \in \mathcal{M}_{2,c}$ such that

$$\langle H \cdot M, N \rangle_{\mathcal{M}_{2,c}} = \mathbb{E}\Big[\int_0^\infty H_s \, d\langle M, N \rangle_s \Big].$$

This proves the theorem. □

Remark 6.7. For any right-continuous martingale $(X_t)_{t \ge 0}$ and any bounded stopping time τ,

$$\mathbb{E}[X_\tau] = \mathbb{E}[X_0].$$

Proof. This is just the optional stopping theorem in the case where X is a continuous-time martingale and τ is bounded. □

Proposition 6.8. *Let $(M_t)_t \in \mathcal{M}_{2,c}$ and $(H_t)_t \in L^2(M)$. Then*

$$\mathbb{E}[(H \cdot M)_\tau N_\tau] = \mathbb{E}\left[\int_0^\tau H_s \, d\langle M, N \rangle_s \right]$$

for every bounded stopping time τ.

Proof. Let τ be bounded. Then

$$\mathbb{E}[(H \cdot M)_\tau N_\tau] = \mathbb{E}[\mathbb{E}[(H \cdot M)_\tau \mid \mathcal{F}_\tau] \, N_\tau] = \mathbb{E}[(H \cdot M)_\tau N_\tau].$$

Applying the martingale identity to

$$C_t = (H \cdot M)_t N_t - \int_0^t H_s \, d\langle M, N \rangle_s$$

and using optional stopping gives the result. □

Remark 6.9. Parameterization extends to the bracket:

$$\langle \int_0^\cdot H_s \, dM_s, \, N \rangle_t = \int_0^t H_s \, d\langle M, N \rangle_s.$$

In particular, if $\int_0^\infty H_s \, dM_s \in \mathcal{M}_{2,c}$, then

$$\mathbb{E}\Big[\int_0^\infty H_s \, dM_s \Big] = 0.$$

Proposition 6.10. *Let $(M_t)_t \in \mathcal{M}_{2,c}$ and $(H_t)_t \in L^2(M)$. If $(N_t)_t \in \mathcal{M}_{2,c}$, then for every $A \in \mathcal{F}_0$,*

$$\int_0^\cdot \mathbf{1}_A H_s \, d\langle M, N\rangle_s = \mathbf{1}_A \int_0^\cdot H_s \, d\langle M, N\rangle_s.$$

Moreover, for $(K_t)_t \in L^2(N)$,

$$\langle H \cdot M, K \cdot N\rangle = \int_0^\cdot H_s K_s \, d\langle M, N\rangle_s.$$

Proof. The first identity is immediate by definition. For the second, we first take $(K_t)_t \in \mathcal{E}(N)$. Then for every $A \in \mathcal{F}_0$,

$$\int_0^\cdot \mathbf{1}_A H_s \, d\langle M, N\rangle_s = \mathbf{1}_A \int_0^\cdot H_s \, d\langle M, N\rangle_s.$$

Hence

$$\langle H \cdot M, K \cdot N\rangle = \int_0^\cdot H_s \, d\langle M, K \cdot N\rangle_s = \int_0^\cdot H_s K_s \, d\langle M, N\rangle_s.$$

The first identity is true by definition for $A \in \mathcal{F}_0$, and for step processes it then extends by linearity. The general case follows by approximation in $L^2(N)$. □

Proposition 6.11. *Let $(M_t)_t \in \mathcal{M}_{2,c}$ and $(H_t)_t, (K_t)_t \in L^2(M)$. Then*

$$((HK) \cdot M)_t = (H \cdot (K \cdot M))_t, \qquad \forall t.$$

Proof. For $N \in \mathcal{M}_{2,c}$, $t \geq 0$, by the characterization already proved,

$$\begin{aligned}\langle (HK) \cdot M, N\rangle_t &= \int_0^t H_s K_s \, d\langle M, N\rangle_s \\ &= \int_0^t H_s \, d\langle K \cdot M, N\rangle_s \\ &= \langle H \cdot (K \cdot M), N\rangle_t.\end{aligned}$$

Thus

$$((HK) \cdot M)_t = (H \cdot (K \cdot M))_t, \qquad \forall t \geq 0.$$

□

Proposition 6.12. *Let $(M_t)_t \in \mathcal{M}_{2,c}$, $(H_t)_t \in L^2(M)$, and let τ be a stopping time. Then*

$$(H \cdot M)^\tau_t = \left(H \, \mathbf{1}_{[0,\tau]} \cdot M\right)_t.$$

Proof. Clearly $H_s \mathbf{1}_{[0,\tau]}(s) \in L^2(M)$. In addition, for all $N \in \mathcal{M}_{2,c}$ and all t,

$$\begin{aligned}\langle (H \cdot M)^\tau, N\rangle_t &= \langle H \cdot M, N\rangle_{t\wedge\tau} \\ &= \int_0^{t\wedge\tau} H_s \, d\langle M, N\rangle_s \\ &= \int_0^t H_s \mathbf{1}_{[0,\tau]}(s) \, d\langle M, N\rangle_s \\ &= \int_0^t H_s \, d\langle M, N^\tau\rangle_s \\ &= \langle (H\mathbf{1}_{[0,\tau]}) \cdot M, N\rangle_t.\end{aligned}$$

Since $\langle M, N^\tau\rangle = \langle M^\tau, N\rangle$, the claim follows.

$$M^\tau_t = \int_0^t dM^\tau_s = (\mathbf{1}_{[0,\tau]} \cdot M)_t$$

is the special case $H = 1$. □

6.2 Integration with Respect to Continuous Local Martingales

Definition 6.13. *For $(M_t)_t \in \mathcal{M}_{c,\mathrm{loc}}$, continuous with $M_0 = 0$, we denote by $L^2_{\mathrm{loc}}(M)$ the set of progressively measurable processes $(H_t)_t$ such that*

$$\int_0^t H_s^2 \, d\langle M\rangle_s < \infty \qquad \text{a.s. for every } t \geq 0.$$

Theorem 6.14. *Let $(M_t)_t \in \mathcal{M}_{c,\mathrm{loc}}$, continuous, with $M_0 = 0$.*

a) For $H \in L^2_{\mathrm{loc}}(M)$, there exists a continuous local martingale, denoted $(H \cdot M)_t$, such that

$$\langle H \cdot M, N\rangle_t = \int_0^t H_s \, d\langle M, N\rangle_s, \qquad \forall t,$$

for every $N \in \mathcal{M}_{c,\mathrm{loc}}$, continuous, null at 0.

b) If $K \in L^2_{\mathrm{loc}}(N)$, then

$$\langle H \cdot M, K \cdot N\rangle_t = \int_0^t H_s K_s \, d\langle M, N\rangle_s.$$

c) If $N \in \mathcal{M}_{2,c}$ and $H \in L^2(M)$, then this definition coincides with the stochastic integral previously defined.

Proof. Uniqueness. This is similar to the $\mathcal{M}_{2,c}$ case.

Existence. Let

$$\tau_n = \inf\left\{t : \ \langle M\rangle_t + \int_0^t H_s^2 \, d\langle M\rangle_s \geq n\right\}.$$

By definition, $\langle M\rangle_{\tau_n} \leq n$, so $(M_t^{\tau_n})_t$ is a martingale bounded in L^2. Furthermore,

$$\int_0^t H_s^2 \, d\langle M^{\tau_n}\rangle_s = \int_0^{t\wedge\tau_n} H_s^2 \, d\langle M\rangle_s,$$

so the stopped process belongs to $L^2(M^{\tau_n})$. It follows that for every n we can consider $(H \cdot M^{\tau_n})_t$.

It follows that for every n we can consider $(H \cdot M^{\tau_n})_t$. We claim that for $m < n$,

$$(H \cdot M^{\tau_n})^{\tau_m} = H \cdot M^{\tau_m}.$$

Indeed, for every $N \in \mathcal{M}_{2,c}$,

$$\begin{aligned}
\langle (H \cdot M^{\tau_n})^{\tau_m}, N\rangle_t &= \langle H \cdot M^{\tau_n}, N\rangle_{t\wedge\tau_m} \\
&= \int_0^{t\wedge\tau_m} H_s \, d\langle M^{\tau_n}, N\rangle_s \\
&= \int_0^t H_s \, d\langle M^{\tau_n}, N\rangle_{s\wedge\tau_m} \\
&= \int_0^t H_s \, d\langle M^{\tau_m\wedge\tau_n}, N\rangle_s \\
&= \langle H \cdot M^{\tau_m}, N\rangle_t.
\end{aligned}$$

Hence the stopped processes are compatible. If $\tau_\infty = \sup_n \tau_n$, then $\tau_\infty = \infty$ almost surely, and we may define

$$(H \cdot M)_t = (H \cdot M^{\tau_n})_t \quad \text{on } \{t \leq \tau_n\}$$

gives a continuous adapted process, and by construction it is a continuous local martingale.

Now let $N \in \mathcal{M}_{c,\mathrm{loc}}$, continuous, null at 0, and define

$$\sigma_n = \tau_n \wedge \inf\{t : \ |N_t| \geq n\}.$$

Then $N^{\sigma_n} \in \mathcal{M}_{2,c}$. For every n,

$$\begin{aligned}\langle H \cdot M, N\rangle_t^{\sigma_n} &= \langle (H \cdot M)^{\tau_n}, N^{\sigma_n}\rangle_t \\ &= \langle H \cdot M^{\tau_n}, N^{\sigma_n}\rangle_t \\ &= \int_0^t H_s \, d\langle M^{\tau_n}, N^{\sigma_n}\rangle_s \\ &= \int_0^{t\wedge\sigma_n} H_s \, d\langle M, N\rangle_s.\end{aligned}$$

Since $\sigma_n \to \infty$ almost surely, by continuity we get

$$\langle H \cdot M, N\rangle_t = \int_0^t H_s \, d\langle M, N\rangle_s.$$

This proves a).

Proof of b). The proof is the same as for the $\mathcal{M}_{2,c}$ case, now with localization and continuity.

Proof of c). If $M \in \mathcal{M}_{2,c}$ and $H \in L^2(M)$, then

$$\mathbb{E}[\langle H \cdot M\rangle_\infty] = \mathbb{E}\left[\int_0^\infty H_s^2 \, d\langle M\rangle_s\right] < \infty.$$

Hence $(H \cdot M) \in \mathcal{M}_{2,c}$, and by uniqueness it coincides with the stochastic integral previously defined. □

Remark 6.15. If τ is a stopping time such that

$$\mathbb{E}\Big[\int_0^\tau H_s^2 \, d\langle M\rangle_s\Big] < \infty,$$

then

$$(H \cdot M)^\tau \in \mathcal{M}_{2,c},$$

hence

$$\mathbb{E}\Big[\int_0^\tau H_s \, dM_s\Big] = 0, \qquad \mathbb{E}\Big[\Big(\int_0^\tau H_s \, dM_s\Big)^2\Big] = \mathbb{E}\Big[\int_0^\tau H_s^2 \, d\langle M\rangle_s\Big].$$

6.3 Integration with Respect to Continuous Semimartingales

We extend these integrals to semimartingales. To this end, we consider as possible integrands those progressively measurable processes $(H_t)_t$ that are locally bounded, i.e.

$$\mathbb{P}\text{-a.s.}, \qquad \forall t, \ \sup_{[0,t]} |H_s| < \infty.$$

Continuous processes are locally bounded. Furthermore, for any process $(V_t)_t$ of bounded variation, any locally bounded process is integrable with respect to $|dV|$ on $[0,t]$ almost surely, for all t. Similarly, any locally bounded process belongs to $L^2_{\text{loc}}(M)$ for any continuous local martingale $(M_t)_t$.

Definition 6.16. *Consider $(X_t)_t$ a semimartingale with canonical decomposition*

$$X_t = X_0 + M_t + V_t.$$

We define the integral of a locally bounded progressively measurable process $(H_t)_t$ as the process

$$((H \cdot X)_t)_t = \left(\int_0^t H_s \, dX_s\right)_t$$

defined, for every t, by

$$\int_0^t H_s \, dX_s = \int_0^t H_s \, dM_s + \int_0^t H_s \, dV_s.$$

Proposition 6.17. *Let $(X_t)_t$ be a semimartingale with canonical decomposition $X_t = X_0 + M_t + V_t$, and let $(H_t)_t$, $(K_t)_t$ be two locally bounded processes.*

a)
$$\{(H \cdot (K \cdot X))_t : t \geq 0\} = \{((HK) \cdot X)_t : t \geq 0\}.$$

b) For every stopping time τ,
$$(H \cdot X)^\tau = (H\mathbf{1}_{[0,\tau]}) \cdot X = H \cdot X^\tau.$$

c) The map $((H_t)_t, (X_t)_t) \mapsto ((H \cdot X)_t)_t$ is bilinear.

d) If $V_t \equiv 0$, then $\left(\int_0^t H_s \, dX_s\right)_t$ is a local martingale.

e) If
$$H_t(\omega) = \sum_{i=0}^{p} H_{t_i^n}(\omega)\mathbf{1}_{(t_i, t_{i+1}]}(t),$$
then
$$\int_0^t H_s \, dX_s = \sum_{i=0}^{p-1} H_{t_i^n}(X_{t_{i+1}\wedge t} - X_{t_i \wedge t}).$$

Proof. It is an easy consequence of the previous results and definitions. □

Proposition 6.18. *Let $(X_t)_t$ be a semimartingale with canonical decomposition $X_t = X_0 + M_t + V_t$, and let $(H_t)_t$ be continuous and adapted. For every t and every mesh $\{t_i^n\}_{i=0}^{p_n}$,*
$$\sum_i H_{t_i^n}(X_{t_{i+1}^n \wedge t} - X_{t_i^n \wedge t}) \xrightarrow[n\to\infty]{\mathbb{P}} \int_0^t H_s \, dX_s.$$

Proof. The only difficult part is the continuous local martingale part, so we first suppose $X_t = M_t$. Define
$$K_t^n = \begin{cases} H_{t_i^n}, & \text{if } t_i^n < t \leq t_{i+1}^n, \\ 0, & \text{otherwise.} \end{cases}$$
Then
$$\sum_i H_{t_i^n}(X_{t_{i+1}^n \wedge t} - X_{t_i^n \wedge t}) = \int_0^t K_s^n \, dX_s.$$
If first $\sup_{[0,t]}(|H_s| + \langle X\rangle_t) \leq R$ almost surely for some finite R, then
$$\left(\int_0^t (K_s^n - H_s) \, dX_s\right)_{t \leq t}$$
is a martingale bounded in L^2, and
$$\mathbb{E}\left[\left(\int_0^t (K_s^n - H_s) \, dX_s\right)^2\right] = \mathbb{E}\left[\int_0^t (K_s^n - H_s)^2 \, d\langle X\rangle_s\right] \to 0$$
by dominated convergence. Hence
$$\int_0^t K_s^n \, dX_s \to \int_0^t H_s \, dX_s \qquad \text{in } L^2.$$

For the general case, define
$$\tau_n = \inf\{u \geq 0 : \sup_{r \in [0,u]} |H_r| + \langle X\rangle_u \geq n\}.$$

Then for every $\varepsilon > 0$,

$$\mathbb{P}\left(\left|\int_0^t K_s^n\,dX_s - \int_0^t H_s\,dX_s\right| > \varepsilon\right) \le \mathbb{P}(\tau_m \le t) + \mathbb{P}\left(\left|\int_0^t K_s^{n,m}\,dX_s^{\tau_m} - \int_0^t H_s \mathbf{1}_{[0,\tau_m]}\,dX_s^{\tau_m}\right| > \varepsilon\right),$$

where $K^{n,m}$ is defined as above with H^{τ_m}. By the first case, the second term goes to 0 as $n \to \infty$, and then letting $m \to \infty$ concludes the proof. □

Remark 6.19. In the previous result, in contrast to the Stieltjes integral,

$$\sum_i H_{t_{i+1}^n}(X_{t_{i+1}^n} - X_{t_i^n}) - \sum_i H_{t_i^n}(X_{t_{i+1}^n} - X_{t_i^n})$$

does not in general converge to 0 in probability. For example, one may take $H_t = X_t = B_t$, a Brownian motion.

We conclude this chapter with an easy instance of Itô's formula.

Proposition 6.20. *Let $(X_t)_t$ and $(Y_t)_t$ be two continuous semimartingales. Then*

$$X_t Y_t = X_0 Y_0 + \int_0^t X_s\,dY_s + \int_0^t Y_s\,dX_s + \langle X, Y\rangle_t, \qquad \forall t.$$

Proof. Let $t \ge 0$ and $\{t_i^n\}$, $n \in \mathbb{N}$, be a mesh of $[0,t]$. Then, as shown above,

$$\sum_i X_{t_i^n}(Y_{t_{i+1}^n} - Y_{t_i^n}) \to \int_0^t X_s\,dY_s \qquad \text{in probability,}$$

$$\sum_i Y_{t_i^n}(X_{t_{i+1}^n} - X_{t_i^n}) \to \int_0^t Y_s\,dX_s \qquad \text{in probability,}$$

and

$$\sum_i (X_{t_{i+1}^n} - X_{t_i^n})(Y_{t_{i+1}^n} - Y_{t_i^n}) \to \langle X, Y\rangle_t \qquad \text{in probability.}$$

On the other hand, summing these three terms gives

$$\sum_i (X_{t_{i+1}^n} Y_{t_{i+1}^n} - X_{t_i^n} Y_{t_i^n}) = X_t Y_t - X_0 Y_0,$$

and the proof is complete. □

Remark 6.21. This result shows that the product of two semimartingales is a semimartingale.

Remark 6.22. This result allows one to identify $M_t^2 - \langle M\rangle_t$ when $(M_t)_t$ is a continuous martingale with $M_0 = 0$: it is

$$2\int_0^t M_s\,dM_s.$$

Example 6.23. Let $(B_t)_t$ be a Brownian motion. For $m, n \ge 1$,

$$\langle B^m, B^n\rangle_t = mn\int_0^t B_s^{m+n-2}\,ds.$$

Proof. We can proceed by induction on $m+n$. Using the previous proposition,

$$B_t^{m+n} = \int_0^t B_s^n\,dB_s^m + + \int_0^t B_s^m\,dB_s^n + \langle B^m, B^n\rangle_t.$$

This implies

$$\langle B^m, B^n\rangle_t = \langle B^m, B^{m+n}\rangle_t = \int_0^t B_s^n\,d\langle B^m, B\rangle_s + + \int_0^t B_s^m\,d\langle B^n, B\rangle_s.$$

By the induction hypothesis,

$$\langle B^m, B\rangle_t = m\int_0^t B_s^{m-1}\,ds, \qquad \langle B^n, B\rangle_t = n\int_0^t B_s^{n-1}\,ds,$$

so

$$\langle B^m, B^n\rangle_t = \{m+n\}\,m\int_0^t B_s^{m+n-1}\,ds$$

and therefore

$$\langle B^m, B^n\rangle_t = mn\int_0^t B_s^{m+n-2}\,ds.$$

For the initialization,

$$\langle B, B\rangle_t = t = mn\int_0^t B_s^{m+n-2}\,ds \qquad \text{when } m = n = 1.$$

□

7 Itô Formula and Consequences

In all this chapter, we work on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ satisfying the usual conditions.

7.1 Itô Formula

Theorem 7.1. *1)* (One-dimensional case.) *Let $(X_t)_t$ be a continuous semimartingale valued in $\mathbb{R}$, and let $f : \mathbb{R} \to \mathbb{R}$ be C^2. Then $(f(X_t))_t$ is also a semimartingale and, almost surely for every t,*

$$f(X_t) = f(X_0) + \int_0^t f'(X_s)\, dX_s + \frac{1}{2}\int_0^t f''(X_s)\, d\langle X\rangle_s.$$

2) (Multidimensional case.) *Let $(X_t)_t$ be a $\mathbb{R}^d$-valued process such that its components $(X_t^i)_{t\geq 0}$, $i \in \{1, \dots, d\}$, are continuous semimartingales. Let $F : \mathbb{R}^d \to \mathbb{R}$ be C^2. Then, almost surely for every t,*

$$F(X_t) = F(X_0) + \sum_i \int_0^t \frac{\partial F}{\partial x_i}(X_s)\, dX_s^i + \frac{1}{2}\sum_{i,j}\int_0^t \frac{\partial^2 F}{\partial x_i \partial x_j}(X_s)\, d\langle X^i, X^j\rangle_s.$$

Remark 7.2. We denote

$$\sum_i \int_0^t \frac{\partial F}{\partial x_i}(X_s)\, dX_s^i \qquad \text{by} \qquad \int_0^t \nabla F(X_s)^\top\, dX_s.$$

Remark 7.3. In particular, if $(X_t)_t$ is a diffusion, i.e. if ξ is bounded on $[0,1]$, and $(B_t)_t$ is a Brownian motion such that

$$X_t = x + \int_0^t b_s\, ds + \int_0^t \sigma_s\, dB_s,$$

where $(B_t)_t$ is a d-dimensional Brownian motion and $\int_0^t \sigma_s\, dB_s$ is the vector with i-th component $\sum_j \int_0^t \sigma_s^{i,j}\, dB_s^j$, then

$$F(X_t) = F(X_0) + \int_0^t \nabla F(X_s)^\top b_s\, ds + \int_0^t \nabla F(X_s)^\top \sigma_s\, dB_s + \frac{1}{2}\sum_{i,j}\int_0^t \frac{\partial^2 F}{\partial x_i \partial x_j}(X_s)\, d\langle X^i, X^j\rangle_s.$$

The last term writes

$$\frac{1}{2}\sum_{i,j}\int_0^t \frac{\partial^2 F}{\partial x_i \partial x_j}(X_s)\, d\langle X^i, X^j\rangle_s = \frac{1}{2}\int_0^t \mathrm{Tr}\big(\sigma_s \sigma_s^\top \nabla^2 F(X_s)\big)\, ds,$$

using

$$\langle X^i, X^j\rangle_t = \sum_k \int_0^t \sigma_s^{i,k}\sigma_s^{j,k}\, ds.$$

Hence, in dimension d, if $a_s = \sigma_s \sigma_s^\top$, then

$$F(X_t) = F(X_0) + \int_0^t \nabla F(X_s)^\top b_s\, ds + \int_0^t \nabla F(X_s)^\top \sigma_s\, dB_s + \int_0^t \mathrm{Tr}(a_s \nabla^2 F(X_s))\, ds.$$

Proof. 1) Let $\{t_i^n\}_{i=0}^{p_n}$, $n \in \mathbb{N}$, be a mesh of $[0,t]$. Using Taylor's formula, for every i, n,

$$f(X_{t_{i+1}^n}) = f(X_{t_i^n}) + f'(X_{t_i^n})(X_{t_{i+1}^n} - X_{t_i^n}) + \int_0^1 f''\big(X_{t_i^n} + u(X_{t_{i+1}^n} - X_{t_i^n})\big)(1-u)\, du\, (X_{t_{i+1}^n} - X_{t_i^n})^2.$$

Summing over i, we get

$$f(X_t) - f(X_0) = \sum_i f'(X_{t_i^n})(X_{t_{i+1}^n} - X_{t_i^n}) + \sum_i I_i(t_i^n, t_{i+1}^n)(X_{t_{i+1}^n} - X_{t_i^n})^2,$$

where

$$I_i(t_i^n, t_{i+1}^n) = \int_0^1 f''(X_{t_i^n} + u(X_{t_{i+1}^n} - X_{t_i^n}))(1-u)\,du.$$

The first sum converges in probability to

$$\int_0^t f'(X_s)\,dX_s.$$

Regarding the second sum,

$$\left|\sum_i I_i(t_i^n, t_{i+1}^n)(X_{t_{i+1}^n} - X_{t_i^n})^2 - \sum_i \frac{1}{2} f''(X_{t_i^n})(X_{t_{i+1}^n} - X_{t_i^n})^2\right|$$

$$\leq \sup_i \left|I_i(t_i^n, t_{i+1}^n) - \frac{1}{2} f''(X_{t_i^n})\right| \sum_i (X_{t_{i+1}^n} - X_{t_i^n})^2.$$

Since $\sum_i (X_{t_{i+1}^n} - X_{t_i^n})^2 \to \langle X\rangle_t$ in probability, and

$$\sup_i \left|I_i(t_i^n, t_{i+1}^n) - \frac{1}{2} f''(X_{t_i^n})\right| \to 0 \qquad \text{almost surely,}$$

we get

$$\sum_i I_i(t_i^n, t_{i+1}^n)(X_{t_{i+1}^n} - X_{t_i^n})^2 \to \frac{1}{2}\int_0^t f''(X_s)\,d\langle X\rangle_s$$

in probability. Combining the previous convergences proves the result.

2) The proof follows exactly the same way, using the d-dimensional Taylor formula:

$$F(X_{t_{i+1}^n}) = F(X_{t_i^n}) + \nabla F(X_{t_i^n})^\top (X_{t_{i+1}^n} - X_{t_i^n}) + \left[(X_{t_{i+1}^n} - X_{t_i^n})^\top \nabla^2 F(\cdots)(X_{t_{i+1}^n} - X_{t_i^n})\right](1-u)\,du.$$

□

Remark 7.4. If $(X_t)_t$ is a process valued in an open $O \subset \mathbb{R}$, the one dimensional Itô formula is still valid assuming only that $f \in C^2(O)$.

Proposition 7.5. *If $(V_t)_t = (V_t^1, \ldots, V_t^d)$ is such that $(V_t^i)_t$ is of finite variation and $(X_t)_t = (X_t^1, \ldots, X_t^d)$ is a d-dimensional semimartingale, then the previous proposition extends: for*

$$F : \mathbb{R} \times \mathbb{R}^d \to \mathbb{R}$$

of class C^2, denoting $Z_t = (V_t, X_t)$, almost surely for every t,

$$F(Z_t) = F(Z_0) + \int_0^t \nabla F(Z_s)^\top\,dZ_s + \frac{1}{2}\sum_{i,j}\int_0^t \frac{\partial^2 F}{\partial x_i \partial x_j}(Z_s)\,d\langle X^i, X^j\rangle_s.$$

Remark 7.6. In particular, for $V_t = t$ and

$$X_t = x + \int_0^t b_s\,ds + \int_0^t \sigma_s\,dB_s,$$

we get

$$F(t, X_t) = F(0, X_0) + \int_0^t \frac{\partial F}{\partial s}(s, X_s)\,ds + \int_0^t \nabla F(s, X_s)^\top b_s\,ds + \frac{1}{2}\int_0^t \mathrm{Tr}(a_s \nabla_x^2 F(s, X_s))\,ds + \int_0^t \nabla F(s, X_s)^\top \sigma_s\,dB_s$$

Example 7.7. Let $(X_t)_t$ be a continuous semimartingale and $\lambda \in \mathbb{R}$.

1) The process $\mathcal{E}(X_t)$, defined by

$$\mathcal{E}(X_t) = \exp\left(X_t - \frac{1}{2}\langle X\rangle_t\right),$$

satisfies

$$\mathcal{E}(X_t) = \mathcal{E}(X_0) + \int_0^t \mathcal{E}(X_s)\, dX_s.$$

2) More generally, if $(X_t)_t$ is a continuous semimartingale satisfying

$$X_t = X_0 + \lambda t + \int_0^t \frac{\lambda^2}{2}\, d\langle X\rangle_s,$$

then $(e^{\lambda X_t})_t$ is a continuous local martingale.

Proof. 1) Itô's formula applied to $(x, y) \mapsto e^{x-y/2}$ gives

$$e^{X_t - \frac{1}{2}\langle X\rangle_t} = e^{X_0} + \int_0^t e^{X_s - \frac{1}{2}\langle X\rangle_s}\, dX_s - \frac{1}{2}\int_0^t e^{X_s - \frac{1}{2}\langle X\rangle_s}\, d\langle X\rangle_s + \frac{1}{2}\int_0^t e^{X_s - \frac{1}{2}\langle X\rangle_s}\, d\langle X\rangle_s,$$

hence the claim.

2) Apply the previous point to λX, noting that

$$\mathcal{E}(\lambda X_t) = e^{\lambda X_t - \frac{\lambda^2}{2}\langle X\rangle_t}.$$

□

7.2 Consequences of Itô's Formula

Lévy's characterization. Let $(B_t)_t = (B_t^1, \dots, B_t^d)$ be a d-dimensional standard Brownian motion. We have seen that

$$\langle B^i, B^j\rangle_t = \delta_{i,j} t.$$

We show that this characterizes Brownian motion.

Theorem 7.8. *Let $(M_t)_t = (M_t^1, \dots, M_t^d)$ be a d-dimensional continuous local martingale, null at 0, such that*

$$\langle M^i, M^j\rangle_t = \delta_{i,j} t.$$

Then $(M_t)_t$ is a d-dimensional standard Brownian motion with respect to $(\mathcal{F}_t)$.

Proof. We first show that for $s \le t$,

$$M_t - M_s \sim \mathcal{N}(0, (t-s) I_d) \qquad \text{conditionally on } \mathcal{F}_s.$$

To this end, fix $a \in \mathbb{R}^d$ and consider

$$N_t = a^\top M_t.$$

Then $(N_t)_t$ is a continuous local martingale with bracket

$$\langle N\rangle_t = \sum_{i,j} a_i a_j \langle M^i, M^j\rangle_t = \|a\|^2 t.$$

Moreover, we have seen that

$$\mathcal{E}(N_t) = \exp\left(N_t + \frac{1}{2}\langle N\rangle_t\right) = \exp\left(a^\top M_t + \frac{1}{2}\|a\|^2 t\right)$$

is a continuous local martingale that, being bounded on every finite interval, is a true martingale. Therefore,

$$\mathbb{E}[e^{a^\top M_t} \mid \mathcal{F}_s] = e^{a^\top M_s} e^{\frac{1}{2}\|a\|^2(t-s)}.$$

This identifies the conditional law of $M_t - M_s$ as $\mathcal{N}(0, (t-s)I_d)$, hence $M_t - M_s \perp \mathcal{F}_s$. Thus the process has independent Gaussian increments. Standard Gaussian maximal estimates then show continuity of paths, completing the proof. □

Example 7.9. If $(B_t)_t$ is a one-dimensional Brownian motion, then

$$\left(\int_0^t \operatorname{sign}(B_s)\, dB_s\right)_t$$

is again a Brownian motion.

Dambis–Dubins–Schwarz and Knight theorem.

Theorem 7.10. *Let $(M_t)_t$ be a one-dimensional continuous local martingale such that $M_0 = 0$, and such that $\langle M\rangle_t \to +\infty$ as $t \to \infty$. Then there exists a one-dimensional Brownian motion $(B_t)_t$ such that*

$$M_t = B_{\langle M\rangle_t}.$$

Theorem 7.11. *Let $(M_t)_t$ be a d-dimensional continuous local martingale such that $M_0 = 0$, and such that*

$$\langle M^i, M^j\rangle_t = 0 \qquad \forall i \neq j.$$

Then there exists a d-dimensional Brownian motion $(B_t)_t$ such that

$$M_t^i = B^i_{\langle M^i\rangle_t}.$$

7.3 Girsanov's Theorem(s)

While we can construct in full generality a filtered space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ satisfying the usual conditions, it may be difficult to use for concrete computations. It is often convenient to work on the canonical Wiener space seen in Chapter 1, namely

$$\Omega = C(\mathbb{R}_+, \mathbb{R}^d), \qquad \mathcal{F} = \mathcal{B}(\Omega),$$

with $(\mathcal{F}_t)$ the augmented filtration associated with the canonical process

$$W_t : (w_s)_{s\geq 0} \longmapsto w_t.$$

Roughly, Girsanov's theorem(s) allow one to:

1) compare local martingale or semimartingale laws under a change of measure on Ω,

2) relate distributions of solutions of SDEs, and in particular derive useful tools for studying sampling and generative stochastic models.

In these notes, we introduce a "standard" Girsanov theorem. We should mention that many extensions exist, in particular ones that allow to compare solutions of SDEs. We start with a few remarks.

Let $(L_t)_{t\geq 0}$ be a continuous local martingale. We define the associated exponential local martingale by

$$\mathcal{E}(L)_t = \exp\left(L_t - \frac{1}{2}\langle L\rangle_t\right).$$

We have seen that non-negative local martingales are supermartingales, so $(\mathcal{E}(L)_t)_{t\geq 0}$ is a supermartingale. The following criterion ensures that it is a uniformly integrable martingale.

Theorem 7.12. *Let $(D_t)_t$ be a non-negative continuous supermartingale. Let $D_\infty = \lim_{t\to\infty} D_t$ almost surely, and suppose*

$$\mathbb{E}[D_\infty] = \mathbb{E}[D_0] = 1.$$

Then $(D_t)_t$ is uniformly integrable.

Proof. First, for $s \geq t$,

$$\mathbb{E}[D_s \mid \mathcal{F}_t] \leq D_t.$$

Taking $s \to \infty$ and using Fatou gives

$$\mathbb{E}[D_\infty \mid \mathcal{F}_t] \leq D_t \qquad \text{a.s.}$$

We now show that this is in fact an equality. Indeed, by assumption,

$$0 = \mathbb{E}\big[\mathbb{E}[D_\infty \mid \mathcal{F}_t] - D_t\big]\mathbf{1}_{\{\mathbb{E}[D_\infty|\mathcal{F}_t]-D_t\leq 0\}},$$

which forces

$$\mathbb{E}[D_\infty \mid \mathcal{F}_t] = D_t \qquad \text{a.s. for all } t.$$

Therefore $(D_t)_t$ is a martingale and, being equal to the conditional expectation of D_∞, it is uniformly integrable. □

If $\mathbb{E}[\mathcal{E}(L)_\infty] = 1$, we can then consider the change of measure, i.e. the measure $\mathbb{Q}$ on $(\Omega, \mathcal{F})$ with density

$$\frac{d\mathbb{Q}}{d\mathbb{P}}(\omega) = \mathcal{E}(L)_\infty(\omega).$$

Girsanov's theorem relates local martingales under $\mathbb{P}$ and under $\mathbb{Q}$.

Theorem 7.13. *Let $(L_t)_t$ be a continuous local martingale such that*

$$\mathbb{E}[\mathcal{E}(L)_\infty] = 1.$$

Then $\mathbb{Q}$, with density $d\mathbb{Q}/d\mathbb{P} = \mathcal{E}(L)_\infty$, is a probability measure on $(\Omega, \mathcal{F})$, and if $(M_t)_t$ is any continuous local martingale under $\mathbb{P}$, then

$$\big(M_t - \langle M, L\rangle_t\big)_{t\geq 0}$$

is a continuous local martingale under $\mathbb{Q}$.

Theorem 7.14. *Let $(L_t)_t$ be a continuous local martingale such that*

$$\mathbb{E}[\mathcal{E}(L)_T] = 1, \qquad T \geq 0.$$

Then $\mathbb{Q}^L$, with density $d\mathbb{Q}^L/d\mathbb{P} = \mathcal{E}(L)_T$, is a probability measure equivalent to $\mathbb{P}$ on $(\Omega, \mathcal{F}_T)$. In addition, for any continuous local martingale $(M_t)_t$ under $\mathbb{P}$,

$$\big(M_t - \langle M, L\rangle_t\big)_{t\leq T}$$

is a continuous local martingale under $\mathbb{Q}^L$.

Remark 7.15. Note that the assumptions above are quite restrictive, and this is the first time we have used a Brownian motion in the argument. Indeed, such a result without a Brownian motion is in general wrong. Consider the equation

$$x_t = t + \int_0^t \mathbf{1}_{\{x_s>0\}}\, ds.$$

One can show that there is no solution on any interval. If this were true, consider

$$\tau = \inf\{t \in [0,T] : x_t > 0\},$$

with the convention $\inf \varnothing = \infty$. If $\tau = \infty$, then $x_t = 0$ for all t, absurd. If $\tau < \infty$, by continuity $x_\tau = 0$, and hence

$$x_t = \int_\tau^t \mathbf{1}_{\{x_s > 0\}}\, ds = 0 \qquad \forall t \geq \tau,$$

again absurd.

Theorem 7.16. *Let $(B_t)_t$ be a Brownian motion and let $f \in L^2([0,T], dt)$, $T > 0$. For any measurable bounded $\Phi : C([0,T], \mathbb{R}) \to \mathbb{R}$,*

$$\mathbb{E}\left[\Phi\left(\left(B_t + \int_0^t f(s)\, ds\right)_{t\in[0,T]}\right)\right] = \mathbb{E}\Big[L_T\, \Phi\Big((B_t)_{t\in[0,T]}\Big)\Big],$$

where

$$L_t = \exp\left(\int_0^t f(s)\, dB_s - \frac{1}{2}\int_0^t f(s)^2\, ds\right).$$

Proof. We consider

$$L_t = \mathcal{E}\left(\int_0^\cdot f(s)\, dB_s\right)_t.$$

By assumption,

$$\left\langle \int_0^\cdot f(s)\, dB_s \right\rangle_T = \int_0^T f(s)^2\, ds < \infty.$$

Therefore $\mathbb{E}[L_T] = 1$, and Girsanov's theorem implies that

$$\widetilde{B}_t := B_t - \int_0^t f(s)\, ds$$

is a Brownian motion under $\mathbb{Q}^L$ with

$$\frac{d\mathbb{Q}^L}{d\mathbb{P}} = L_T.$$

Therefore, for any bounded measurable Ψ,

$$\mathbb{E}^{\mathbb{P}}\Big[\Psi\Big((B_t)_{t\in[0,T]}\Big)\Big] = \mathbb{E}^{\mathbb{Q}^L}\left[\Psi\left(\left(\widetilde{B}_t + \int_0^t f(s)\, ds\right)_{t\in[0,T]}\right)\right].$$

Applying this with $\Psi = \Phi$ and transferring $\mathbb{Q}^L$ back to $\mathbb{P}$ yields the claim. □

Remark 7.17. For $T > 0$, $t \geq 0$ finite, and $A \in \mathcal{F}_t$,

$$\mathbb{Q}^L(A) = \mathbb{E}[\mathbf{1}_A \mathcal{E}(L)_T].$$

Girsanov's theorem implies that semimartingales under $\mathbb{P}$ are also semimartingales under $\mathbb{Q}^L$, and that a result which is true $\mathbb{P}$-almost surely is also $\mathbb{Q}^L$-almost surely true since $\mathbb{Q}^L \ll \mathbb{P}$.

Remark 7.18. If $(B_t)_t$ is a $(\mathbb{P}, (\mathcal{F}_t))$-Brownian motion, then

$$\big(B_t - \langle L, B\rangle_t\big)_t$$

is a continuous local martingale under $\mathbb{Q}^L$ with the same bracket as $(B_t)_t$. Hence, by Lévy's characterization, it is a $(\mathbb{Q}^L, (\mathcal{F}_t))$-Brownian motion.

Proposition 7.19. *Let $(L_t)_t$ be a continuous martingale with $L_0 = 0$. Then the following implications hold for $T \in \mathbb{R}_+ \cup \{\infty\}$:*

(i) (Novikov criterion)
$$\mathbb{E}\Big[e^{\frac{1}{2}\langle L\rangle_T}\Big] < \infty.$$

(ii) (Kazamaki criterion) $(e^{L_t/2})_{t\in[0,T]}$ *is uniformly integrable and*
$$\mathbb{E}\Big[e^{L_T/2}\Big] < \infty.$$

(iii)
$$\mathbb{E}[\mathcal{E}(L)_T] = 1.$$

Proof. The proof can be skipped in a first reading. We only consider the case $T = \infty$. □

Theorem 7.20. *A family of real random variables $\{X_i : i \in I\}$ is uniformly integrable if and only if*
$$\sup_{i\in I} \mathbb{E}[|X_i|] < \infty$$
and for every $\varepsilon > 0$ there exists $\delta > 0$ such that for any $A \in \mathcal{F}$, if $\mathbb{P}(A) \le \delta$, then
$$\sup_{i\in I} \mathbb{E}[|X_i|\mathbf{1}_A] \le \varepsilon.$$

Application of Girsanov's theorem.

Proposition 7.21. *Let $b : \mathbb{R}_+ \times \mathbb{R}^d \to \mathbb{R}^d$ be measurable and bounded. Then there exists a probability space*
$$(\overline{\Omega}, \overline{\mathcal{F}}, (\overline{\mathcal{F}}_t), \overline{\mathbb{Q}})$$
supporting a Brownian motion $(B_t)_{t\ge0}$ and a continuous adapted process $(X_t)_{t\ge0}$ such that
$$X_t = x_0 + B_t + \int_0^t b(s, X_s)\, ds, \qquad \forall t \ge 0.$$

Proof. We only restrict the proof to a finite horizon. Let $T > 0$ and consider a Brownian motion $(X_t)_{t\in[0,T]}$ started at x_0. Define, for $t \le T$,
$$L_t = -\int_0^t b(s, X_s) \cdot dX_s.$$
Then
$$\langle L\rangle_t = \int_0^t \|b(s, X_s)\|^2\, ds \le t \sup_{s,x} \|b(s, x)\|^2,$$
so by boundedness of b,
$$\mathbb{E}[\mathcal{E}(L)_t] = 1.$$
Under the measure $\mathbb{Q}^L$ defined by
$$\frac{d\mathbb{Q}^L}{d\mathbb{P}} = \mathcal{E}(L)_T,$$
the process
$$B_t := X_t - x_0 - \int_0^t b(s, X_s)\, ds$$
is a continuous local martingale. Moreover, under $\mathbb{Q}^L$,
$$\langle B^i, B^j\rangle_t = \langle X^i, X^j\rangle_t = t\,\delta_{i,j}.$$
Hence $(B_t)_{t\in[0,T]}$ is a Brownian motion under $\mathbb{Q}^L$. □

Remark 7.22. To extend the previous proof to $\mathbb{R}_+$, one needs a stronger result ensuring that for a family of probability measures $(\mathbb{P}_t)_{t\geq 0}$ on the canonical space there exists $\mathbb{Q}$ such that $\mathbb{Q}|_{\mathcal{F}_t} = \mathbb{P}_t$ for all $t \geq 0$. One also needs to extend Girsanov's theorem to this setting; the notes refer to Revuz and Yor, Chapter VIII.

8 Stochastic Differential Equations

8.1 SDE: Existence and Pathwise Uniqueness

Example 8.1 (Linear SDE). Let $(B_t)_t$ be a m-dimensional Brownian motion and $A \in \mathbb{R}^{d\times d}$, $C \in \mathbb{R}^{d\times m}$. The d-dimensional equation

$$X_t = X_0 + \int_0^t AX_s\,ds + \int_0^t C\,dB_s$$

is called a linear SDE and is often written as

$$dX_t = AX_t\,dt + C\,dB_t.$$

We can easily verify by Itô's formula that a solution is given by

$$X_t = e^{At}X_0 + \int_0^t e^{A(t-s)}C\,dB_s.$$

Furthermore, pathwise uniqueness holds, i.e. if $(Y_t)_t$ is another solution to the linear SDE, then

$$X_t - Y_t = \int_0^t A(X_s - Y_s)\,ds.$$

So differentiating with respect to t, we get almost surely for all t,

$$\frac{d}{dt}(X_t - Y_t) = A(X_t - Y_t), \qquad X_0 - Y_0 = 0.$$

By uniqueness of solutions of linear ODEs, it follows that almost surely for all t, $X_t = Y_t$.

We extend the notation of SDEs as follows.

Definition 8.2. *Let*

$$b : \mathbb{R}_+ \times \mathbb{R}^d \to \mathbb{R}^d \qquad \text{and} \qquad \sigma : \mathbb{R}_+ \times \mathbb{R}^d \to \mathbb{R}^{d\times m}$$

be measurable maps. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space endowed with a m-dimensional Brownian motion $(B_t)_t$ with augmented filtration $(\mathcal{F}_t)_{t\geq 0}$. Let X_0 be a d-dimensional random variable.

Definition 8.3. *We say that $(X_t)_{t\geq 0}$ is a strong solution of*

$$\mathcal{E}(X_0, \Omega, (\mathcal{F}_t), \mathbb{P}, (B_t), b, \sigma)$$

if it is $(\mathcal{F}_t)$-adapted, almost surely continuous, and

$$X_t = X_0 + \int_0^t b(s, X_s)\,ds + \int_0^t \sigma(s, X_s)\,dB_s.$$

Equivalently, its components satisfy almost surely for all t and every i,

$$X_t[i] = X_0[i] + \int_0^t b(s, X_s)[i]\,ds + \sum_{j=1}^m \int_0^t \sigma(s, X_s)[i, j]\,dB_s[j].$$

Definition 8.4. *We say that pathwise uniqueness holds for $\mathcal{E}_{X_0}(b, \sigma)$ if for any other strong solution*

$$\mathcal{E}(X_0, \Omega, (\mathcal{F}_t), \mathbb{P}, (B_t), b, \sigma),\ (Y_t)_{t\geq 0},$$

it holds that almost surely for all t, $X_t = Y_t$, i.e. $(X_t)_t$ and $(Y_t)_t$ are indistinguishable.

Definition 8.5. *We say that there is weak existence of solutions for $\mathcal{E}_{X_0}(b,\sigma)$ if there exists a filtered probability space $(\Omega,\mathcal{F},\mathbb{P},(\mathcal{F}_t))$ endowed with a $(\mathcal{F}_t)$-Brownian motion $(B_t)_t$, and a process $(X_t)_t$, such that X is a solution of*

$$\mathcal{E}(X_0,\Omega,(\mathcal{F}_t),\mathbb{P},(B_t),b,\sigma).$$

Remark 8.6. While it is intuitive to think that $(X_t)_t$ is adapted to the filtration generated by $(B_t)_t$, this is sometimes wrong. That is why we distinguish the two types of solutions. We denote from now on by $(\mathcal{F}_t^{X,B})_t$ the augmented canonical filtration associated with X and $(B_t)_t$.

Proposition 8.7. *Let $T\subset\mathbb{R}_+$ or $T=[0,\infty)$. We suppose that there exist L and $M\geq 0$ such that, for every $t\in T$, or for every $t\in T$ with $t<\infty$, and every x,y,*

$$\|b(t,x)-b(t,y)\|+\|\sigma(t,x)-\sigma(t,y)\|\leq L\|x-y\|,$$

and

$$\|b(t,0)\|+\|\sigma(t,0)\|\leq M.$$

Remark 8.8. In fact, we prove existence of $(\mathcal{F}_t^B)_t$-adapted solutions under the assumption above.

The existence and pathwise uniqueness of solutions can be derived similarly to existence of solutions to ODEs, using Picard iteration methods on an appropriate Banach space. It is then not surprising to consider Lipschitz conditions of the form below.

We consider the space

$$L_T^2:=L^2([0,T]\times\Omega,\mathcal{P},dt\otimes\mathbb{P}),$$

where $\mathcal{P}$ is the progressive σ-field, and similarly

$$L_T^2(\mathbb{R}^d\times\Omega,\mathcal{P},dt\otimes\mathbb{P}).$$

We recall that for $T\in\mathbb{R}_+\cup\{\infty\}$, L_T^2 is a Hilbert space with inner product

$$\langle Y,Z\rangle_{L_T^2}=\int_0^T\mathbb{E}[\langle Y_t,Z_t\rangle]\,dt.$$

Theorem 8.9. *Suppose the previous assumption holds, and let $(\Omega,\mathcal{F},\mathbb{P},(\mathcal{F}_t))$ be endowed with a m-dimensional Brownian motion and its augmented canonical filtration $(\mathcal{F}_t)$. Then there exists a unique pathwise solution $(X_t)_{t\geq 0}$ of $\mathcal{E}_{X_0}(b,\sigma)$ such that*

$$(X_t)_t,\ (b(t,X_t))_t,\ (\sigma(t,X_t))_t\in L_T^2.$$

Proof. Existence. We prove existence for $T<\infty$. For $T=\infty$, this implies existence and uniqueness on every bounded interval and hence on $\mathbb{R}_+$.

We follow closely the usual proof of existence for ODEs. First define

$$\psi:(Y_t)_t\in L_T^2\longmapsto\psi((Y_t)_t)=\left(X_0+\int_0^t b(s,Y_s)\,ds+\int_0^t\sigma(s,Y_s)\,dB_s\right)_t.$$

We show that:

1) ψ is well-defined on L_T^2;

2) starting from $Y^0:=X_0\in L_T^2$, the sequence $(Y^{n+1})_t=\psi((Y^n)_t)$ is Cauchy in L_T^2, hence converges to $(X_t)_t$;

3) finally $(X_t)_{t\geq 0}$ is a solution of $\mathcal{E}_{X_0}(b,\sigma)$.

1) First, using the assumption, it is easy to see that for all s, y, there exists $C > 0$ such that

$$\|b(s,y)\| + \|\sigma(s,y)\| \le C(1 + \|y\|).$$

Therefore, if we denote $(Z_t)_t = \psi((Y_t)_t)$,

$$\|Z\|^2_{L^2_T} \le C \int_0^T \mathbb{E}[\|X_0\|^2 + T(1 + \|Y_t\|)^2]\, dt$$

for some constant $C_T > 0$, which implies that $\psi((Y_t)_t) \in L^2_T$.

2) Define $(Y^n_t)_{t\ge 0}$, $n \in \mathbb{N}$, as announced. Using the isometry and Cauchy–Schwarz inequality, for every $\widetilde{T} \le T$,

$$\|Y^{n+1} - Y^n\|^2_{L^2_{\widetilde{T}}} \le 2 \int_0^{\widetilde{T}} \mathbb{E}\left[\left\| \int_0^t \{b(s, Y^n_s) - b(s, Y^{n-1}_s)\}\, ds \right\|^2 + \left\| \int_0^t \{\sigma(s, Y^n_s) - \sigma(s, Y^{n-1}_s)\}\, dB_s \right\|^2 \right] dt.$$

Hence,

$$\|Y^{n+1} - Y^n\|^2_{L^2_{\widetilde{T}}} \le 4L^2 \widetilde{T} \int_0^{\widetilde{T}} \int_0^t \mathbb{E}[\|Y^n_s - Y^{n-1}_s\|^2]\, ds\, dt.$$

By induction on n, this implies

$$\|Y^{n+1} - Y^n\|^2_{L^2_{\widetilde{T}}} \le \frac{(4L^2T)^n}{n!} \|Y^1 - Y^0\|^2_{L^2_T}.$$

Therefore

$$\sum_n \|Y^{n+1} - Y^n\|_{L^2_T} < \infty,$$

which implies that $(Y^n)_n$ is Cauchy and so converges in L^2_T to some $(X_t)_{t\ge 0}$.

3) To prove that $(X_t)_{t\ge 0}$ is a solution of $\mathcal{E}_{X_0}(b, \sigma)$, we show that ψ is continuous. In fact, for any $(Y_t)_t$ and $(Z_t)_t$ in L^2_T and any $\widetilde{T} \le T$,

$$\|\psi(Y) - \psi(Z)\|^2_{L^2_{\widetilde{T}}} \le 4L^2T \|Y - Z\|^2_{L^2_{\widetilde{T}}},$$

and therefore ψ is Lipschitz, hence continuous. Since $Y^{n+1} = \psi(Y^n)$, passing to the limit gives $\psi(X) = X$.

Uniqueness. Now if $(X_t)_t$ and $(\widetilde{X}_t)_t$ are solutions of $\mathcal{E}_{X_0}(b, \sigma)$, we have, following the same estimates as above, for any $\widetilde{T} \le T$ and every n,

$$\|\psi^n(X) - \psi^n(\widetilde{X})\|^2_{L^2_{\widetilde{T}}} \le \frac{(4L^2T)^n}{n!} \|X - \widetilde{X}\|^2_{L^2_T}.$$

But $\psi^n(X) = X$ and $\psi^n(\widetilde{X}) = \widetilde{X}$. Taking $n \to \infty$, we conclude that

$$(X_t)_t = (\widetilde{X}_t)_t \qquad \text{in } L^2_T,$$

hence by continuity they are indistinguishable.

The only remaining detail is that any solution of $\mathcal{E}_{X_0}(b, \sigma)$ must belong to L^2_T and be continuous. Let $(Y_t)_t$ be such a solution and set

$$\tau_R = \inf\{t : \|Y_t\| \ge R\}.$$

Following the previous bounds, for any $\widetilde{T} \le T$ and R,

$$\mathbb{E}\left[\int_0^{\widetilde{T}} \|Y^{\tau_R}_t\|^2\, dt \right] \le 8\widetilde{T}\, \mathbb{E}[\|X_0\|^2] + 8(1 + \widetilde{T}) \int_0^{\widetilde{T}} \mathbb{E}[\|Y^{\tau_R}_t\|^2]\, dt.$$

Setting

$$u_{\widetilde{T}}^R = \int_0^{\widetilde{T}} \mathbb{E}[\|Y_t^{\tau_R}\|^2]\,dt,$$

we may write

$$u_{\widetilde{T}}^R \le 8(1+\widetilde{T}) \int_0^{\widetilde{T}} u_t^R\,dt + 8\widetilde{T}\,\mathbb{E}[\|X_0\|^2].$$

Gronwall's lemma implies

$$u_{\widetilde{T}}^R \le 8\widetilde{T}\,\mathbb{E}[\|X_0\|^2] \exp\bigl(8(1+T)T\bigr).$$

Hence

$$\mathbb{E}\left[\int_0^T \|Y_t^{\tau_R}\|^2\,dt\right] \le 8T\,\mathbb{E}[\|X_0\|^2] \exp\bigl(8(1+T)T\bigr).$$

Fatou's lemma completes the fact that necessarily $(Y_t)_t \in L_T^2$. Finally, if $(Y_t)_t$ is a solution in L_T^2, then it is continuous since both

$$t \longmapsto \int_0^t H_s\,ds \qquad \text{and} \qquad t \longmapsto \int_0^t H_s\,dB_s$$

are continuous almost surely on $[0,T]$. □

Remark 8.10. Let $(\mathcal{F}_t^B)_{t\ge 0}$ be the usual augmentation of $\sigma(X_0, B_s : s \le t)$. By construction, the solution $(X_t)_{t\ge 0}$ is $(\mathcal{F}_t^B)_{t\ge 0}$-adapted. We say that $(X_t)_t$ is a strong solution.

8.2 Weak Uniqueness and Strong Solutions

Definition 8.11. *We say that weak uniqueness holds for $\mathcal{E}_{X_0}(\sigma, b)$ if for any two solutions $(X_t)_t$ and $(\widetilde{X}_t)_t$ of $\mathcal{E}_{X_0}(\sigma, b)$ defined on possibly different probability spaces, the induced laws on the Wiener space coincide. More precisely, if*

$$\mu(A) = \mathbb{P}\bigl((X_t)_t \in A\bigr), \qquad \widetilde{\mu}(A) = \widetilde{\mathbb{P}}\bigl((\widetilde{X}_t)_t \in A\bigr), \qquad A \in \mathcal{W},$$

then

$$\mu = \widetilde{\mu}.$$

Example 8.12. We consider the SDE

$$dX_t = \operatorname{sign}(X_t)\,dB_t.$$

We can construct easily a solution starting from a one-dimensional Brownian motion $(X_t)_{t\ge 0}$ on $(\Omega, \mathcal{F}, \mathbb{P}, (\mathcal{F}_t))$. Then

$$\beta_t = \int_0^t \operatorname{sign}(X_s)\,dX_s$$

is a Brownian motion by Lévy's characterization, since $\langle \beta \rangle_t = t$, and

$$\int_0^t \operatorname{sign}(X_s)\,d\beta_s = \int_0^t \operatorname{sign}(X_s)^2\,dX_s = X_t - X_0.$$

In addition, weak uniqueness holds, since the distribution is that of a Brownian motion. Finally, pathwise uniqueness fails, since $(-X_t)_t$ is also a solution and

$$\int_0^t \mathbf{1}_{\{X_s=0\}}\,d\beta_s = 0 \qquad \forall t$$

(exercise).

Example 8.13. Let $b : \mathbb{R}_+ \times \mathbb{R}^d \to \mathbb{R}^d$ be measurable and bounded, and let $\sigma = \mathrm{Id}$. One can show that $\mathcal{E}_{x_0}(1, b)$ admits solutions, for instance by applying Girsanov's theorem. Following the same proof, one can also deduce weak uniqueness. Exercise.

Definition 8.14. *Let $(X_t)_t$ be a solution of $\mathcal{E}(\Omega, \mathcal{F}, \mathbb{P}, (\mathcal{F}_t)_t, X_0, (B_t)_t, b, \sigma)$. We say that it is a strong solution of $\mathcal{E}_{X_0}(b, \sigma)$ if it is adapted to the canonical augmented filtration generated by $(B_t)_t$.*

Remark 8.15. Solutions of $\mathcal{E}_{X_0}(b, \sigma)$ are not necessarily strong and, motivated by the previous terminology, they are then called weak solutions. However, this notion is a bit suspicious here and we will only talk about solutions when no confusion is possible.

Example 8.16. 1) Under the previous Lipschitz assumption, solutions of $\mathcal{E}_x(b, \sigma)$ are strong, based on the Yamada–Watanabe theorem below.

2) If we consider the equation

$$dX_t = \mathrm{sign}(X_t)\, dB_t,$$

the constructed solution is not strong. Indeed, one can show, using the introduced notion, that

$$\mathcal{F}_t^B \subset \mathcal{F}_t^{|X|} \neq \mathcal{F}_t^X,$$

so $(X_t)_t$ is not $(\mathcal{F}_t^B)$-adapted. Indeed, for $\varphi_\varepsilon(x) = (\varepsilon + x^2)^{1/2}$, Itô's formula gives

$$\varphi_\varepsilon(X_t) = \varepsilon + \int_0^t \varphi_\varepsilon'(X_s)\, dX_s + \frac{1}{2}\int_0^t \varphi_\varepsilon''(X_s)\, ds.$$

As φ_ε and φ_ε'' are even,

$$\left(\int_0^t \varphi_\varepsilon'(X_s)\, dX_s\right)_t$$

is $(\mathcal{F}_t^{|X|})$-adapted, so taking $\varepsilon \downarrow 0$, it converges to $(\beta_t)_t$, which is then $(\mathcal{F}_t^{|X|})$-adapted.

Theorem 8.17. *Suppose pathwise uniqueness holds for $\mathcal{E}_x(b, \sigma)$. Then:*

1) *weak uniqueness holds;*

2) *for any filtered probability space $(\Omega, \widetilde{\mathcal{F}}, \widetilde{\mathbb{P}}, (\widetilde{\mathcal{F}}_t)_t)$ endowed with a Brownian motion $(B_t)_t$, there exists a strong solution of $\mathcal{E}_x(\Omega, \widetilde{\mathcal{F}}, \widetilde{\mathbb{P}}, (\widetilde{\mathcal{F}}_t)_t, (B_t)_t, x, b, \sigma)$.*

8.3 Markov Property and Kolmogorov Equation

Definition 8.18 (Continuation)**.** *$(X_t)_t$ is said to satisfy the Markov property, with respect to $(\mathcal{F}_t)$, if for every measurable bounded $g : \mathbb{R}^d \to \mathbb{R}$,*

$$\mathbb{E}[g(X_t) \mid \mathcal{F}_s] = \mathbb{E}[g(X_t) \mid X_s] \qquad \forall s \leq t.$$

Theorem 8.19. *Suppose the previous Lipschitz assumption holds. Then for $x \in L^2$, the unique solution $(X_t)_t$ of $\mathcal{E}_x(b, \sigma)$ is $(\mathcal{F}_t)$-Markov.*

Proof. Note that we have for any $t \geq s$,

$$X_t = X_s + \int_s^t b(u, X_u)\, du + \int_s^t \sigma(u, X_u)\, dB_u.$$

Setting $Y_u = X_{s+u}$ and $\widetilde{B}_u = B_{s+u} - B_s$, with $\widetilde{B} \perp \mathcal{F}_s$, this yields

$$Y_t = Y_0 + \int_0^t \widetilde{b}(u, Y_u)\, du + \int_0^t \widetilde{\sigma}(u, Y_u)\, d\widetilde{B}_u,$$

where $\widetilde{b}(u,y) = b(u+s,y)$ and $\widetilde{\sigma}(u,y) = \sigma(u+s,y)$. By the Lipschitz assumption, we deduce that $(Y_t)_{t\geq 0}$ is the unique solution of $\mathcal{E}_{X_s}(\widetilde{b},\widetilde{\sigma})$, adapted to the usual augmentation generated by $(X_s, \widetilde{B}_t : t \leq s)$. Therefore for every $t \geq s$, there exists a measurable function $\psi_{s,t}$ such that

$$Y_{t-s} = \psi_{s,t}(X_s, (\widetilde{B}_r)_{r\in[0,t-s]}).$$

Since X_s is $\mathcal{F}_s$-measurable and $\widetilde{B}$ is independent of $\mathcal{F}_s$, for any measurable bounded g,

$$\begin{aligned}\mathbb{E}[g(X_t) \mid \mathcal{F}_s] &= \mathbb{E}[g(Y_{t-s}) \mid \mathcal{F}_s] \\ &= \mathbb{E}[g(\psi_{s,t}(X_s, (\widetilde{B}_r)_{r\in[0,t-s]})) \mid \mathcal{F}_s] \\ &= P_{s,t}g(X_s),\end{aligned}$$

where

$$P_{s,t}g(x) = \mathbb{E}\big[g(\psi_{s,t}(x, (\widetilde{B}_r)_{r\in[0,t-s]}))\big].$$

This is precisely the Markov property. □

We can refine a bit this property in case b and σ do not depend on time.

Theorem 8.20. *Suppose the previous assumption holds and b and σ do not depend on time. Then, for any measurable bounded g and $s \leq t$, there exists a measurable ψ^g_{t-s} such that*

$$\mathbb{E}[g(X^x_t) \mid \mathcal{F}_s] = \mathbb{E}[g(X^{X_s}_{t-s})] = \psi^g_{t-s}(X_s),$$

where $(X^x_u)_u$ and $(X^{X_s}_u)_u$ are solutions of $\mathcal{E}_x(b,\sigma)$ and $\mathcal{E}_{X_s}(b,\sigma)$, respectively.

Proof. The proof follows the same lines as the previous one and is left to the reader. □

Definition 8.21. *Assume the previous assumption, and denote by $(X^x_t)_t$ any solution of $\mathcal{E}_x(b,\sigma)$. We set, for any starting point $x \in \mathbb{R}^d$, bounded measurable g, and $t \geq 0$,*

$$P_t g(x) = \mathbb{E}[g(X^x_t)].$$

Remark 8.22. It is called the Markov semi-group associated with $\mathcal{E}(b,\sigma)$.

From the previous theorem, we obtain easily that $(P_t)_{t\geq 0}$ is a semi-group.

Corollary 8.23. *Under the previous assumption, for all $t, s \geq 0$,*

$$P_t \circ P_s = P_{t+s},$$

where $P_t \circ P_s$ denotes the composition of the two operators.